CP100

SAP Integration Suite and SAP Extension Suite, Introduction

PARTICIPANT HANDBOOK
INSTRUCTOR-LED TRAINING

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Typographic Conventions

American English is the standard used in this handbook.

The following typographic conventions are also used.

This information is displayed in the instructor's presentation	
Demonstration	
Procedure	
Warning or Caution	
Hint	
Related or Additional Information	
Facilitated Discussion	
User interface control	Example text
Window title	Example text

Contents

Course Overview

TARGET AUDIENCE

This course is intended for the following audiences:

- Application Consultant
- Development Consultant
- Super / Key / Power User
- Business User
- Business Process Architect
- Business Process Owner/Team Lead/Power User
- Developer
- Enterprise Architect
- Solution Architect

UNIT 1 The Intelligent Enterprise Strategy of SAP

UNIT OBJECTIVES

- Explain the levels: business process, applications and technology
- Explain the ideas and innovations of an experience management of an industry cloud
- Explain, that the SAP BTP covers two areas: integration and extension

Unit 1
Lesson 1

Adopting an Intelligent Enterprise Strategy

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain the levels: business process, applications and technology

The Strategy of an Intelligent Enterprise

In this lesson we will cover the following topics:

- SAP Intelligent Enterprise strategy in a nutshell.
- A deeper look at the three levels of Intelligent Enterprise.

SAP Intelligent Enterprise strategy in a nutshell

Intelligent enterprises apply:

- Advanced technologies
- Best practices
- Within agile and integrated business processes.
- This makes them resilient, successful, and sustainable.

A deeper look at the three levels of Intelligent Enterprise

Intelligent enterprises consists of 3 levels:

1. Business Processes
2. Applications
3. Technology

Let us take a closer look at the three individual areas in the following overview picture.

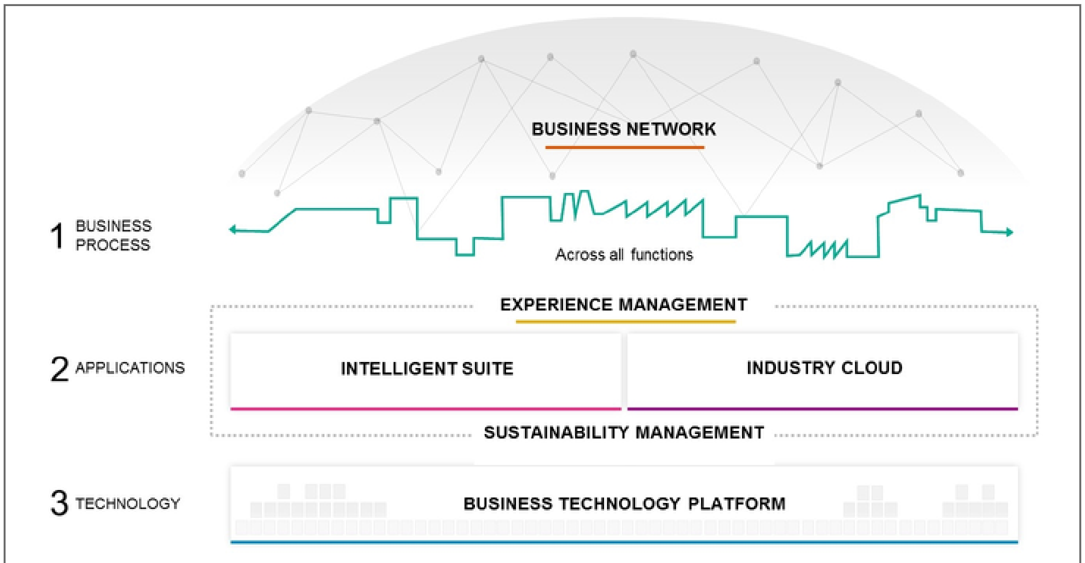


Figure 1: The Three Levels of the Intelligent Enterprise

Business Processes

The Level Business Process includes the Business Network. The SAP business network will help you to digitalize cross-company business processes. The network builds on current procurement, travel, and contingent workforce solutions to help intelligent enterprises work together to create flexible value chains.

Applications

The Applications includes the Experience Management, the Intelligent Suite and the Industrie Cloud. Experience Management understanding what people want, and how they feel is critical to making the right decisions. Experience management solutions give insight on the sentiments and feelings of customers, employees, and other business stakeholders.

Intelligent Suite

SAP offers an integrated suite of applications - the intelligent suite- that support your end-to-end business processes. These would be SAP S/4HANA with different deployment options, SAP 3 and the various SaaS solutions such as SAP SuccessFactors, SAP Concur, SAP Fieldglass and many more. SAP's industry cloud will enable you to discover and deploy vertical solutions from SAP and partners. Industry means „Core Business".

Industry Cloud

SAP's industry cloud will enable you to discover and deploy vertical solutions from SAP and partners. Industry - Means Core Business.

Technology

The Technology includes Business Technology Platform (BTP). Being best-run means running sustainably. SAP solutions for sustainability will help customers understand and manage their impact on people and the environment.

The BTP provides data management and analytics, supports application development and integration, and allows you to use intelligent technologies - such as artificial intelligence, machine learning, and the Internet of Things - to drive innovation.

Furthermore, we only want to take a closer look at the Industry Cloud and the Business Technology Platform.

Learn more about Business Network, Experience Management, and Business Suite on https://sap.com .

To the Point

Focused statement: Intelligent enterprises consists of 3 levels:

- Business Processes
- Applications
- Technology

Learn more: https://www.sap.com/products/intelligent-enterprise.html

LESSON SUMMARY
You should now be able to:

- Explain the levels: business process, applications and technology

Explaining the Experience Management of an Industry Cloud

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain the ideas and innovations of an experience management of an industry cloud

Ideas and Innovations of the Experience Management of an Industry Cloud

In this lesson we will cover the following topics:

- X-and O Data
- Industry - Means Core Business
- New API Framework of the Intelligent Enterprise – SAP Graph and its new Business Model
- Business Process Model
- Business and Technology Services
- Vertical Edge
- The Four Main Use Cases of SAP

X-and O Data

Facts about x-and o data:

- To get a **full view** of the **customers**, to improve your business, you need not only the (o) **opaque data** but also the (x) **behavioral data** of these customers.

This is summarized again in the following picture.

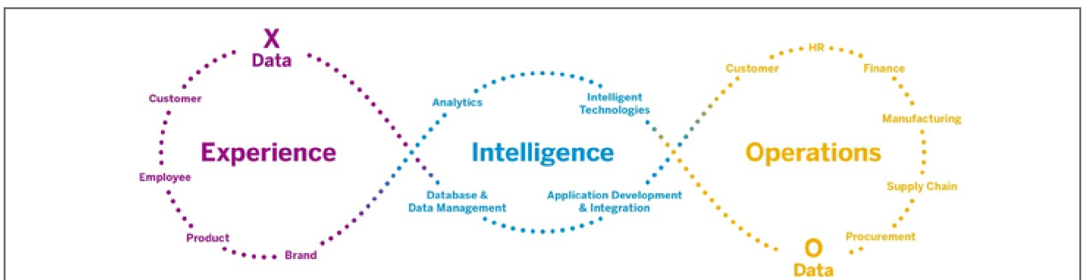


Figure 2: Intelligent Enterprises - Overview

On the left side you can see the experience data that describes the behavior of the business partner, employee, customer or product.

On the right side you can see the operational data, such as functional data of the different modules, as well as the technical data. In between is the Business Technology Platform which combines both data to a complete view of the customer.

In short:

(O) Operational data are:

All business data

All technical data

(X) Experience data are:

All data that directly affects the customer, emotions, opinions, perceptions and many more.

The NEW one:

is now to connect this information. This requires a Technology Platform that allows this, in a simple and wiseable way.

The technology platform is the SAP BTP.

It is a portfolio of integrated solutions that accelerate transformation of data into business value. It includes:

Database and data management

Application development

Integration

intelligent technologies

All technologies from on-premise to the cloud.

We will deal with the BTP in detail later.

Industry - Means Core Business

Industry cloud is our strategy:

To extend the intelligent suite with innovative industry cloud solutions.

Built by SAP and our partners in the cloud.

Based on SAP BTP.

Running on the cloud infrastructure of SAP and our hyperscaler partners.

The intelligent suite offers best practices and support for end to end processes for all industries as the foundation for:

efficient operations of finance,

material management,

supply chains,

or asset management.

The following picture illustrates the situation

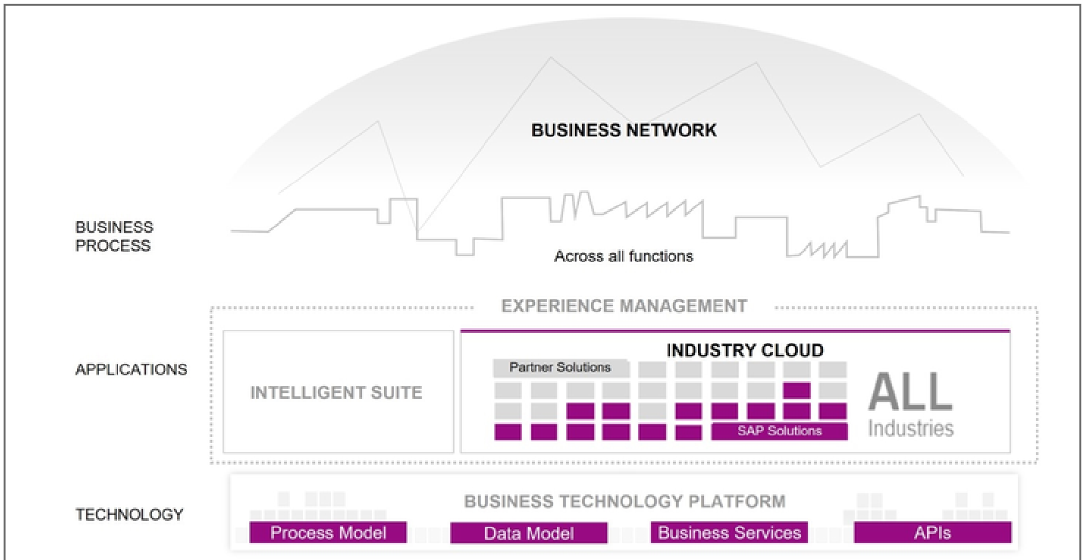


Figure 3: Ideas and Innovations of the Experience Management of an Industry Cloud

In the area of applications, you can see the innovative extensions to the right of the Intelligent Suite. At the moment the focus is on the 4 preliminary processes:

Design-to-Operate

Total Workforce Management

Lead-to-Cash

Source-to-Pay

The Intelligent Suite and also the innovative extensions run on the technology layer represented by the BTP.

The open business and technology platform supports in particular in the following areas:

Process Modell

Data Modell

Business Services

APIs

Let us take a closer look at the individual areas.

New API Framework of the Intelligent Enterprise – SAP Graph and its new Business Model

In addition to the well-known API's such as REST, OData and SOAP, there is a new innovative API framework based on OData.

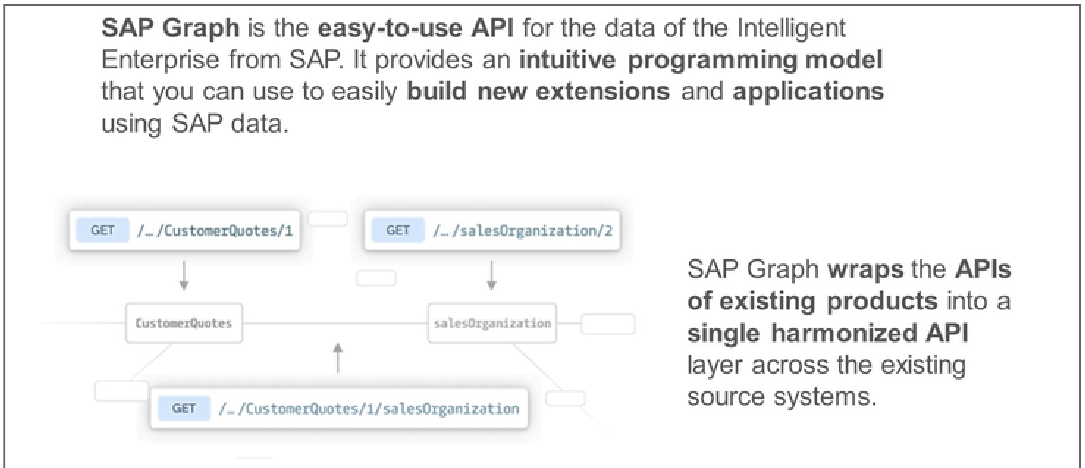


Figure 4: SAP Graph

More Details:

SAP Graph is an API for accessing SAP-managed data. It exposes the SAP ODM, a unified graph-like domain model of business objects (domain entities, like Products, Sales Orders and Customers) and relationships (links) to users of client applications, hence its name.

Application developers use SAP Graph as a RESTful service to navigate and access the data, regardless of where (in which SAP business or industry-specific system) this data resides, on premise or in the cloud.

SAP Graph uses open standards like OData v.4 as a single access protocol and OAuth for unified authentication. By unifying the data access to a single interface, the entry barrier for extending the Intelligent Enterprise gets lowered significantly. Overall, SAP Graph leads to higher productivity and a better development experience.

If the ODM provides all necessary data objects for your industry cloud solution, you must use SAP Graph. SAP Graph is designed for developers of extension applications.

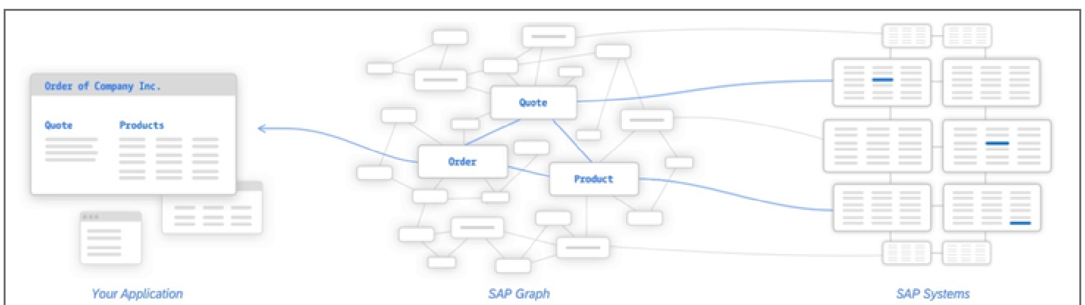


Figure 5: Schematic Overview of SAP Graph

The figure shows a schematic overview of SAP Graph.

SAP Graph combines the data from different business domains. Here -for example- products, quotas and orders and makes them available as OData. It is more than just views of technical data as it is based on a new business domain model.

Business Modell of SAP Graph

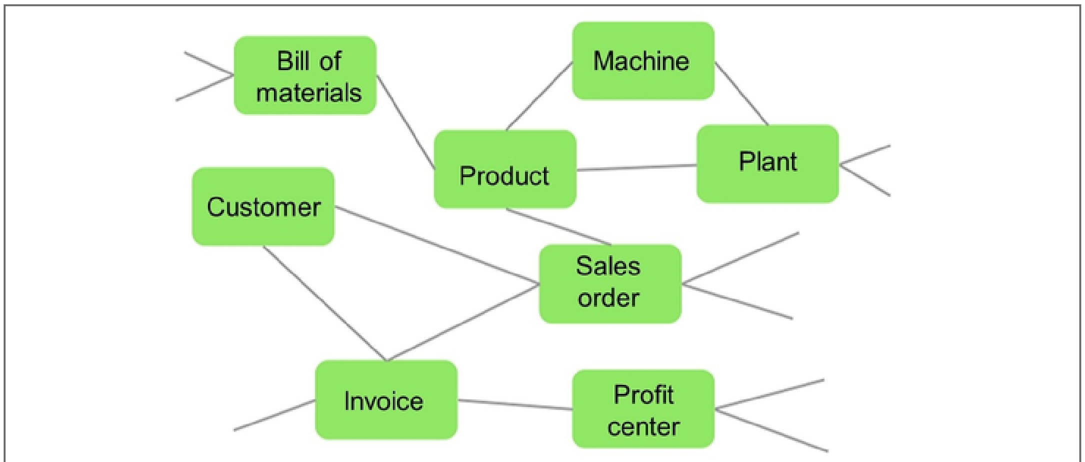


Figure 6: Schematic Overview of the new Business Model of SAP Graph

The model no longer consists of just one professional domain. Combined models are created along defined processes. This graph approach is used , to connect very large sets of distributed data, to create a dynamic schema and to determine the interconnections between the online actions for the Intelligent Enterprise processes.

SAP focuses on the following processes:

- Design-to-Operate
- Total Workforce Management
- Lead-to-Cash
- Source-to-Pay
- And several cross entities

Examples of cross entities are :

- Bank Identifiers
- Budget Periods
- Business Partners
- Checks
- Company Codes
- Cost Centers
- Country Codes
- Currency Codes
- And more- see **https://beta.graph.sap**

The Business Process Model

Business Process Model

The data model, domain model, and APIs are important to give industry cloud solutions access to the end-to-end process support in the intelligent suite. Understanding the effects of invoking an API that creates, deletes, or changes an object in the intelligent suite requires a business process model that describes the side effects of those operations.

SAP provides the business architecture framework serving as the foundation that helps developers design, build, test, and validate industry cloud solutions.

Each industry cloud solution also comes with content from SAP Model Company services and content that describes how to configure the intelligent suite to enable plugging in the solution.

Business and Technology Services

Industry cloud solutions have ready access to a broad range of business and technology services that make life easy for developers.

These range from simple business services such as currency or unit conversion to Leading edge artificial intelligence and machine learning libraries. With user experience libraries, solution designers can give users a seamless experience as they run end-to-end processes that span multiple industry cloud solutions, the intelligent suite, and business networks.

Subscription management, invoicing, usage monitoring, and application support are also managed through integrated central services.

Vertical Edge

Every path needs goals and waypoints. On the way to the Intelligent Enterprise, SAP describes goals and waypoints for several industries. The goals are known as vertical edges.

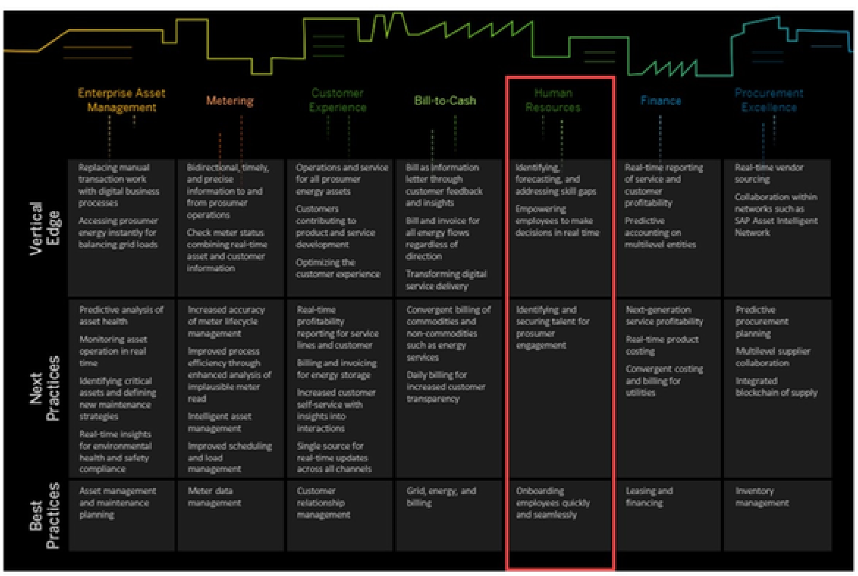


Figure 7: Innovation Spaces for the Industries

The individual industrial sectors are divided into columns. The journey begins from below. Taking Human Resources as an example, this looks like this.

Further Facts:

Best Practices - Now

Onboarding employees quickly and seamlessly.

Next Practices - Next Step

Identifying and securing talent for prosumer engagement.

Vertical Edge - Goal

Identifying, forecasting, and addressing skill gaps Empowering employees to make decisions in real time.

The Four Main Use Cases of SAP

In order to make the many innovations, ideas and innovations available to the customer, SAP applies this to 4 selected process.

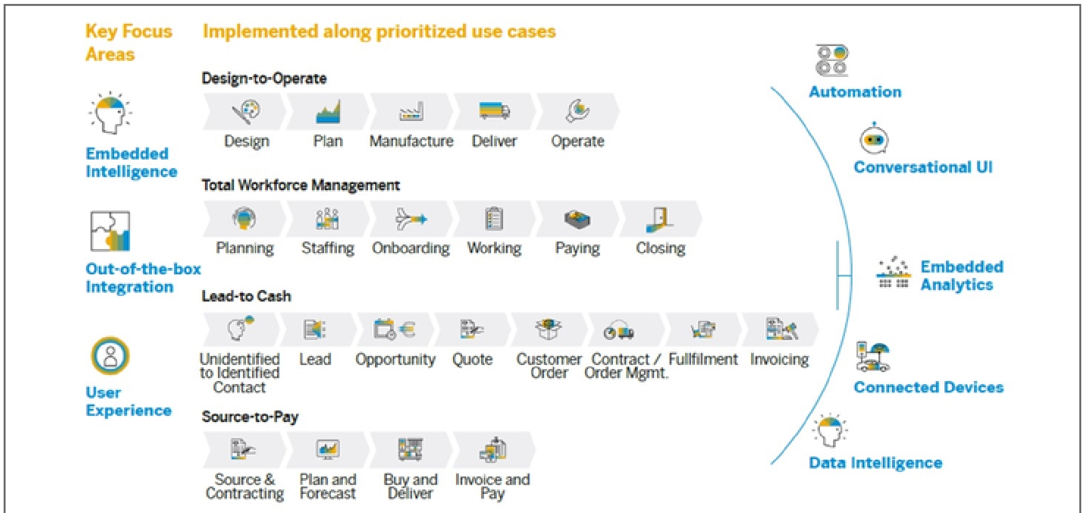


Figure 8: Main Processes in the Intelligent Enterprise

On the picture you can see the four main customer processes that are currently in the focus. The processes are divided into uses cases.

The processes are :

Design-to-Operate

Total Workforce Management

Lead-to-Cash

Source-to-Pay

Get Deeper

Design-to-Operate (D2O)

Is the Digital Supply Chain & Manufacturing contribution to the Intelligent Enterprise Suite and delivers a seamlessly integrated Supply Chain which is enriched by intelligent technologies across the entire Digital Supply Chain portfolio:

including Design,

Plan,

Manufacture,

Deliver,

and Operate solutions,

primarily from a Manufacturer and Operator perspective, with a principle focus on manufacturing industries in the initial delivery.

Total Workforce Management

Manages all aspects of the total workforces (employees and external workers) in line with company's business objectives and a clear line of sight into the financial impact. It enables the end-to-end business process of planning, sourcing, and managing the total workforce, and facilitates a holistic talent strategy to achieve agility and game-changing business outcomes.

The Total Workforce Management includes all the activities needed to maintain a productive workforce, such as:

the Core HR management,

performance and training,

management,

but also recruiting,

forecasting,

etc.

Lead to Cash

Typically one of the most important business process for many companies, because it is the process ultimately responsible for driving revenue for an organization. From a high level perspective, the Lead-to-Cash process connects a customer's interest to buy (SAP C/4HANA) to a company's realization of revenue (in the back-end SAP S/4HANA system), bringing the front office together with back office.

Source-to-Pay

While companies often focus on revenue, sourcing and procurement savings can increase profits about five times more. In the Intelligent Enterprise process definition, both the steps - Sourcing and Procurement are comprised - with Sourcing being the strategic process of ensuring, for example that the right suppliers get chosen. And Procurement being the operational process that takes care of the actual procurement of direct material, indirect materials, or services.

To the Point

Focused statement: Definition of o and x data:

(O) Operational data are:

All business data

All technical data

(X) Experience data are:

All data that directly affects the customer, emotions, opinions, perceptions and many more.

Focused statement: The 4 Process Templates where we are pioneers:

- Design-to-Operate
- Total Workforce Management
- Lead-to-Cash
- Source-to-Pay

Learn more:https://www.sap.com/industries/industry-cloud.html

LESSON SUMMARY
You should now be able to:

- Explain the ideas and innovations of an experience management of an industry cloud

Unit 1
Lesson 3

Explaining the SAP Business Technology Platform (BTP)

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain, that the SAP BTP covers two areas: integration and extension

SAP Business Technology Platform

The Business Technology Platform is the central cloud-based development platform.

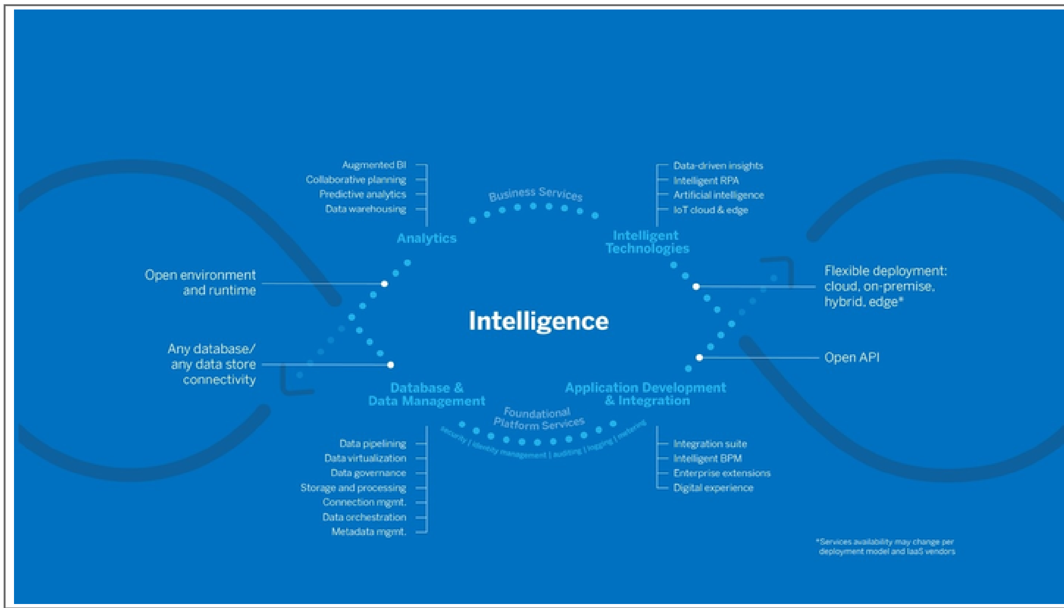


Figure 9: Bus Technology Platform

The figure illustrates, how business intelligence is in the center of all processes.

We will take a closer look at the following parts:

- Analytics
- Intelligent Technologies
- Application Development & Integration
- Database & Data Management
- Foundational Platform Services

Get Deeper

Business Analytics

Uncover deep insights and simplify access to critical information to drive better business outcomes and accelerate growth.

It focusses on:

Business Intelligence

Collaborative Planning

Data Warehousing

Intelligent Technologies

Ignite innovation in any area of your business and unlock the intelligent enterprise. Sometimes this is called the Innovation System.

The focus is on:

Artificial Intelligence

Internet of Things (IoT)

Blockchain

And More

Application Development and Integration

This is where the Integration Suite and Extension Suite are located.

Extend, integrate, and build enterprise applications with scalable, intelligent, and insightful Capabilities.

The focus is on:

Integration

Enterprise Extensions

Database and Data Management

Manage your enterprise data efficiently to optimize analytics and insights and improve business outcomes.

The focus is on:

SAP HANA and Databases

Data Intelligence and Orchestration

Master Data Management

Information Governance and Metadata Management

Data Quality and Integration

Foundational Platform Services

Services that maximize the flexibility and adaptability of your application investment with a multicloud, future-proof foundation that embraces all major hyperscaler infrastructures and cloud-based, native technologies.

The focus is on:

Authentication

Authorisation

Logging

Monitoring

And more

The two Main Areas of the Business Technology Platform

The Business Technology Platform focuses on Application Development and Integration

If we look at the Application Development and Integration section more closely, we can identify two basic use cases. The integration and expansion of heterogeneous landscapes, services and processes.

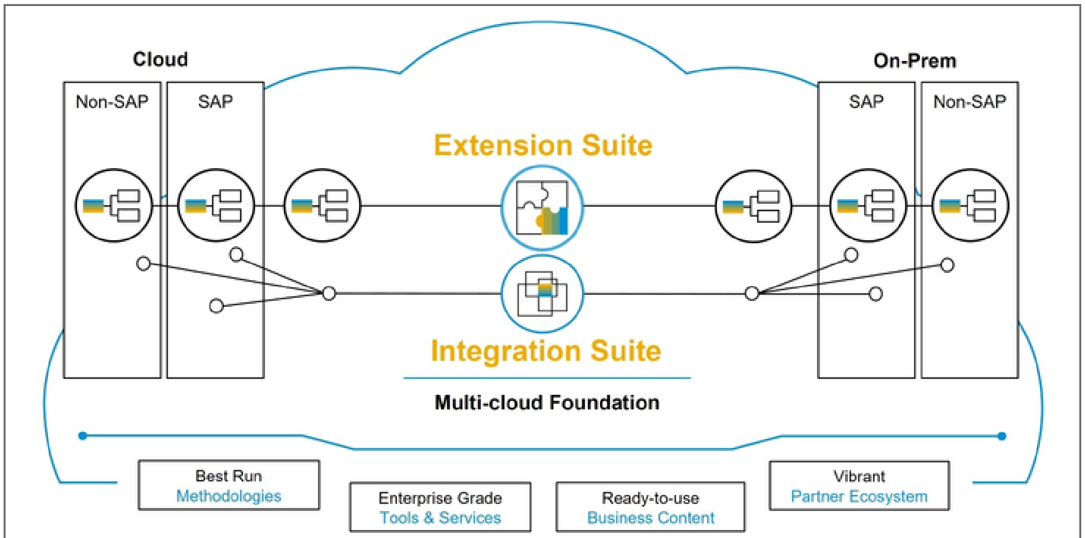


Figure 10: The two Main Areas of the SAP Business Technology Platform

The SAP BTP consists of two main areas:

The Integration Suite

The Extension Suite

On the left-hand side, you see all the cloud systems that are out there in the world, may it be an SAP system or a non SAP system.

And on the right-hand side, you see systems on premise. So here SAP systems with SAP S/4HANA or an SAP S/4 Enterprise Suite, for instance, but it could also be non-SAP systems.

The Integration Suite:

makes sure, that you have the right data at the right time in the right system whenever you need that.

The Extension Suite comes:

With best run methodologies,

And guidance,

And guidelines,

For the customers on how to implement a good user experience, a good process integration, and more.

So we don't just put the tools in place, but we also provide the guidance for the customers to leverage those tools and services and to achieve their goals much faster. Also, on the right-hand side, you see the ready-to-use business content.

So the SAP BTP is not just a set of tools, it also provides ready-to-use business content that helps you to integrate and extend in the best possible way and in the fastest way.

We will work intensively with the integration and extension suite in the following Units and Lessons.

To the Point

Focused statement: two suites are part of the SAP BTP:

The Integration Suite

The Extension Suite

Learn more: https://www.sap.com/products/business-technology-platform.html

LESSON SUMMARY
You should now be able to:

Explain, that the SAP BTP covers two areas: integration and extension

Learning Assessment

1. Which of the following statements is correct for an intelligent enterprise?

   Choose the correct answers.

   - [ ] **A** They use advanced technologies.
   - [ ] **B** They focus on marketing.
   - [ ] **C** They use best practices technologies.
   - [ ] **D** They use agile and integrated business processes.

2. Design-to-Operate is the Digital Planning contribution to the Intelligent Enterprise Suite and delivers a seamlessly integrated Supply Chain which is enriched by intelligent technologies across the entire Digital Supply Chain portfolio.

   Determine whether this statement is true or false.

   - [ ] True
   - [ ] False

3. Which services belong to the foundtional platform services?

   Choose the correct answers.

   - [ ] **A** Monitoring
   - [ ] **B** SAP HANA and Databases
   - [ ] **C** Authentication
   - [ ] **D** Authorisation

Learning Assessment - Answers

1. Which of the following statements is correct for an intelligent enterprise?

   Choose the correct answers.

   - [X] **A** They use advanced technologies.
   - [ ] **B** They focus on marketing.
   - [X] **C** They use best practices technologies.
   - [X] **D** They use agile and integrated business processes.

   This is correct.

2. Design-to-Operate is the Digital Planning contribution to the Intelligent Enterprise Suite and delivers a seamlessly integrated Supply Chain which is enriched by intelligent technologies across the entire Digital Supply Chain portfolio.

   Determine whether this statement is true or false.

   - [ ] True
   - [X] False

   This is correct.

3. Which services belong to the foundtional platform services?

   Choose the correct answers.

   - [X] **A** Monitoring
   - [ ] **B** SAP HANA and Databases
   - [X] **C** Authentication
   - [X] **D** Authorisation

   This is correct.

UNIT 2

The SAP Business Technology Platform (SAP BTP)

UNIT OBJECTIVES

- Introduce the basic components of SAP BTP
- Explain the commercial model of SAP BTP
- Explain the account model in Cloud Foundry
- Explain the difference between platform and business users
- Explore the most important components of the SAP BTP, ABAP environment
- Explain, what Kyma is, what components Kyma is made of and what you can do with it

Unit 2
Lesson 1

Introducing SAP BTP

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Introduce the basic components of SAP BTP

Basic Components of SAP BTP

In this lesson we will cover the following topics:

- SAP BTP in a nutshell
- SAP BTP Cockpit
- The Environments
- Which services are available?
- SAP BTP Reference Architecture

SAP BTP in a nutshell

SAP BTP is an enterprise platform-as-a-service (enterprise PaaS) that provides:

- Comprehensive application development services and capabilities.
- Which lets you build, extend, and integrate business applications in the cloud.

SAP BTP offers two types of global accounts, Trial accounts and enterprise accounts. In the following we will only deal with the enterprise accounts.

SAP BTP Cockpit

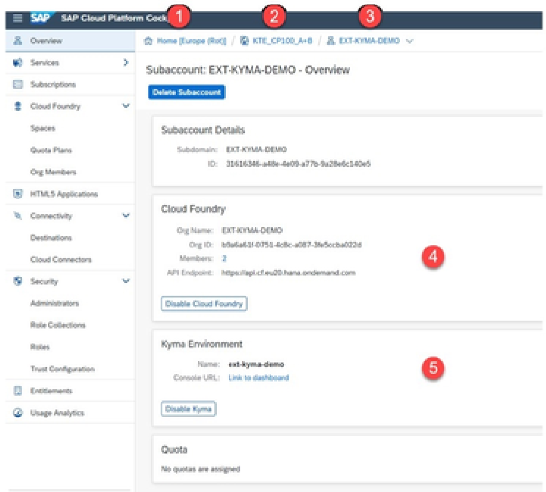


Figure 11: SAP BTP Cockpit of an Enterprise Account

On the screenshot you will see the following:

1. Region of the global account - here Europe Rot.
2. Global Account - here KTE_CP100_A+B.
3. Subaccount - here EXT-KYMA-DEMO.
4. Environment Cloud Foundry with organization and endpoint - here: https://api.cf.eu20.hana.ondemand.com (Netherlands Europe).
5. Environment Kyma with name ext-kyma-demo.

The Environments

Currently, the following environments are available:

Cloud Foundry (fully managed by SAP)

ABAP (platform as a Service)

Kyma (open Build-on approach)

More to come.

Cloud Foundry

SAP BTP, Cloud Foundry environment is an open Platform-as-a-Service (PaaS) targeted at microservice development and orchestration.

SAP Cloud Foundry should be used if you prefer a Managed Build-on approach. Cloud Foundry is fully managed by SAP and you benefit from a high level of abstraction from the underlying infrastructure.

You don't have to deal with technical aspects such as virtual machines, networking, monitoring, and more. It provides out-of-the-box integration into all mandatory kernel services.

Cloud Foundry runs on Kubernates (K8s).

ABAP

The ABAP environment is a **platform as a service** that allows you to extend existing ABAP-based applications and develop ABAP cloud apps decoupled from the digital core. You can leverage your ABAP know-how in the cloud and reuse existing ABAP assets by writing your source code with ABAP Development Tools for Eclipse.

Kyma

The **Kyma environment** (Isteio with K8s) allows you to extend existing SAP systems with your own Functions or microservices.

Kyma is a platform for extending applications with serverless functions and microservices. It provides a selection of cloud-native projects glued together to simplify the creation and management of extensions.

Kyma should be used for an Open Build-on approach. It provides you with more flexibility by using containers and Kubernetes, to build cloud native applications. By using Kubernetes, it will be easier to develop a solution that is highly scalable.

Which services are available?

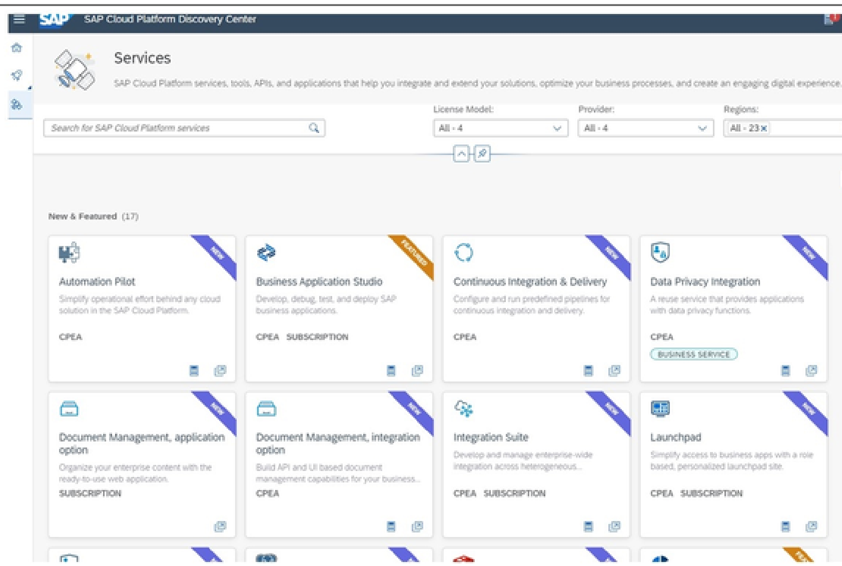


Figure 12: Screenshot of the Discovery Center

All services currently available can be found at https://discovery-center.cloud.sap/viewServices. There they are listed in the derivative of the regions, commercial models and more.

Not all Services are available at KYMA environment.

Note:
ABAP is also a service and not an environment.

SAP BTP Reference Architecture

So that you get an approximate overview of what the SAP BTP is from a technical point of view.

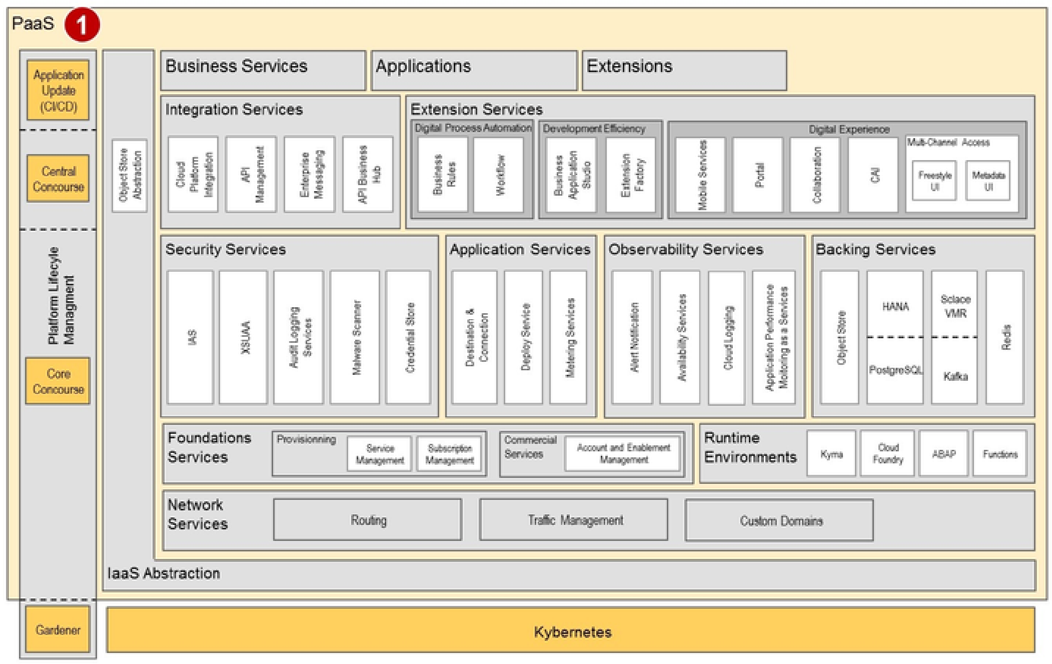


Figure 13: Overview of the SAP Cloud Foundry Reference Architecture, Part 1

Explanations:

1. The PaaS Layer provides all functionality, applications and services we can use direct or indirect. It runs on K8s.
2. The Virtualisation Layer, see figure below.
3. Hardware and Infrastructure (IaaS), see figure below.

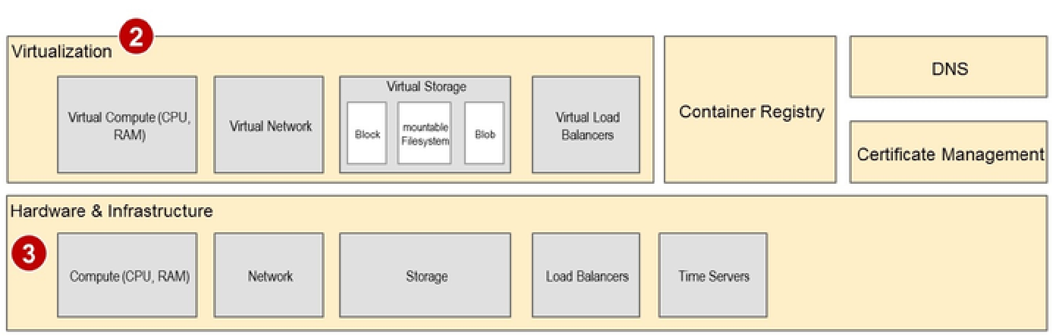


Figure 14: Overview of the SAP Cloud Foundry Reference Architecture, Parts 2 and 3

Explanations:

1. The PaaS Layer provides all functionality, applications and services we can use direct or indirect. It runs on K8s, see figure above.

2. The Virtualization Layer.

3. Hardware and Infrastructure (IaaS).

Get more in Detail

SAP sets the following priorities:

- Security
- Availability
- Operational Efficiency
- Performance
- Development Efficiency

Security

Customers, especially in managed cloud services scenarios, expect maximum security and data protection compliance when moving their business processes to the Cloud. SAP BTP architectures are and should be built in a way that SAP's Product Standard Security is fulfilled.

Availability

Customers also expect an "always-on" behavior for Cloud services. Therefore, SAP BTP targets a 99,9% availability

Operational Efficiency

Optimizing operations isn't just about cost optimization, but about empowering people to safely operate complex distributed systems

Performance

All SAP BTP services and solutions should define relevant performance KPIs, and constantly measure them, optimize them and keep them equal or better than the target values.

Development Efficiency

Hyperscalers do not list development efficiency as a quality, but for SAP BTP it is an important topic.

To the Point

Focused statement: Currently, the following environments are available:

- Cloud Foundry (fully managed by SAP).
- ABAP (platform as a Service).
- Kyma (open Build-on approach).

Learn more: https://help.sap.com/viewer/65de2977205c403bbc107264b8eccf4b/Cloud/en-US/c2fec62b49fa43b8bd945c85ecc2e5bd.html

LESSON SUMMARY
You should now be able to:

- Introduce the basic components of SAP BTP

Unit 2
Lesson 2

Explaining the Commercial Model of SAP BTP

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain the commercial model of SAP BTP

The Commercial Model of SAP BTP

In this lesson we will cover the following topics:

- The nature of Services
- Applications
- SAP BTP: Choice of Commercial Models
- Structure of the Commercial Models
- Consumption-Based Commercial Model: CPEA

The nature of Services

Services constitute SAP BTP as PaaS offering since major function blocks are implemented and exposed as service. Services can be accessed and managed via the Service Manager

Services have a provider and a consumer "view". For service providers it's an entity to "ship" functionality, manage the lifecycle and an operation unit. Consumers see it as re-use entity, for which an entitlement, configuration and access points (API and/or UI) are given. The default model for consuming services today is to separate service consumption from service provisioning completely. From a consumer perspective, this is ideal because the service is "provided as a Service" often-times with an SLA.

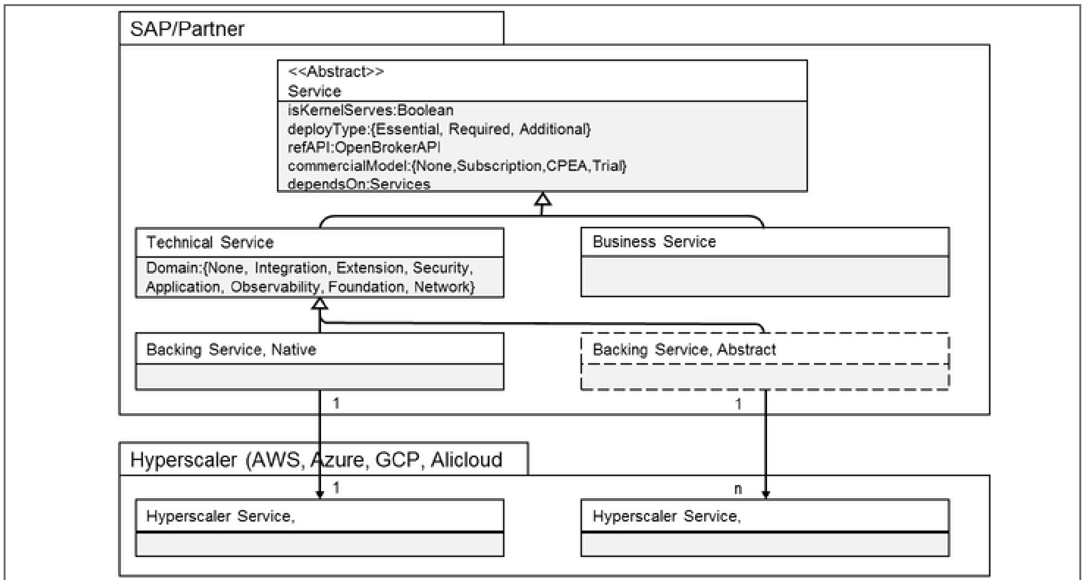


Figure 15: Provider View of a Service

Description of the figure:

A service is a Technical Service or a Business Service

Backing Services are either abstracted and exposed as Technical Service or it is natively consumed without any abstraction except the access via Service Manager (for example: ObjectStore, MS Embrace project with OSB implementation).

A service can be a Kernel Service

A service has a deploy type. Essential services are deployed and exposed in all SAP BTP data centers or availability zones respectively. Required services are deployed with the essential services since the essential services depend on them. Additional services are deployed on selected data centers or availability zones as needed on request. SAP BTP offers approximately 70 essential or required technical services today, which are deployed on all SAP BTP data centers and availability zones as defined in the service availability matrix at the discovery center.

Applications

Applications building on or integrating with SAP BTP can use the Service Discovery and Management Kernel Service (22), also known as Service Manager, for the provisioning/deprovisioning of service instances listed in the marketplace and creating/accessing/updating/deleting credentials for these service instances.

The actual implementations of these operations are service specific, and services implement them in service brokers that they register with the Service Manager.

SAP BTP: Choice of Commercial Models

Basically, there are two models:

The **Subscription model** licenses only the services you need.

In the **Consumption model** you can use all services but only pay for the use.

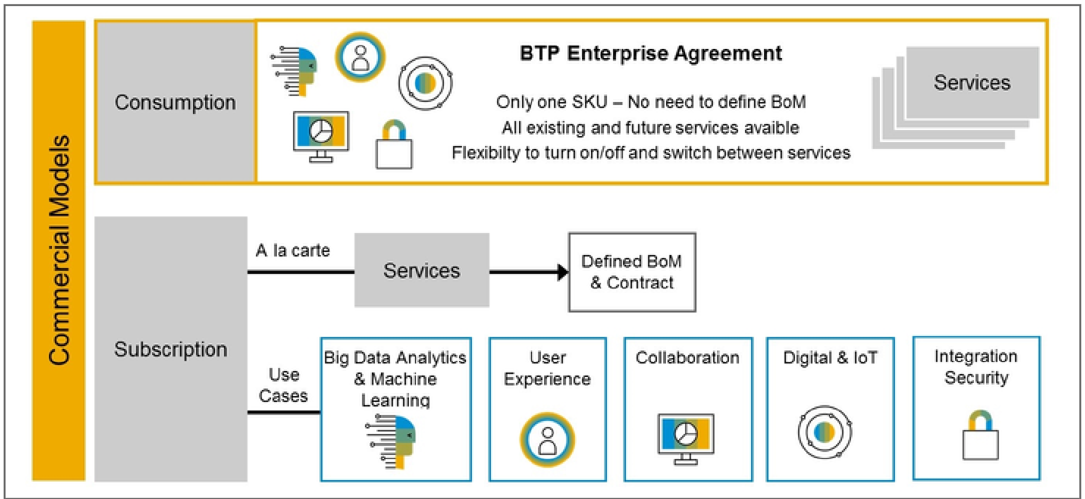


Figure 16: Commercial Models: Overview

Explanations about the figure:

Subscription

You plan and pay in advance for every service separately (high touch),whether you use it or not.

Certain contract duration (three month or more, typical three years!).

Coarse granularity for capacity (blocks of for example 100 users, 5.000 visits).

Consumption-based

You get access to all eligible services, without quotas or limitations.

Self-service activation and de-activation ("low touch").

Only pay for what you use.

Cloud Credits

Cloud credits are a pre-paid commitment for the consumption of SAP Cloud Services in a defined time period à unused cloud credits expire at the end of the phase and contract year.

Cloud credits are subject to discount.

Purchasing of top-up cloud credits at any point of time.

Structure of the Commercial Models

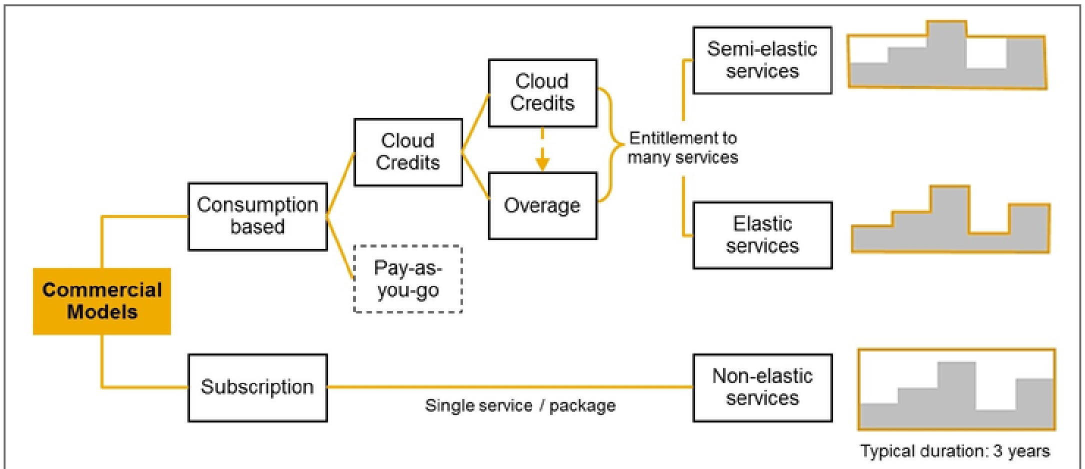


Figure 17: Structure of the Commercial Models

As you can see on the figure, the Consumption based Model is much more flexible. It offers Elastic services. The picture on the far right describes the ability of the model to adapt to load changes. The middle picture shows a completely flexible adjustment of the costs to the actual usage.

Consumption-Based Commercial Model - CPEA

CPEA is a Consumption-Based Commercial Model for SAP BTP.

Facts about the CPEA Consumption-Based Commercial Model:

- You get access to all eligible services, without quotas or limitations.
- Self-service activation and de-activation (“low touch”)
- Only pay for what you use.

Go deeper:

The consumption-based model decouples:

- Commercial agreements and
- Technical adoption

to help customers easily to:

- Find,
- Try,
- Buy,
- Consume and
- Pay

Facts about Cloud Credits:

Cloud credits are a pre-paid commitment for the consumption of SAP Cloud Services in a defined time period à unused cloud credits expire at the end of the phase and contract year.

Cloud credits are subject to discount.

Purchasing of top-up cloud credits at any point of time.

Cloud credits entitle you to flexible usage of all consumption-based services in the portfolio, as well as future (yet to be introduced) services.

Service usage is "debited" from the cloud credits.

Excess usage ("overage") is invoiced.

Overview - Price List

Under https://cloudplatform.sap.com/price-lists, you will find the currently valid price site, limited to countries.

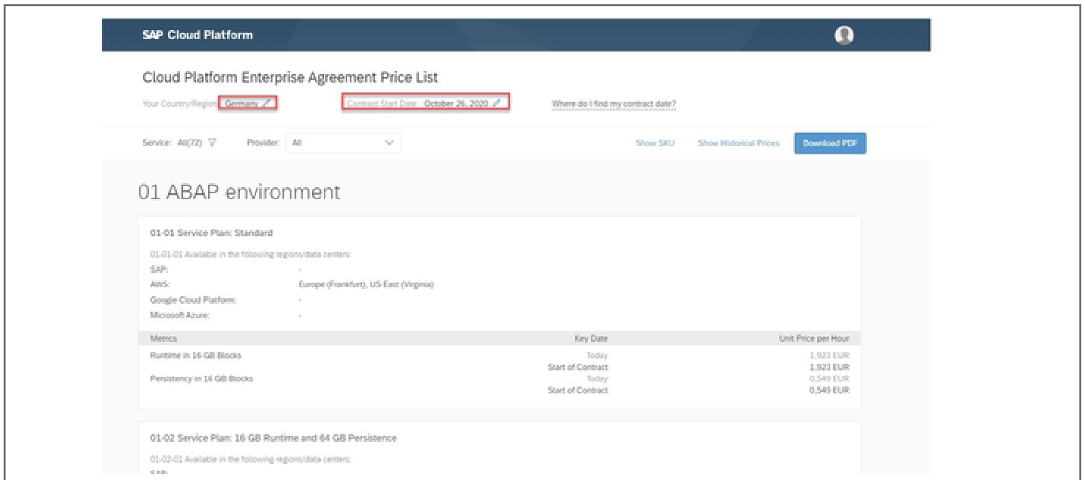


Figure 18: An example for the ABAP service was in October 2020

In the upper area you can see the previously selected region and the date. A little deeper under the Show Historical Prices tab you can see the price changes and many more on this side.

To the point

Focused statement: basically, there are two models:

The Subscription model licenses only the services you need.

In the Consumption model. (CPEA) you can use all services but only pay for the use:

- Cloud Credits
- Pay-as-you-go

Learn more: https://help.sap.com/viewer/65de2977205c403bbc107264b8eccf4b/Cloud/en-US/263d40009a5a4237a62e8f5c05ee641e.html

LESSON SUMMARY

You should now be able to:

- Explain the commercial model of SAP BTP

Unit 2
Lesson 3

Explaining the Runtime of SAP BTP, Cloud Foundry

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain the account model in Cloud Foundry
- Explain the difference between platform and business users

SAP BTP, Cloud Foundry Account Model

In this lesson we will cover the following topics:

- SAP BTP, Cloud Foundry in a Nutshell
- SAP BTP, Cloud Foundry Account Model
- Global Account
- Subaccounts
- Orgs, and Spaces
- Directories (Beta) [Feature Set B]
- Custom Properties [Feature Set B]
- User Management
- Platform Users
- Business Users

SAP BTP, Cloud Foundry in a Nutshell

SAP BTP, Cloud Foundry:

- should be used if you prefer a Managed Build-on approach. Cloud Foundry is **fully managed** by SAP and you benefit from a high level of abstraction from the underlying infrastructure.
- is the **industry-standard open source** cloud application platform for developing and deploying enterprise cloud applications.

You don't have to deal with technical aspects such as virtual machines, networking, monitoring, and more. It provides out-of-the-box integration into all mandatory kernel services.

SAP BTP, Cloud Foundry Account Model

In the following the account model.

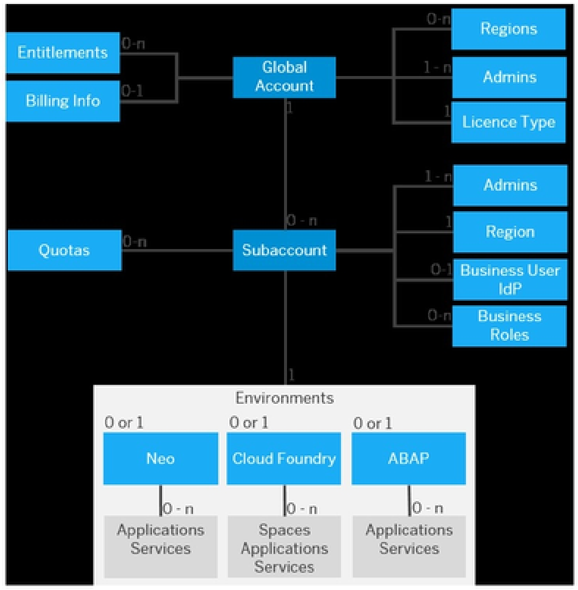


Figure 19: Account Model Overview

A global account has:

- 0.n Entitlements
- 0-n Regions (based on the subaccounts)
- 1-n Members with the role admins
- O-n license Types

On licence Type

A subaccount has:

- 0-n Quotas
- 1-n Business Users within the IDP
- 0-n Business Roles
- One Region

Cloud Foundry environment has:

- 0-n Spaces
- 0-n Applications
- 0-n Services

Global Account

Facts about the Global Account:

It is the realization of a **contract** you made **with SAP**.

Global accounts are region- and environment-independent.

Within a global account, you manage all of your subaccounts, which in turn are specific to one region.

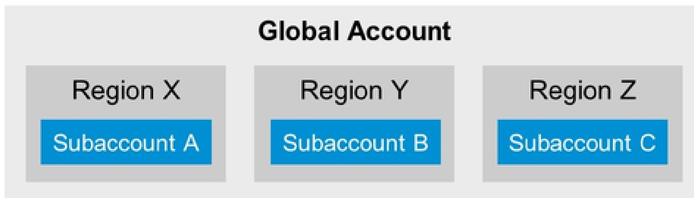


Figure 20: Global Account

A global account is used to manage subaccounts, members, entitlements and quotas. You receive entitlements and quotas to use platform resources per global account and then distribute the entitlements and quotas to the subaccount for actual consumption. There are two types of global accounts: enterprise accounts (paid) and trial accounts (free). The type determines pricing, conditions of use, resources, available services, scope of the functionality that you can use, and the level of support you can receive.

Sample

The following shows a real sample of an global account.

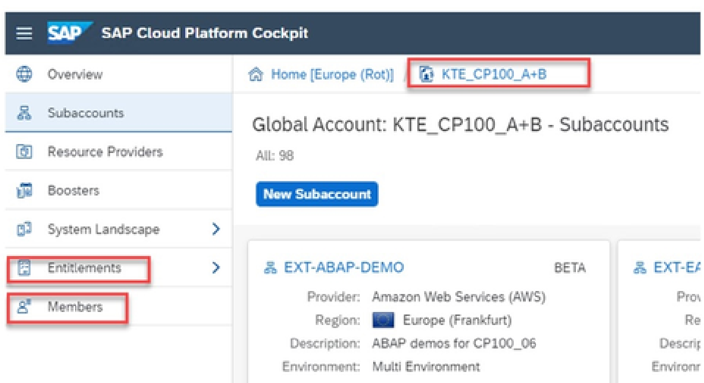


Figure 21: A Global Account Named KTE_CP100_A+B With Entitlements and Members

Here you can see the cockpit of a global account. At the top you can see the breadcrumb navigation with the name of the global account. On the left side the navigation with the global members and entitlements on global level.

Subaccounts

Facts about subaccounts:

Subaccounts let you **structure a global account** according to your organization's and project's requirements with regard to members, authorizations, and entitlements.

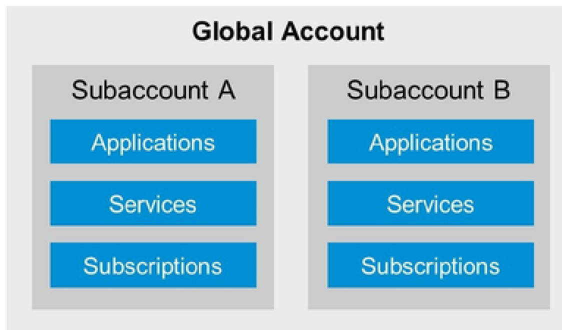


Figure 22: Subaccount

A global account can contain one or more subaccounts in which you deploy applications, use services, and manage your subscriptions. Subaccounts in a global account are independent from each other. This is important to consider with respect to security, member management, data management, data migration, integration, and so on, when you plan your landscape and overall architecture.

Sample

The following shows a real sample of an sub account.

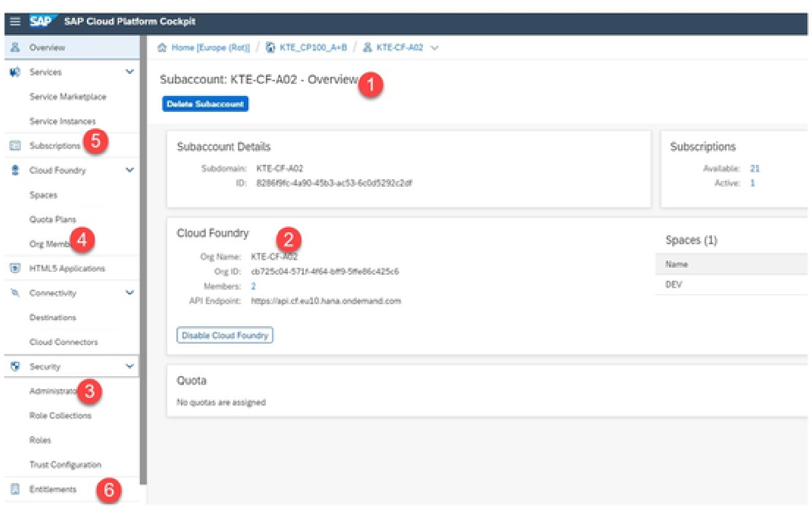


Figure 23: A Real Subaccount with Name KTE-CF-A02

Subaccount with:

1. Name

2. Runtime - here Cloud foundry
3. Security Admin environment independent
4. Org Member Cloud Foundry specifically
5. Apps as Subscriptions
6. Entitlements and Quotas

Orgs, and Spaces

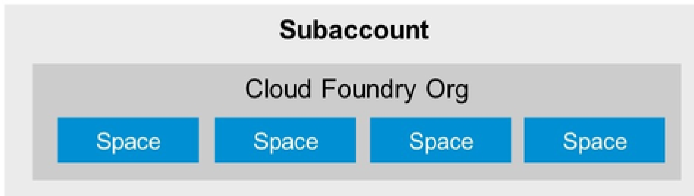


Figure 24: Orgs and Spaces

The subaccount and the org have a 1:1 relationship and the same navigation level in the cockpit (even though they may have different names). You can create spaces within that Cloud Foundry Org. Spaces let you further break down your account model and use services and functions in the Cloud Foundry environment.

Sample

In the following, a sample of an org and spaces of an real enterprise is shown.

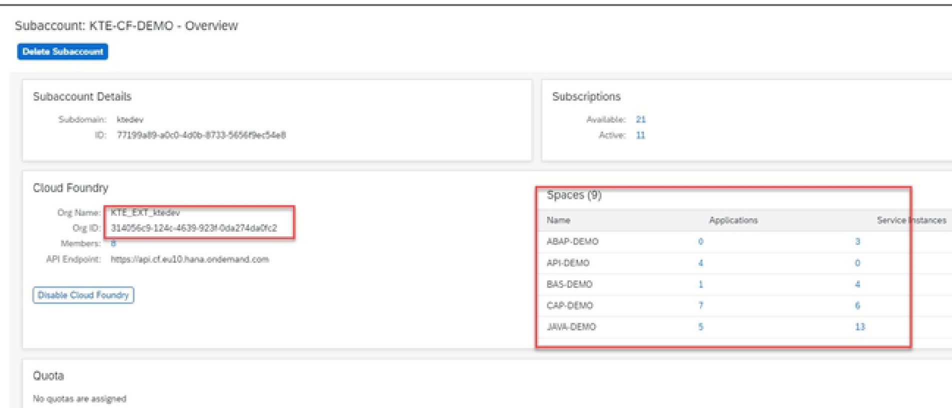


Figure 25: A Real Sample of Orgs and Spaces

Subaccount with:

Org Name and ID

With 9 spaces with deployed Apps

Directories (Beta) [Feature Set B]

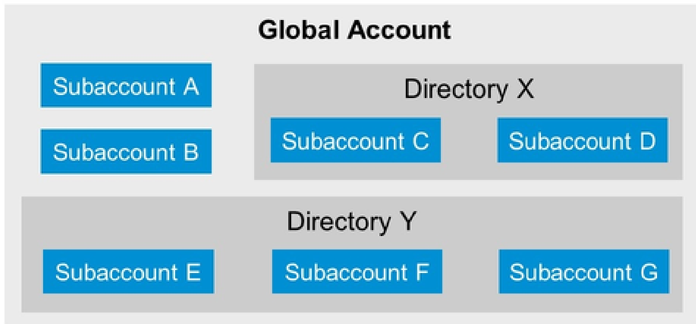


Figure 26: Use of Directories

Directories allow you to organize and manage your subaccounts according to your technical and business needs.

A directory can contain one or more subaccounts. It cannot contain other directories. Using directories to group subaccounts is optional - you can still create subaccounts directly under your global account.

Custom Properties [Feature Set B]

Use of Custom Properties:

Table 1: Use of Custom Properties

Custom Property (Name)	Property Values
Landscape	Dev, Test, Production
Department	Hr, IT, Finance, Sales
Cost Center	000001134789, 000002155534, To be defined
Flagged for Deletion	(no values)
Important	(no values)

Custom properties allow you to label or tag your directories and subaccounts according to your own business and technical needs. This makes organizing and filtering your directories and subaccounts easier within your global account.

You create and assign custom properties when you create or edit a directory or subaccount. Using custom properties is optional.

Users in SAP BTP, Cloud Foundry

A user account corresponds to a particular user in an identity provider, such as the SAP ID service and consists, for example, of an SAP user ID (S-user or P-user) and password.

User accounts enable users to log on to SAP BTP and access subaccounts and use services according to the permissions given to them.

It's important to understand the difference between the 2 types of users we are referring to:

Platform users

Business users

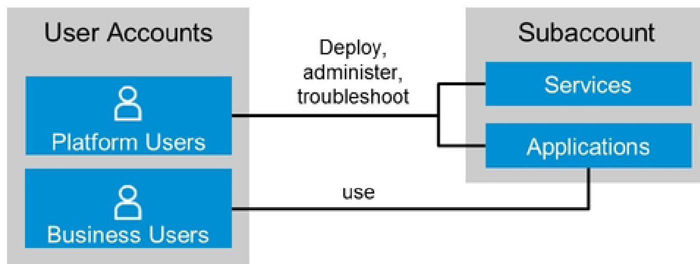


Figure 27: Different Types of Users

Platform users deploy , adminster and create apps and uses services to integrate or extend business functionality. Business User use this created apps- Both need to authenticate on a central place, for example SAP Identity Provider.

Platform Users

Facts about Users in SAP BTP, Cloud Foundry:

Platform users are usually developers, administrators or operators who deploy, administer, and troubleshoot applications and services.

They're the users that you give certain permissions for instance at global account or subaccount level, either by adding them as members with certain permissions or by assigning role collections to them .

Platform users who were added as members and who have administrative permissions can view and/or manage the list of global accounts, subaccounts, and Cloud Foundry orgs and spaces that are available to them, and access them using the cockpit or the command line interface.

For platform users, the default identity provider is SAP ID Service, but if you want to have subaccount members from your own user base, you can use your own identity authentication tenant with SAP Cloud Identity Authentication Service.

Sample - Authentication of platform users

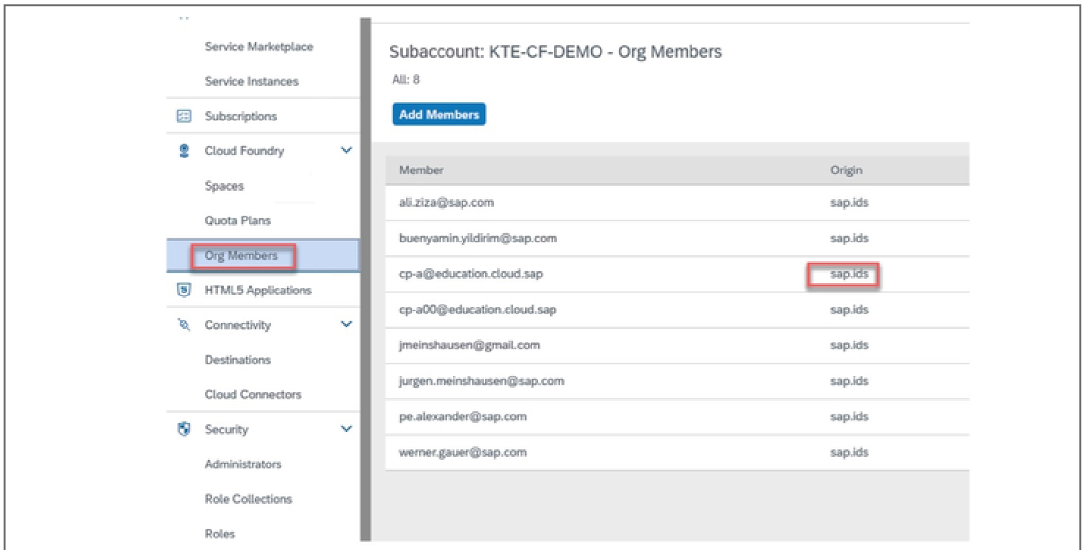


Figure 28: A Real Subaccount Which has Users Authenticated by the SAP ID Service

Org Members - Platform Users authenticated within the SAP ID Service.

Sample - Autorisation of platform users

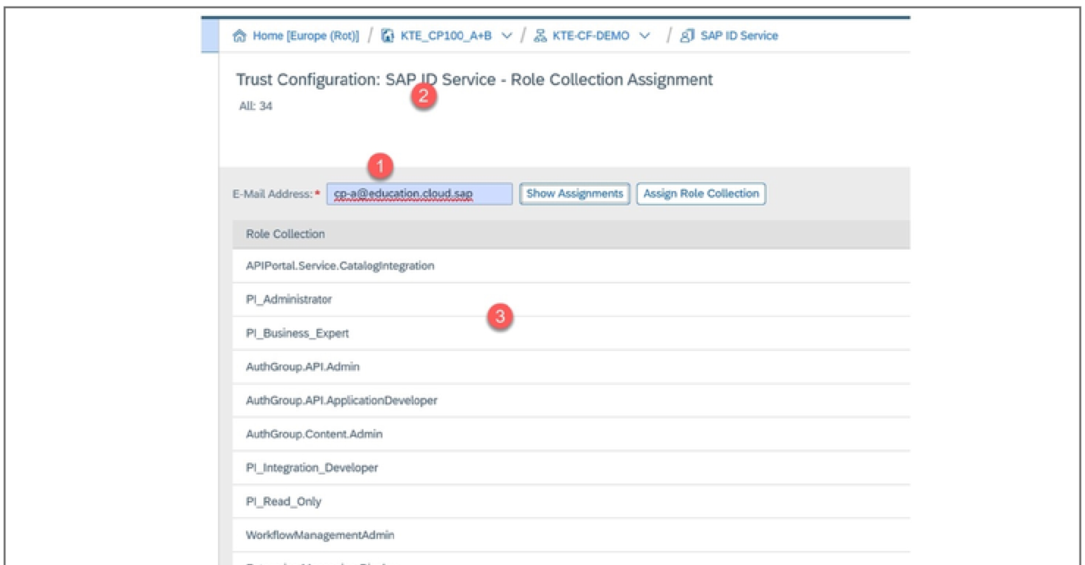


Figure 29: Authorisation of a Platform User With Roles Coming From Applications

Configured Platform users with role collections coming from applications. Explanations about the numbers:

1. The platform-user name.
2. The platform-user has to authenticated by the SAP ID Service.
3. The roles collections coming from the SaaS Applications like BAS, Integration Suite , IOT and others.

Business Users

Business users use the applications that are deployed to SAP BTP:

For example, the end users of your deployed application or users of subscribed apps or services, such as SAP Business Application Studio or SAP Web IDE, are business users.

In the SAP BTP, Cloud Foundry environment, application developers (platform users) create and deploy application-based security artifacts for business users. Administrators use these artifacts to assign roles, build role collections, and assign these role collections to business users or user groups. In this way, they control the users' permissions in the deployed application.

For business users, the identity provider can be, for example, Identity Authentication or your own, such as Active Directory.

To the point

Focused statements:

A global account can contain one or more subaccounts in which you deploy applications, use services, and manage your subscriptions. Subaccounts in a global account are independent from each other.

The assignment of Role Collections to a user takes place in the Trust Configuration. The execution of this BTP admin task is also controlled by associated roles.

Microservices need the right platform. NEO and SAP Netweaver are not microservice platforms. Microservices can be deployed on Cloud Foundry and on KYMA.

Learn more

https://help.sap.com/viewer/65de2977205c403bbc107264b8eccf4b/Cloud/en-US/8ed4a705efa0431b910056c0acdbf377.html

https://help.sap.com/viewer/65de2977205c403bbc107264b8eccf4b/Cloud/en-US/cc1c676b43904066abb2a4838cbd0c37.html

https://help.sap.com/viewer/65de2977205c403bbc107264b8eccf4b/Cloud/en-US/40a8f8f6f1724e0ca0fd2a8777f45504.html

https://discovery-center.cloud.sap/serviceCatalog/cloud-foundry-runtime?service_plan=cloud-foundry-runtime&region=all

LESSON SUMMARY
You should now be able to:

Explain the account model in Cloud Foundry

Explain the difference between platform and business users

Unit 2
Lesson 4

Exploring SAP BTP, ABAP Environment

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explore the most important components of the SAP BTP, ABAP environment

SAP BTP, ABAP Environment

In this lesson we will cover the following topics:

- Introduction
- The main building blocks of ABAP in the Cloud.
- How are the look and feel.
- Step-by-step sample.

Introduction

ABAP in the cloud:

- Is a Platform as a Service (PaaS) offering for ABAP.
- Develop ABAP cloud apps decoupled from the digital core Leverage your ABAP know how in the cloud Reuse your existing ABAP assets.
- Benefit from newest ABAP Programming Model Exploit SAP HANA capabilities Consume SAP BTP services.

Facts about ABAP in the cloud

The ABAP Platform provides the technology layer for several SaaS applications like SAP S/4HANA Cloud and Integrated Business Planning as well as for on-premise solutions like SAP S/4HANA.

Furthermore, the ABAP Platform is also used as the key pillar of SAP's Platform as a Service offering for ABAP which has the product name SAP BTP, ABAP environment.

Using the same code line for these different flavors enables customers and partners to use the same powerful toolset (ABAP Development Tools in Eclipse) and the same ABAP RESTful Programming Model (RAP) for cloud and on-premise development.

There are two main reasons to develop applications and extensions for the cloud with ABAP: Existing assets (custom code developed in on-premise systems) and developers, experienced in using ABAP. The typical extension scenarios built on top of ABAP are:

- Cloud ERP. Extend SAP S/4HANA Cloud or other SAP cloud offerings with cloud extensions.

Innovation Platform : Develop and run innovative ABAP apps on a PaaS in the Cloud.

Hub-like Usage: Integrate multiple cloud & on-premise systems with SAP & non-SAP cloud services.

The ABAP service runs on the Cloud Foundry Environment and must be configured there on the desired subaccount.

The main building blocks of ABAP in the Cloud

Below is the overview of the main building blocks of ABAP in the Cloud.

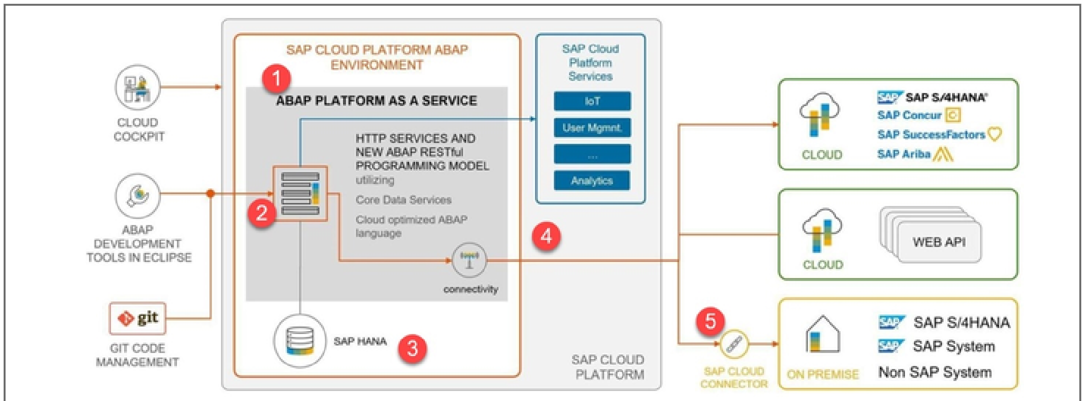


Figure 30: The Main Building Blocks of ABAP in the Cloud

Explanations:

1. An ABAP instance runs on SAP BTP Cloud Foundry as a Service.
2. There is a SaaS Application which connect to the Development tools.
3. You use only a HANA Database with all the advantages of HANA.
4. You can use all available services on the SAP BTP like connectivity or IoT.
5. You can use the cloud connector to connect to back ends.

How are the look and feel

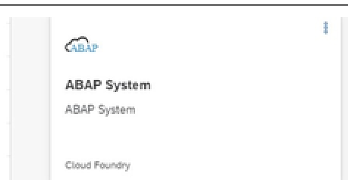


Figure 31: Service ABAP System in Cloud Foundry

ABAP service as a tile in the SAP BTP.

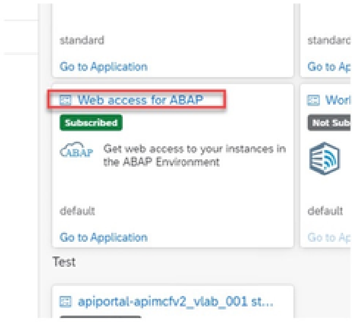


Figure 32: SaaS App Web Access for ABAP in Cloud Foundry

You will see the Web access for ABAP at Subscriptions .

Step-by-step sample

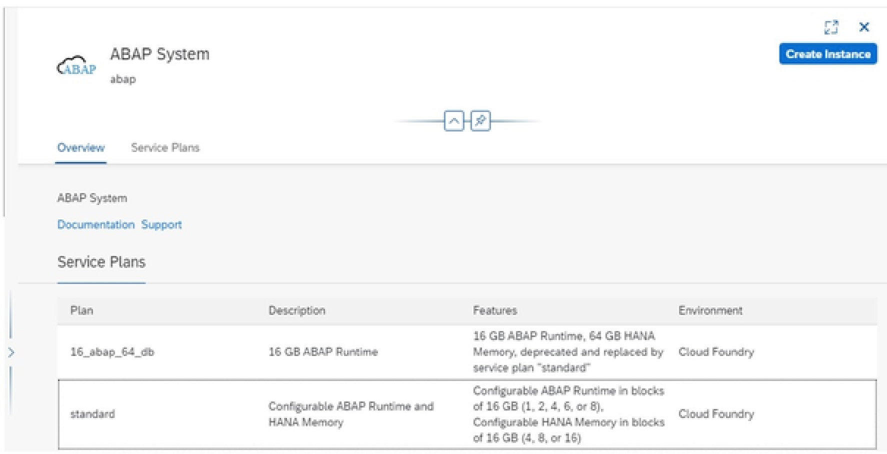


Figure 33: ABAP Service

Explanations:

1. Create and configure your SAP BTP, Cloud Foundry subaccount.

    Assign the ABAP Service.

    Assign the SaaS app Web access for ABAP.

2. Instance your ABAP Service with the instance wizzard

3. Download and open an Eclipse for ABAP.

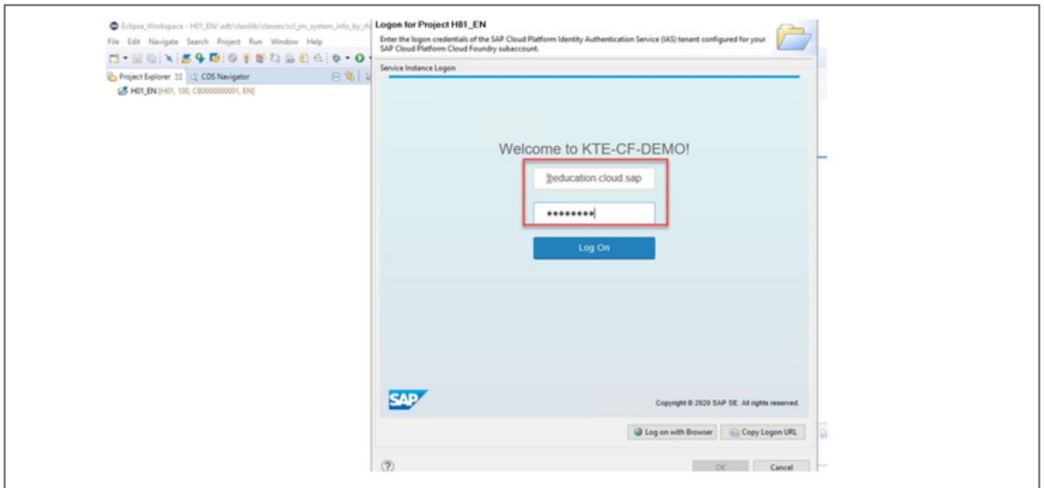


Figure 34: ABAP Service

4. Connect against your configured subaccount with the ABAP service.

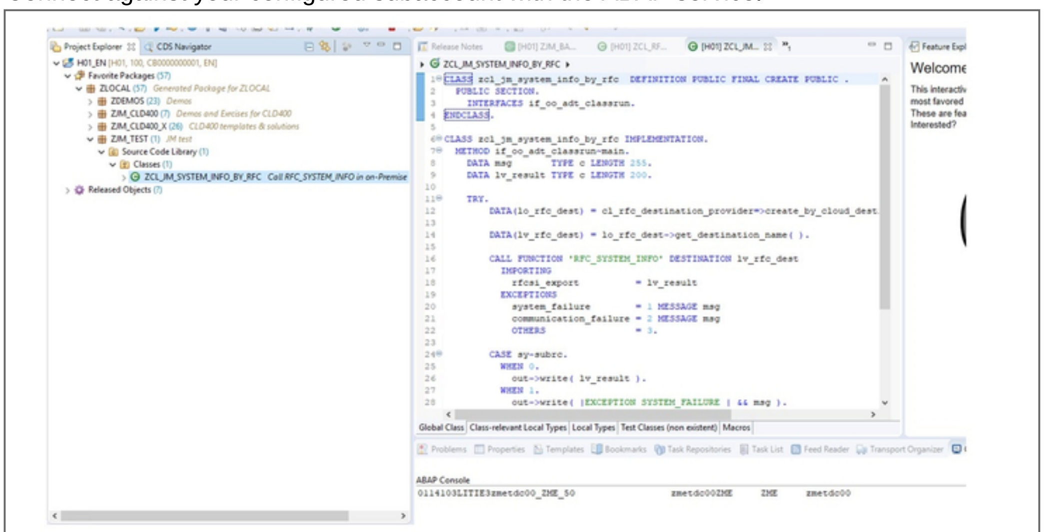


Figure 35: Connecting Against the ABAP Service

5. Code and run your app.

To the point

Focused statement: ABAP in the cloud:

Is a Platform as a Service (PaaS) offering for ABAP.

Develop ABAP cloud apps decoupled from the digital core Leverage your ABAP know how in the cloud Reuse your existing ABAP assets.

Benefit from newest ABAP Programming Model Exploit SAP HANA capabilities Consume SAP BTP services.

Learn more

https://help.sap.com/viewer/65de2977205c403bbc107264b8eccf4b/Cloud/en-US/31367ef6c3e947059e0d7c1cbfcaae93.html

https://discovery-center.cloud.sap/serviceCatalog/abap-environment?region=all

LESSON SUMMARY

You should now be able to:

- Explore the most important components of the SAP BTP, ABAP environment

Unit 2
Lesson 5

Explaining Kyma

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain, what Kyma is, what components Kyma is made of and what you can do with it

Kyma

In this lesson we will cover the following topics:

- Kyma in a nutshell
- Main features
- Key components

Kyma in a Nutshell

Kyma runtime:

- Is a fully managed Kubernetes runtime based on the open-source project "Kyma". This cloud-native solution allows the developers to extend SAP solutions with serverless Functions and combine them with containerized microservices.
- Is available as a runtime within the SAP BTP.

Kyma allows you to extend applications with microservices and Functions. First, connect your application to a Kubernetes cluster and expose the application's API or events securely. Then, implement the business logic you require by creating microservices or Functions and triggering them to react to particular events or calls to your application's API. To limit the time spent on coding, use the built-in cloud services from Service Catalog, exposed by open service brokers from such cloud providers as GCP, Azure, and AWS.

Kyma comes equipped with these out-of-the-box functionalities:

- Service-to-service communication and proxying (Istio-based Service Mesh).
- Built-in monitoring, tracing, and logging (Grafana, Prometheus, Jaeger, Loki, Kiali).
- Secure authentication and authorization (Dex, Ory, Service Identity, TLS, Role Based Access Control).
- The catalog of services to choose from (Service Catalog, Service Brokers).
- The development platform to run lightweight Functions in a cost-efficient and scalable way (Serverless).

The endpoint to register Events and APIs of external applications (Application Connector).

Secure API exposure (API Gateway).

The messaging channel to receive Events, enrich them, and trigger business flows using Functions or services (Event Mesh, NATS).

CLI supported by the intuitive UI (Console).

Asset management and storing tool (Rafter, MinIO).

Backup of Kyma clusters (Kyma Backup).

Main features

Major open-source and cloud-native projects, such as Istio, NATS, Serverless, and Prometheus, constitute the cornerstone of Kyma. Its uniqueness, however, lies in the "glue" that holds these components together. Kyma collects those cutting-edge solutions in one place and combines them with the in-house developed features that allow you to connect and extend your enterprise applications easily and intuitively.

Kyma allows you to extend and customize the functionality of your products in a quick and modern way, using serverless computing or microservice architecture.

The extensions and customizations you create within Kyma are decoupled from the core applications, which means that:

Deployments are quick.

Scaling is independent from the core applications.

The changes you make can be easily reverted without causing downtime of the production system.

Last but not least, Kyma is highly cost-efficient. All Kyma native components and the connected open-source tools are written in Go. It ensures low memory consumption and reduced maintenance costs compared to applications written in other programming languages such as Java.

Key components

Kyma is built of numerous components but these three drive it forward:

Application Connector

Serverless

Service Catalog

Further details:

Application Connector:

Simplifies and secures the connection between external systems and Kyma.

Registers external Events and APIs in the Service Catalog and simplifies the API usage.

Provides asynchronous communication with services and Functions deployed in Kyma through Events.

Manages secure access to external systems.

Provides monitoring and tracing capabilities to facilitate operational aspects.

Serverless:

Ensures quick deployments following a Function approach.

Enables scaling independent of the core applications.

Gives a possibility to revert changes without causing production system downtime.

Supports the complete asynchronous programming model.

Offers loose coupling of Event providers and consumers.

Enables flexible application scalability and availability.

Service Catalog

Connects services from external sources.

Unifies the consumption of internal and external services thanks to compliance with the Open Service Broker standard.

Provides a standardized approach to managing the API consumption and access.

Eases the development effort by providing a catalog of API and Event documentation to support automatic client code generation.

This basic use case shows how the three components work together in Kyma:

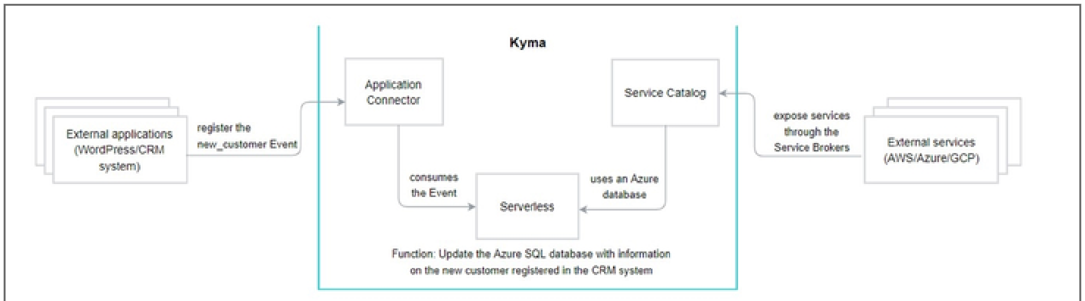


Figure 36: The Main Components Works Together

On the left side is the application, here Wordpress. On the right side the available services of the Hyperscaler. In the middle you see the principle of the application connector which connects the app to be extended with the platform. The serverless runtime which contains the code that is extended by various services from the service catalog.

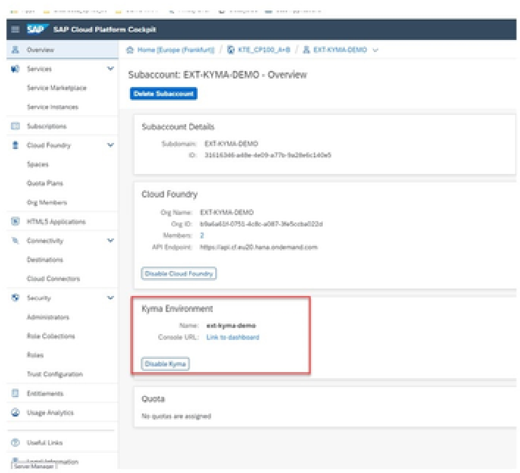


Figure 37: Kyma as Runtime

The figure shows how the Kyma runtime is activated within SAP BTP.

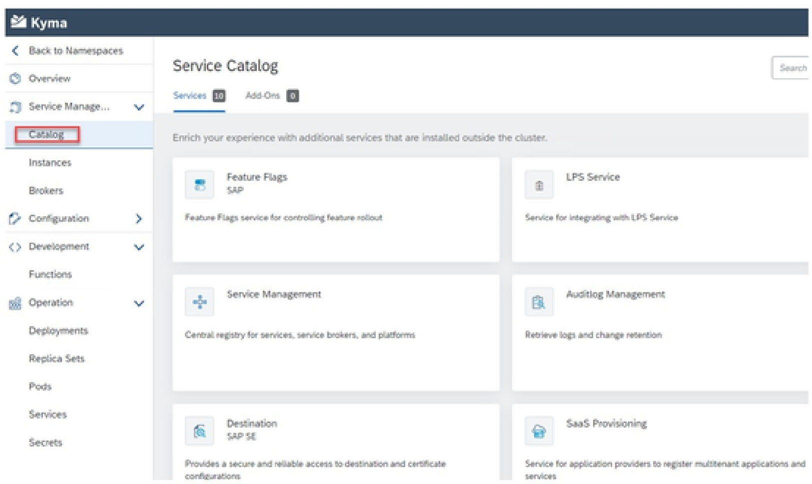


Figure 38: Service Catalog

Within the service catalog in KYMA.

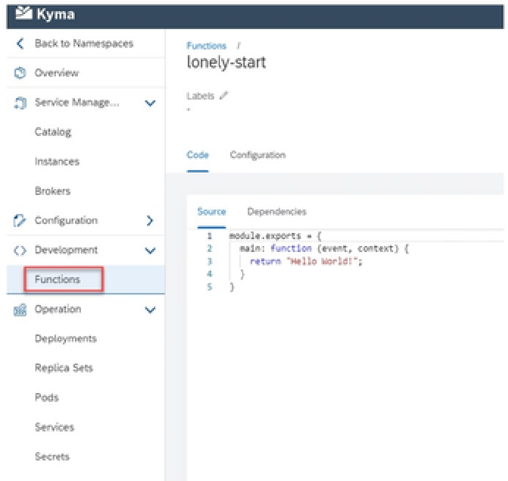


Figure 39: Serverless and Functions

Functions in action.

To the point

Focused statement:

Is a fully managed Kubernetes runtime based on the open-source project "Kyma". This cloud-native solution allows the developers to extend SAP solutions with serverless Functions and combine them with containerized microservices.

Is available as a runtime within the SAP BTP.

Learn more:

https://help.sap.com/viewer/65de2977205c403bbc107264b8eccf4b/Cloud/en-US/606ec610ee4746c09d5d2bef5a85a124.html

https://discovery-center.cloud.sap/serviceCatalog/kyma-runtime?region=all

LESSON SUMMARY
You should now be able to:

Explain, what Kyma is, what components Kyma is made of and what you can do with it

Learning Assessment

1. ABAP is a service, not an environment in the perspective of SAP BTP.

   Determine whether this statement is true or false.

   - [ ] True
   - [ ] False

2. Whch are the commercial models of SAP BTP?

   Choose the correct answers.

   - [ ] **A** The comsumption model.
   - [ ] **B** The rental model.
   - [ ] **C** The subscription model.

3. A Business user uses services and applications in a subaccout.

   Determine whether this statement is true or false.

   - [ ] True
   - [ ] False

4. The ABAP service runs on the Cloud Foundry Environment and must be configured there on the desired subaccount.

   Determine whether this statement is true or false.

   - [ ] True
   - [ ] False

5. Which are the components the Kyma project relies on?

   Choose the correct answers.

   - [ ] **A** Serverless
   - [ ] **B** Runtime free
   - [ ] **C** Service catalog
   - [ ] **D** Application connector

Learning Assessment - Answers

1. ABAP is a service, not an environment in the perspective of SAP BTP.

   Determine whether this statement is true or false.

   [X] True

   [ ] False

   This is correct.

2. Whch are the commercial models of SAP BTP?

   Choose the correct answers.

   [X] **A** The comsumption model.

   [ ] **B** The rental model.

   [X] **C** The subscription model.

   This is correct.

3. A Business user uses services and applications in a subaccout.

   Determine whether this statement is true or false.

   [ ] True

   [X] False

   This is correct.

4. The ABAP service runs on the Cloud Foundry Environment and must be configured there on the desired subaccount.

   Determine whether this statement is true or false.

   [X] True

   [ ] False

   This is correct.

5. Which are the components the Kyma project relies on?

   Choose the correct answers.

   - [x] **A** Serverless
   - [ ] **B** Runtime free
   - [x] **C** Service catalog
   - [x] **D** Application connector

   This is correct.

UNIT 3 SAP Integration Suite

UNIT OBJECTIVES

- Explain the Integration strategy in an overview
- Perform a classification according to ISA-M
- Explain integration comprehensively
- Explore APIs in the SAP Universe
- Explain the use of API Management
- Explain the use of SAP Graph
- Create B2B mappings
- Explain the use and the main components of Cloud Integration
- Explain the features of SAP Enterprise Messaging
- Explain the use of SAP Open Connectors

Unit 3
Lesson 1

Explaining the Integration Strategy

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain the Integration strategy in an overview

Integration Strategy

In this lesson we will cover the following topics:

- Facts in a nutshell
- SAP Integration Suite
- What's in it ?
- Typical use case: Build enterprise digital mobile applications with the SAP Integration Suite
- Typical use case: Monetize your APIs with SAP API Management

Facts in a nutshell

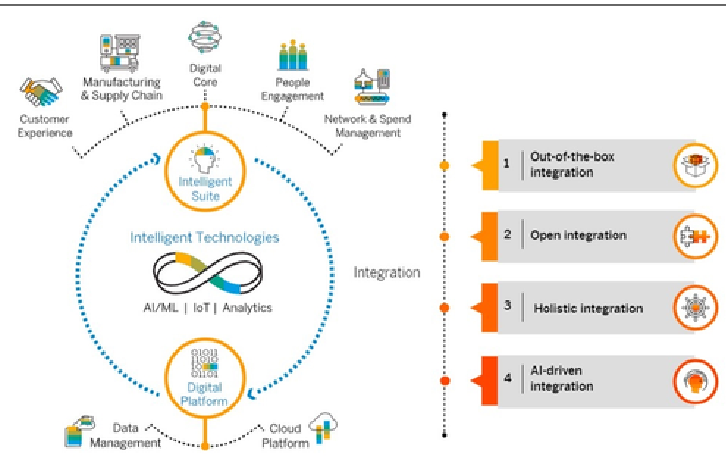


Figure 40: Integration principles of SAP

SAP bases its integration strategy for the Intelligent Enterprise on four principles:

- out-of-the-box integration,
- open integration,
- holistic integration,

AI-driven integration

What exactly does that mean?

Out-of-the-box integration

SAP software supports end-to-end business processes, including out-of-the-box integrations, enabled by a standardized application and technology portfolio.

Prepackaged integration content of the SAP Integration Suite builds the foundation for integrating applications of the Intelligent Enterprise.

Open integration

Besides SAP-to-SAP integrations, SAP is open for any third-party integration as well as custom extensions. The foundation for open integration is based on public APIs. With the Open Connectors service,

SAP provides feature-rich, prebuilt connectors for more than 150 non-SAP applications.

Holistic integration

SAP provides a holistic integration technology portfolio that covers all flavors of integration. Based on the SAP Integration Suite and SAP HANA Data Management Suite, SAP supports all types of integration use cases, ranging from process, data, user, and IoT to analytics-centric integration. A methodology helps enterprise architects shape their integration strategies, which can include integration technologies from SAP and third parties.

AI-driven integration

In addition to bringing intelligence into core business processes, SAP is using AI techniques to simplify the development of integration scenarios. One example is the SAP Integration Advisor service. Its crowd-based machine learning approach enables users to define, maintain,share, and deploy B2B and A2A integration content much faster than building it from scratch. Further uses of artificial intelligence in other areas of integration, such as for integration monitoring, are planned.

SAP Integration Suite

Based on the SAP Integration Suite and SAP HANA Data Management Suite, SAP supports all types of integration use cases, ranging from process, data, user, and IoT to analytics-centric integration

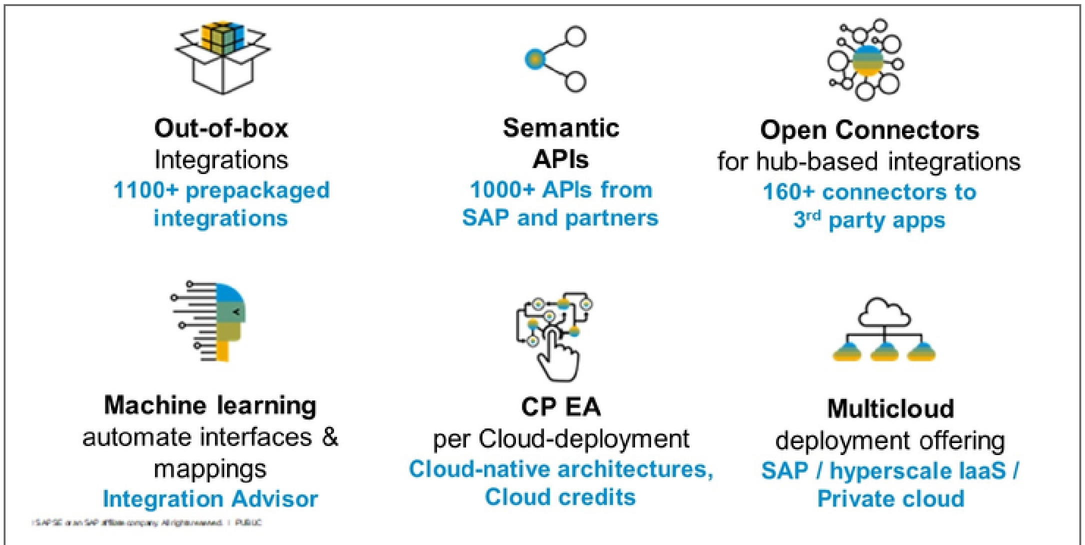


Figure 41: Offering of the Integration Suite

The SAP Integration Suite offers:

Out-of-box Integrations: 1100+ prepackaged Content.

Semantic API's: 1000 APIs from SAP and Partners.

Open Connectors for hub-based integrations: 160+ Connectors to third party apps.

Machine Learning automates interfaces & mappings - Integration Advisor.

Multicloud deployment offerings - SAP, Hyperscaler, private cloud.

And tools, tools …

What's in it ?

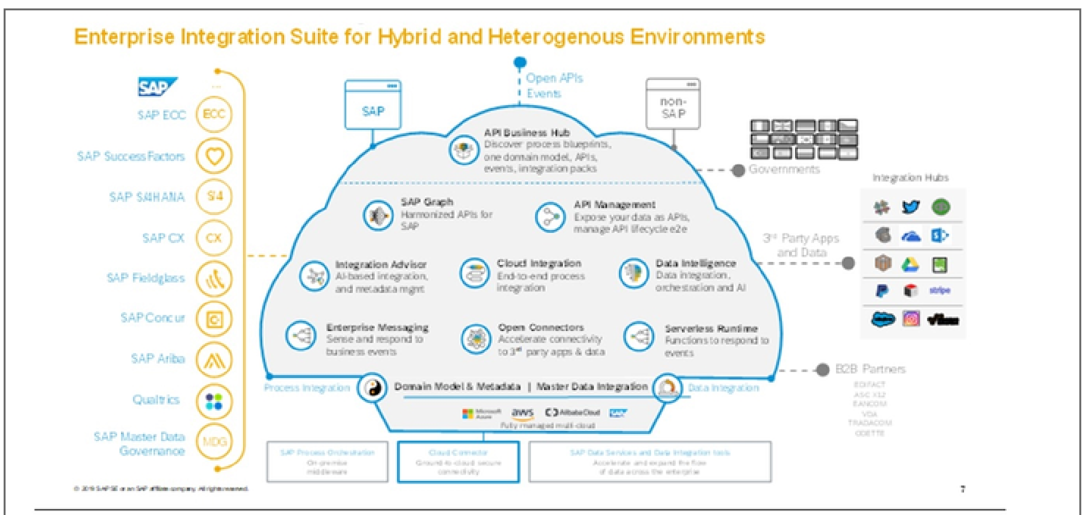


Figure 42: Overview of the SAP Integration Suite

On the left are the technical SaaS applications from SAP. On the right are the professional apps and B2B partners that are available outside the SAP world.

In the middle the tools we offer for simplified enterprise integration for hybrid, huge and heterogenous environments

In the middle below you will find the hyperscalers such as AWS, Azure and other.

With Azure, you can achieve very close interlocking through the Emphrase Project. In concrete terms, this means that they have either SAP only or only Azure as a contractor.

Components

The following components are part of the Integration Suite:

API Business HUB

API Management

SAP Graph EX

Integration Advisor

Cloud Integration

Data Intelligence

Enterprise Messaging

Open Connectors

Serverless Runtimes (belongs also to Extensibility Suite)

We will discuss the individual tools in more detail below.

Typical use case: build enterprise digital mobile applications with the SAP Integration Suite

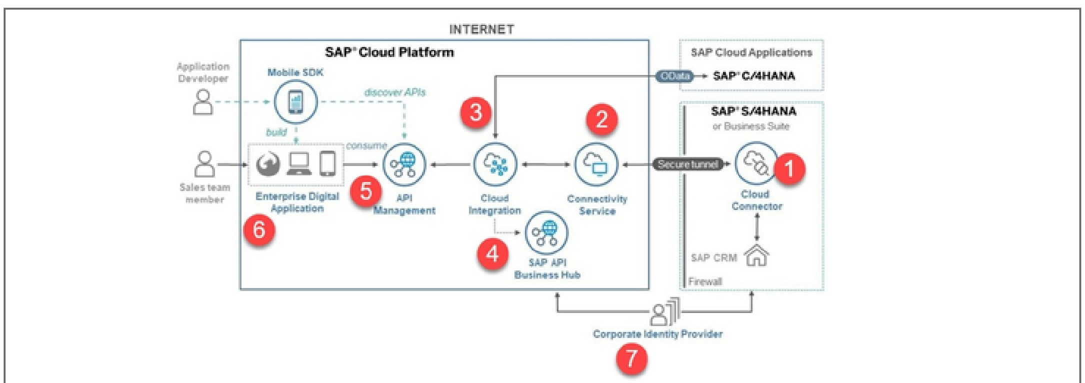


Figure 43: Scenario Overview

1: The backends are connected via the Cloud Connector, which establishes a secure connection between on-premise and Cloud.

2: The connectivity service detects connectivity to the services and apps available on SAP BTP.

3: the data is then processed in the SAP Integration Suite, according to the defined iFlow.

4: Standard API's , such as OData , are used for this purpose.

5: API Management then makes the processed data available as an API.

6: Various channels can then use this data API.

7: A joint authentication and authorization via the identity provider makes the procedure secure.

Typical use case: Monetize your APIs with SAP API Management

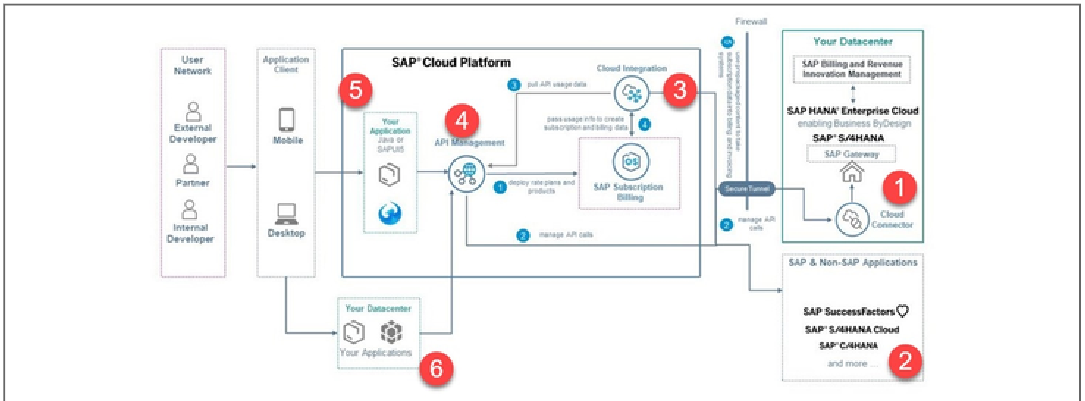


Figure 44: Scenario Overview

Description:

1: The on-premise back end is connected to the Cloud Connector.

2: Other SaaS apps such as SAP SuccessFactors are directly connected.

3: The data is processed via the cloud integration via a configured iFlow.

4: API Management provides an API.

5: My apps can now use this API.

6: Data management is handled by an external data center.

To the point

Focused statement: SAP bases its integration strategy for the Intelligent Enterprise on four principles:

- Out-of-the-box integration
- Open integration
- Holistic integration
- AI-driven integration

Focused statement: The following components are part of the Integration Suite:

- API Business HUB
- API Management
- SAP Graph EX
- Integration Advisor
- Cloud Integration

- Data Intelligence
- Enterprise Messaging
- Open Connectors
- Serverless Runtimes (belongs also to Extensibility Suite)

Learn more:

https://discovery-center.cloud.sap/serviceCatalog/integration-suite?region=all

LESSON SUMMARY
You should now be able to:

- Explain the Integration strategy in an overview

Unit 3
Lesson 2

Performing a Classification According to to ISA-M

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Perform a classification according to ISA-M

Classification According to ISA-M

Introduction

With the ISA-M, customers can systematically and comprehensively determine their integration needs.

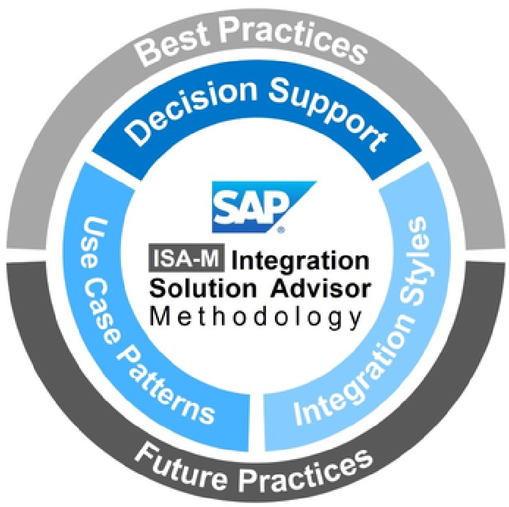


Figure 45: ISA-M Methodology Overview

The figure, ISA-M Methodology Overview, gives an overview of the ISA-M methodology.

The following applications are possible:

- Enables the systematic selection of integration technologies.
- Customer-proven set of core integration patterns.
- Extensible to future use cases and technologies.
- Focus on key dimensions of the problem space.
- Open to SAP & non-SAP middleware.

You can use ISA-M:

As systematic terminology to improve communication between teams.

As governance blueprint to provide integration architecture guidance.

As assessment tool to define and update your integration strategy.

As accelerator for pattern-based interface development (reuse).

Determination of the optimal integration domains

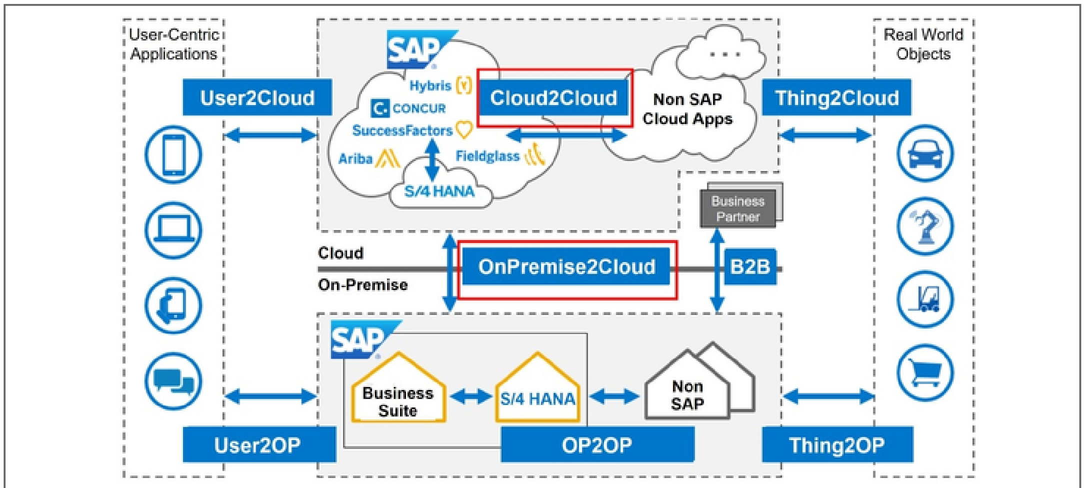


Figure 46: Integration Domains

The figure, Integration Domains, gives an overview of the interdependencies between the different integratiton domains.

The SAP Integration Suite is particularly suitable for the following scenarios:

Cloud2Cloud.

OnPremise2Cloud.

> Note:
> The main role of SAP Cloud Integration is to integrates cloud applications into an application landscape.

Classification in the four fundamental types of application integration – Integration Styles

The SAP Integation Suite is particularly suitable for the process Invocation style.

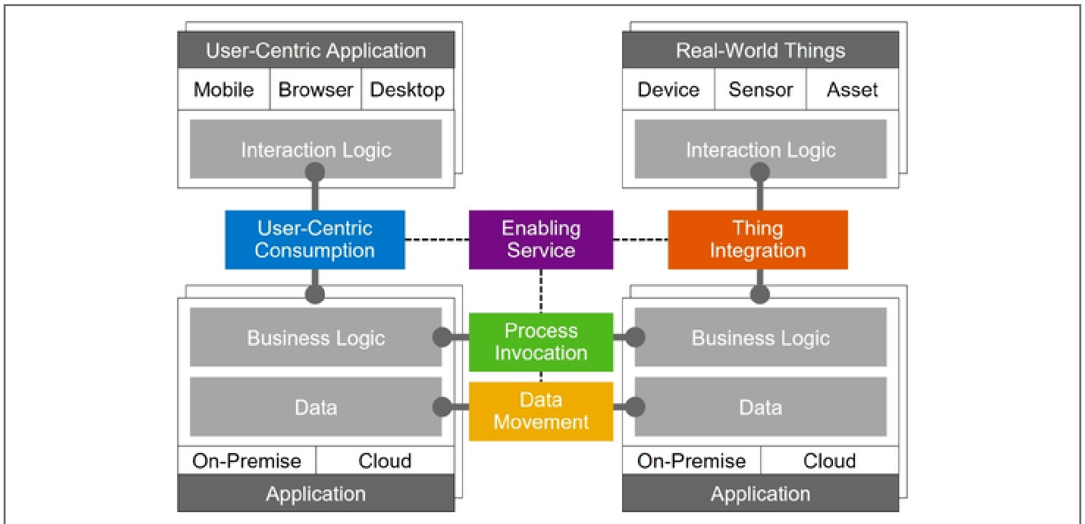


Figure 47: The Four Fundamental Types of Application Integration

The figure, The Four Fundamental Types of Application Integration, shows the four fundamental types of application integration.

The four fundamental types of application integration are:

1. **Process Invocation**- How to chain business processes across business applications?
2. **Data Movement** - How to synchronize data between business applications?
3. **User-Centric Consumption**- How to integrate user-centric applications with business applications.
4. **Thing Integration**- How to integrate real world things with business applications.

Key Characteristics of Process Invocation

The following table shows the key characteristics of process invocation:

Table 2: Key Characteristics of Process Invocation

Objective	**Chaining of business processes**
Interaction Type	System-2-System
Coupling to Application	Process-Level
Primary Trigger	Application Event
Urgency of Completion	(Near) Real-Time
Unit of Data Exchange	Single Objects
Specific Requirements	Transactional Integrity Reliable Messaging Message Orchestration B2B protocol support

To the point

Focused statements:

ISA (Independent Systems Architecture) is a collection of best practices based on experience in particular with microservices and Self-contained Systems and the challenges faced in those projects.

Far too often microservices are adopted but the projects then fail to succeed. This set of best practices ensures that common pitfalls are avoided and the promised benefits of microservices are achieved.

Learn more:

https://isa-principles.org/

LESSON SUMMARY
You should now be able to:

Perform a classification according to ISA-M

Unit 3
Lesson 3

Explaining Integration Comprehensively

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain integration comprehensively

The Comprehensive Integration Approach of SAP

In this lesson we will cover the following topics:

- Intelligent Enterprises are integrated enterprises.
- Integration tools
- Integration content

Intelligent Enterprises are Integrated Enterprises

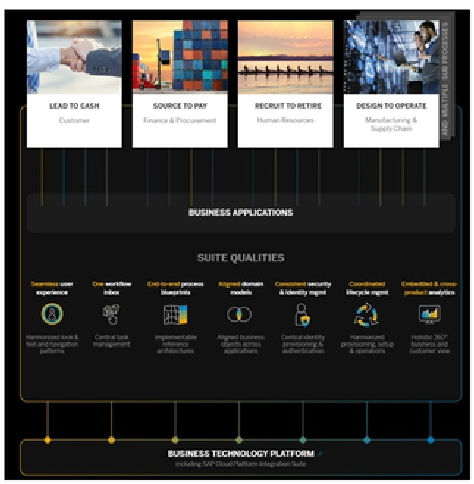


Figure 48: Intelligent Enterprises are Integrated Enterprises

Integration in the sense of SAP goes far beyond purely technical integration. Rather, SAP takes a holistic approach to integration. Described below on the 4 main use cases:

- Lead to Cash
- Source to Pay
- Recruit to Retire
- Design to Operate

The following integration objectives are to be met with this holistic strategy:

Seamless user experience

Harmonized look & feel and navigation patterns.

One workflow inbox

Central task management.

End-to-end process blueprints

Implementable reference architectures.

Aligned Domain models

Aligned business objects across applications.

Consitent security & identity management

Central identity provisioning & authentication.

Coordinated Lifecycle Management

Harmonized provisioning, setup & operations.

Embedded & cross-product analytics

Holistic 360° business and customer view Seamless user experience.

Lead to Cash

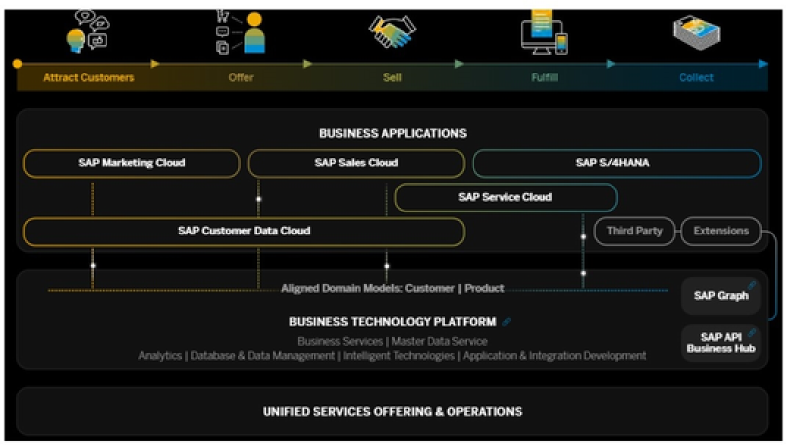


Figure 49: Lead to Cash

Manage all aspects of the customer experience, from the initial interaction to order fulfillment and service delivery. Drive and realize revenue along the customer journey.

Source to Pay

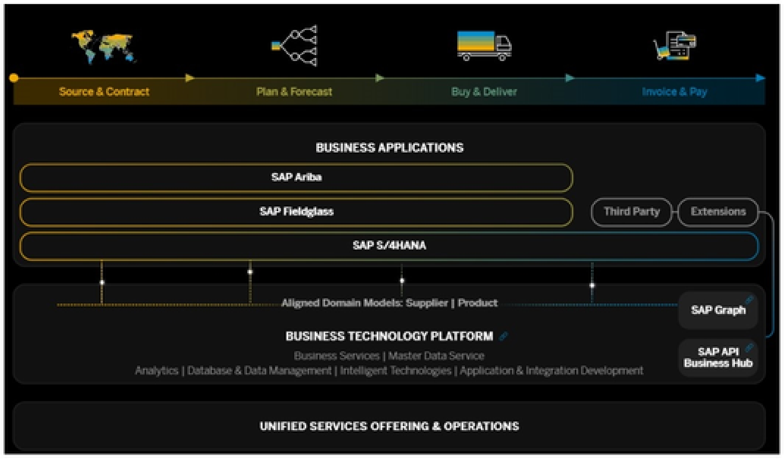


Figure 50: Source to Pay

Manage all purchasing processes from strategic supplier selection to assuring compliance demands inoperational procurement. Analyze and manage spend across business units and drive business value.

Recruit to Retire

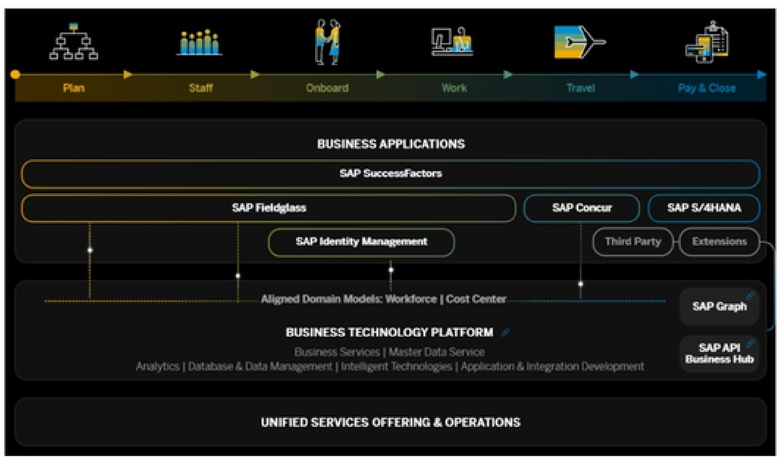


Figure 51: Recruit to Retire

Understand, manage and optimize all aspects of your workforce (employees and external workers) in line-with business objectives and with clear financial impact.

Design to Operate

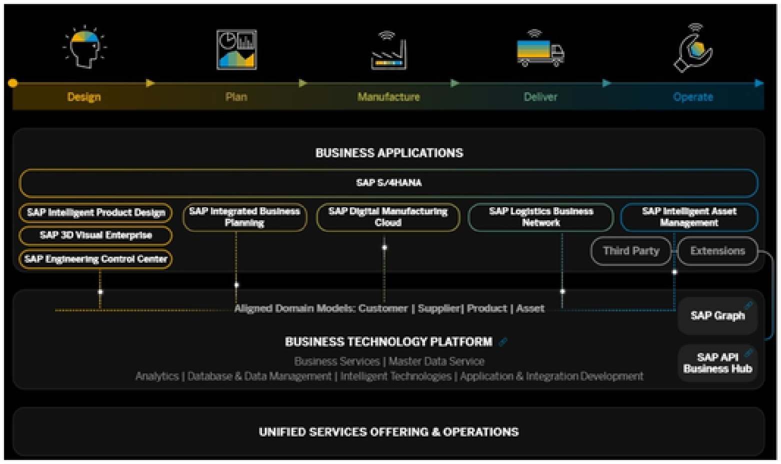


Figure 52: Design to Operate

Create a digital mirror of your entire supply chain - from design to planning, manufacturing, logistics and ongoing maintenance. Embed intelligence and ensure your customers are central to each and every phase of your business.

Integration Content

To simplify the effort between SAP business processes, you can:

Use preconfigured integration Content

Integrate from SAP solutions to other SAP solutions and leverage integrated mission-critical business processes.

Open integration - open Connectors

Integrate from SAP solutions to non-AP solutions and benefit from APIs and more than 160 connectors for third-party solutions and extensions.

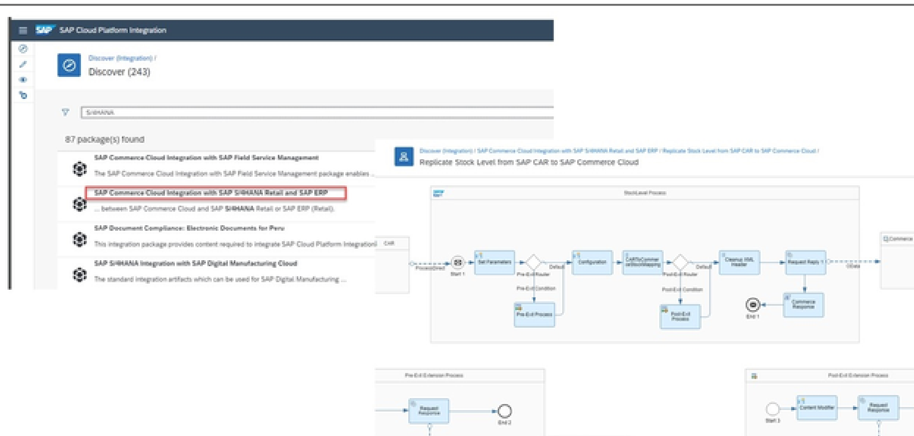


Figure 53: Integration Content

The figure illustrates open connectors.

We will discuss it in the next Lesson.

AI-driven Business Services

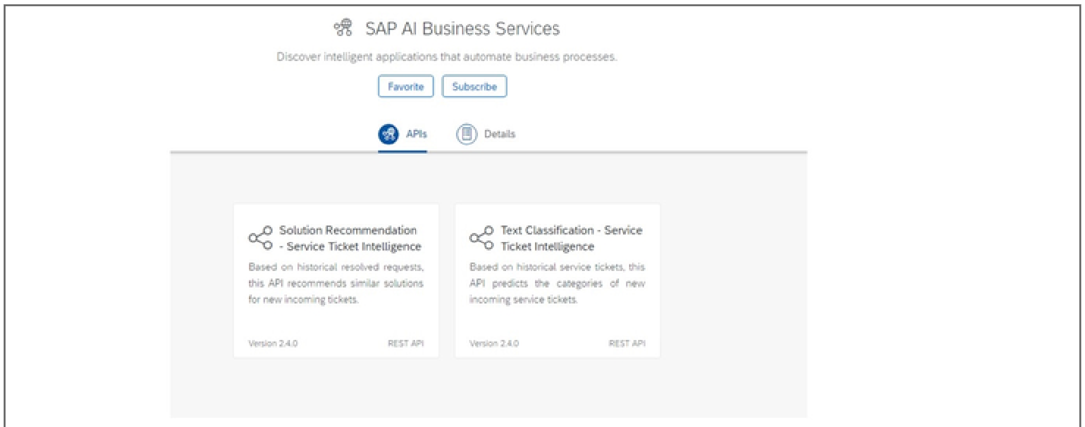


Figure 54: AI-driven Business Services

Integrate artificial Intelligence and simplify integration development for your business.

LESSON SUMMARY

You should now be able to:

- Explain integration comprehensively

Unit 3
Lesson 4

Exploring APIs in the SAP Universe

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explore APIs in the SAP Universe

APIs in the SAP Universe

Overview

The SAP API Business Hub is a web application hosted by SAP to discover, explore, and test SAP and partner APIs (application programming interfaces) that are required to build extensions or process integrations.

APIs in SAP solutions use various protocols, documentation, and access mechanisms. Application and integration developers must have a consistent overview of the APIs available in the relevant SAP systems (deployed on-premise and in the cloud). Testing APIs and building prototypes also involves organizing access to relevant systems and tenants for developers.

SAP API Business Hub addresses these challenges by offering a central catalog of APIs, along with an integrated test environment in the cloud for easy testing. The SAP API Business Hub covers APIs from SAP S/4HANA, SAP SuccessFactors, SAP BTP, SAP Hybris, and more.

SAP API Business Hub therefore simplifies the development process and reduces effort for:

- Application developers when building extensions (for example, when building extensions to applications within the same Line of Business), mobile applications, or Web applications.
- Integration developers when developing process integrations to third-party systems (Application-to-Application [A2A] or Business-to-Business [B2B]).

Our Salient Features

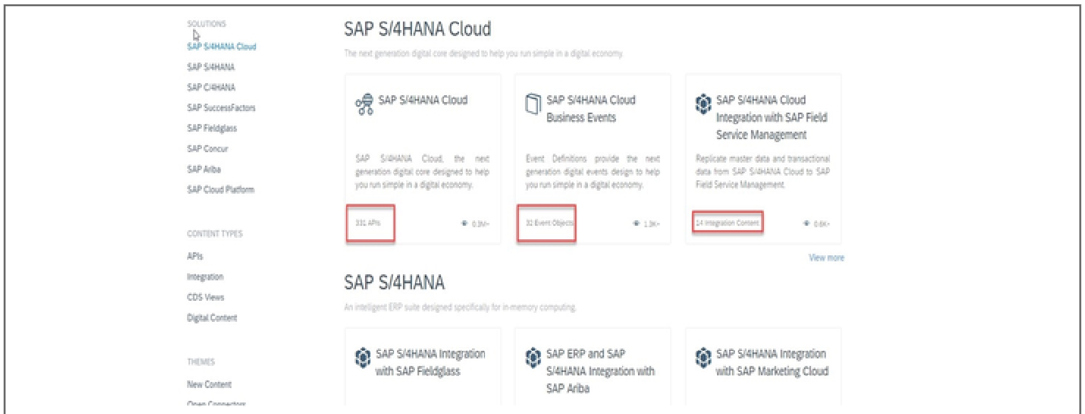


Figure 55: Entry Screen with OData APIs, Events and Integration Content

The figure illustrates the entry screen containing OData APIs, Events and Integration Content.

The following are further explanations about the entry screen:

Simple API discovery and exploration

SAP API Business Hub allows you to find and explore APIs and API packages (a group of APIs) from SAP (SAP S/4HANA, SAP SuccessFactors, SAP Hybris, SAP BTP, and more) and selected partners. You can navigate to individual APIs, their resources, and supported operations. The integrated comprehensive API documentation in accordance with the OpenAPI specification helps you quickly understand the behavior of the APIs.

Collaboration

You can collaborate with other community members by providing comments and ratings for APIs you have used and use comments and ratings provided by other members to help you select your APIs.

Sandbox API testing

You can test-drive selected APIs, either by making calls using the integrated, cloud-based sandbox test environment or by using your application of choice.

Integration with SAP Web IDE

You can consume APIs on the SAP API Business Hub directly on SAP Web IDE to develop SAP Fiori extension apps.

Simple Integration with API Management

As discussed in other lessons, the SAP Business Hub is deeply integrated with API Management.

Build and Provided along openAPI Specification

As discussed in other lessons, the SAP Business Hub is build and provided along openAPI specifications.

Benefits of Using the SAP API Business Hub are:

Simplification

SAP API Business Hub is a one-stop catalog to find, learn about, and consume an ever-increasing number of SAP and selected partner APIs from different areas. As the repository is hosted and operated by SAP, it is always up to date and easy to access.

Streamlined development

SAP API Business Hub enables a streamlined development process by eliminating overhead and technical prerequisites by providing Web tooling and cloud-based services.

Reduced total cost of extension and development

Improves productivity by offering a comprehensive and easy learning and testing environment for developers. Easy consumption of APIs to develop SAP Fiori extension apps is enabled via integration with SAP Web IDE.

The Business Hub, a small tour

As an example of the API Business Partner, we want to show the essential features. We are looking for business partners and find the appropriate interface. As an example of the API Business Partner, we want to show the essential features. We are looking for business partners and find the appropriate interface.

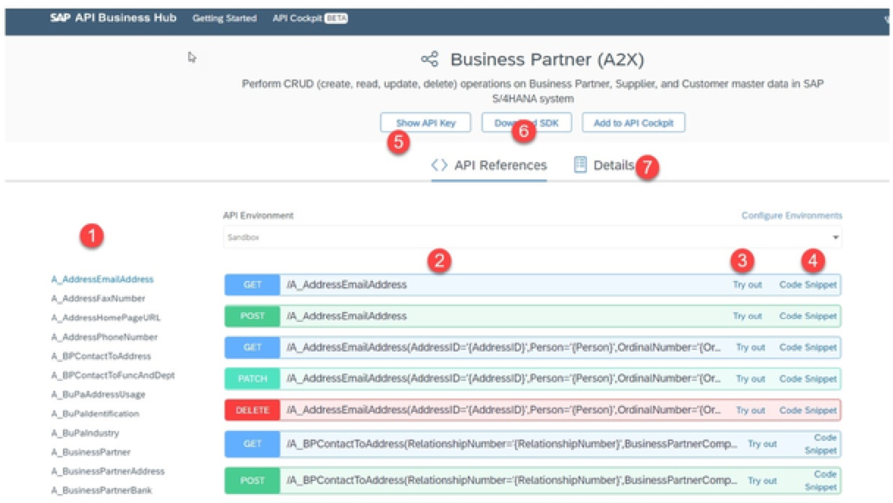


Figure 56: Business Partner Entry Screen

Explanations about the bullets:

1. Here we find the different entities of the OData interface. We are on the A_AddressEmailAddress entity.
2. This entity has the following URLs and http verbs to access the data.
3. Here you can test the interface against a sandbox.

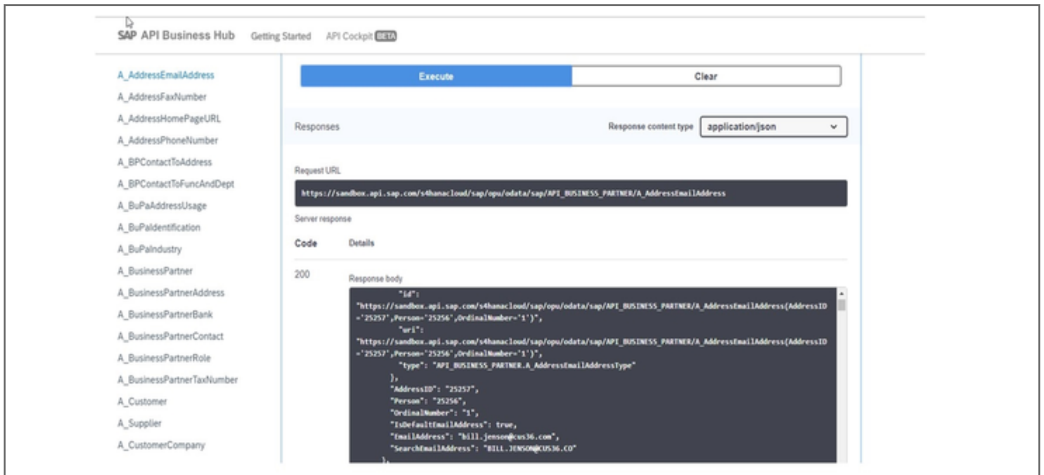


Figure 57: Entry Screen for Testing

4. Here we find code snippets um to call this interface in different languages.

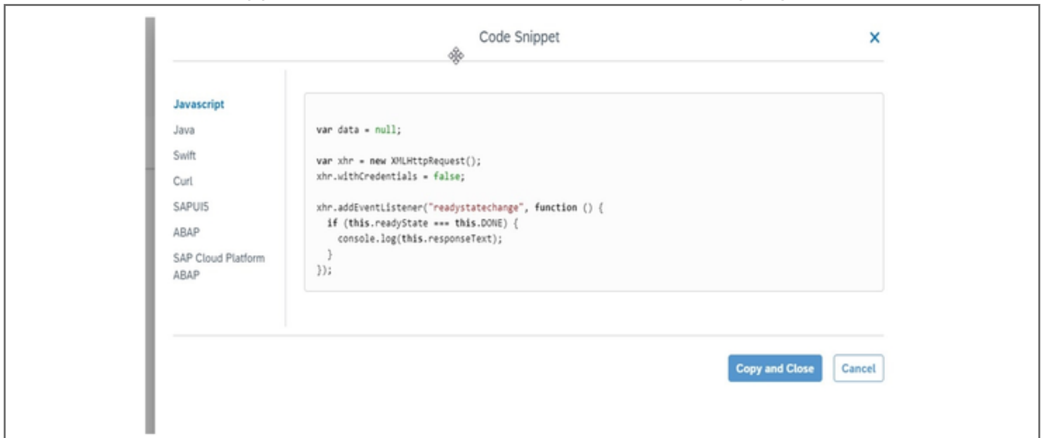


Figure 58: Code Snippets

5. This is the API key to call this interface from the outside.

6. Here we can download a client sdk in Java that was created from the API Doc.

7. Here we find the API documentation according to openAPI.

Interface Call in customer own systems

We can configure the interface, if available on your system, against your own system.

To do this, we use the Link Configure Environments :

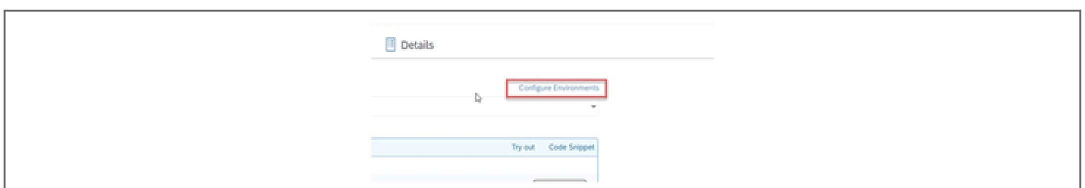


Figure 59: Choosing the Link Configure Environments

Now we can try each verb and URL on this client to retrieve the data from our own system:

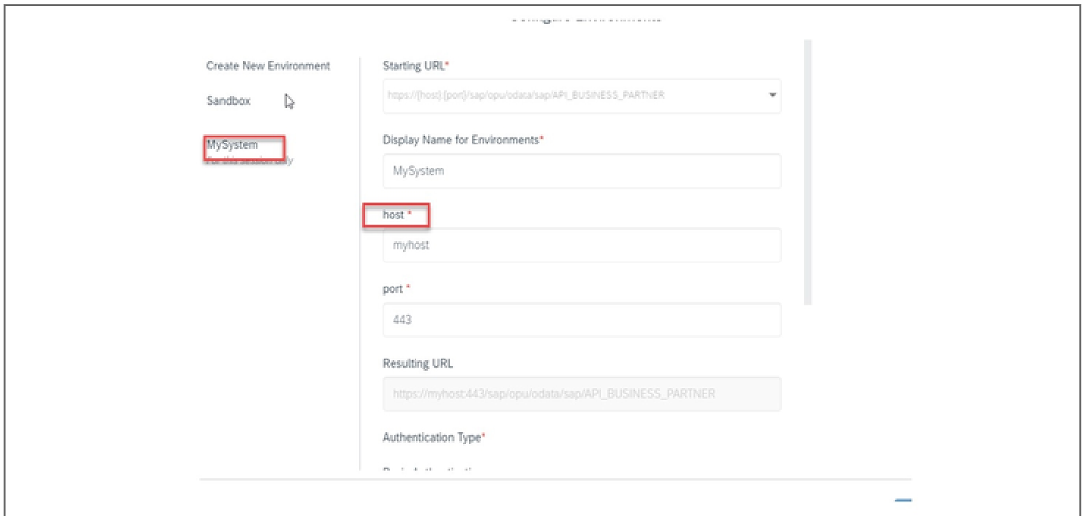


Figure 60: Performing Entries

The figure shows the screen, to enter the required data.

To the point

Focused statement: The following content types are available on the SAP Business Hub:

- Business Processes
- APIs (OData, SOAP, REST)
- Events
- Integration Content
- CDS Views
- Workflow Management Artifacts

Learn more:

https://api.sap.com/

LESSON SUMMARY
You should now be able to:

- Explore APIs in the SAP Universe

Unit 3
Lesson 5

Explaining API Management

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain the use of API Management

API Management

In this lesson we will cover the following topics:

- What can I do with it?
- Components
- SAP API Management key features
- Typical use case
- Steps to provide an custom API

What can I do with it?

You can enable enterprise collaboration and co-innovation that means:

What is SAP API Management?

- Publish, promote, and oversee APIs in a secure and scalable environment with the API Management capability. Empower developer communities to monetize data and digital assets in new channels, devices, and UIs.
- Common platform to define and publish APIs.
- Related APIs bundled and offered as a product
- Comprehensive consumption analytics capabilities.
- Monetization feature for all API providers.

Using an API Management layer will allow companies to:

- Create an Omni-channel experience for customers and employees (mobile, web, devices).
- Implement simplified Integration with various ecosystems (partners, suppliers, marketplaces etc.).
- Expose data with external ecosystems to generate new revenue streams (Data-as-a-Service).

Components

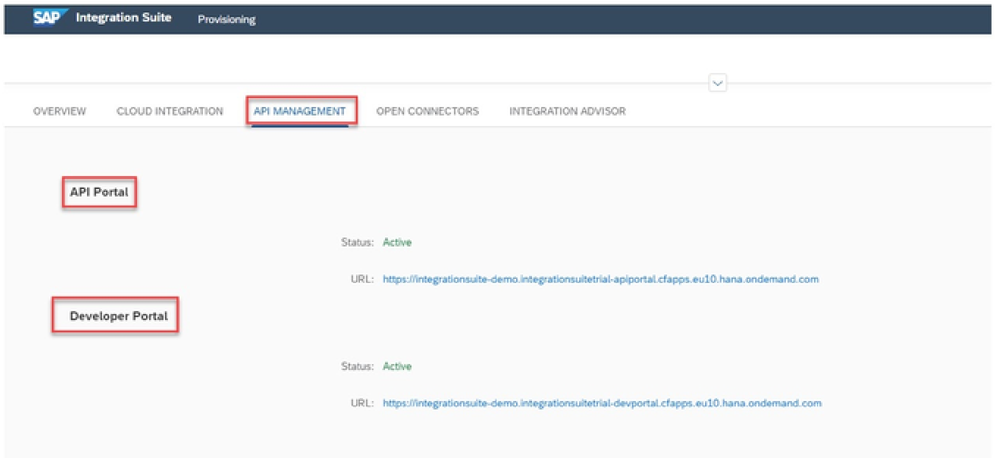


Figure 61: Components of SAP API Management Used Within the Integration Suite

The API Management belongs to the SAP Integration Suite.

It has two parts:

- API Portal
- Developer Portal

SAP API Management key features are:

- Configuration instead of coding
- Efficient API Publishing
- Broad analytics

Further details:

Configuration instead of coding

Managing APIs consists of applying standard features in the areas of security, traffic management and mediation. All these features are pre-defined within SAP API Management, and simply need to be adapted to your needs.This allows to centrally simplify and streamline the management of APIs, in order to increase agility and to enforce central governance easily.

Efficient API Publishing

One of the most important aspects of API Management, is to simplify API consumption for the Application Developer. Hence API Management provides a self-service and integrated Developer Portal, featuring self-service capabilities, searching, documentation, testing and subscription capabilities. This allows to reduce the time and effort of onboarding Developers, allowing for better time-to-market and reduced costs.

Broad analytics

API usage is not limited to internal consumption, hence monitoring and analytics are even more relevant than before. API Management provides central, technical and business-driven analytic capabilities. This allows to quickly solve issues for increased user satisfaction and to steer your API program for an optimal return on investment of your digitalization projects.

How does it work ?- What do I do for it?

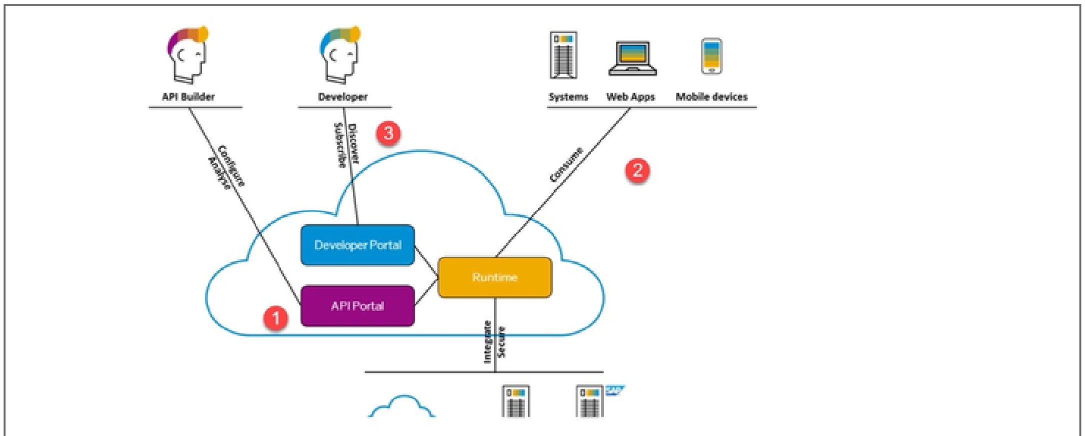


Figure 62: How does it work ?- What do I do for it?

From a high-level perspective, SAP API Management is divided in three main areas:

1: Designing APIs through the API Portal.

3: Consume it with Internet Technologies.

3: Providing APIs through the Developer Portal.

Typical use case

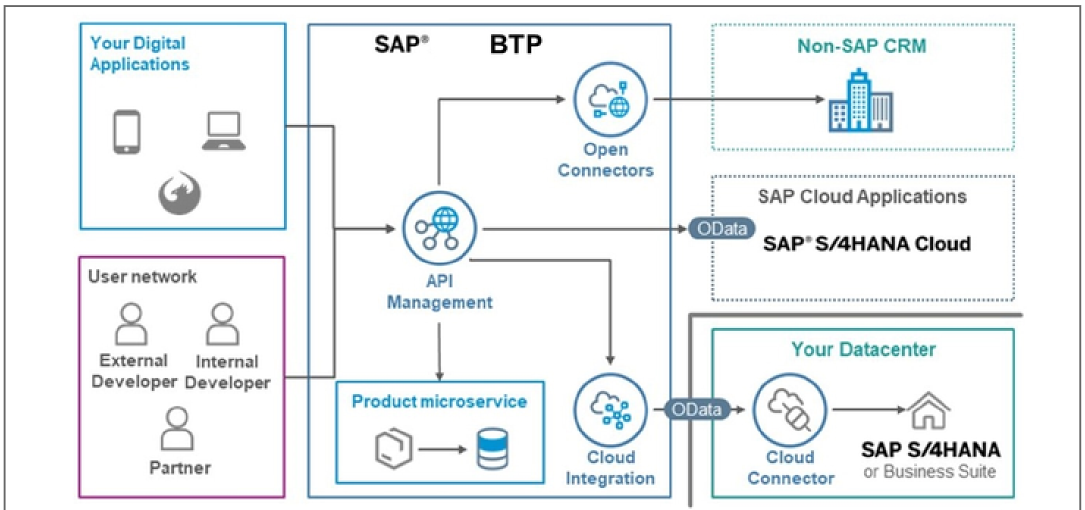


Figure 63: Typical Use Case

API management is in the middle between APIs delivery and APIs consumers.

Steps to provide an custom API are:

1: Create an API Provider artifact based on an API resource located elsewhere.

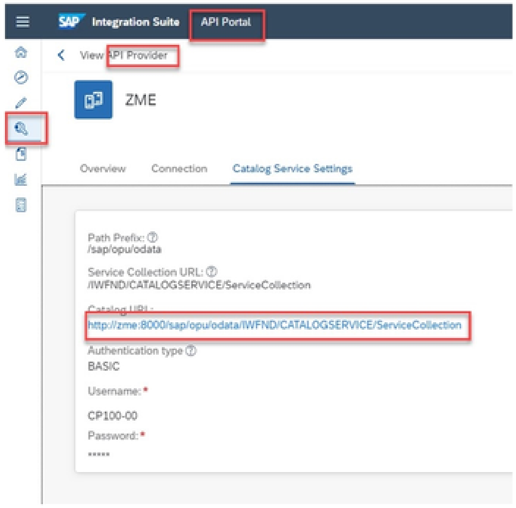


Figure 64: First Step to Provide an Custom API

2: Create an API Proxy artefact based on the API Provider artefact.

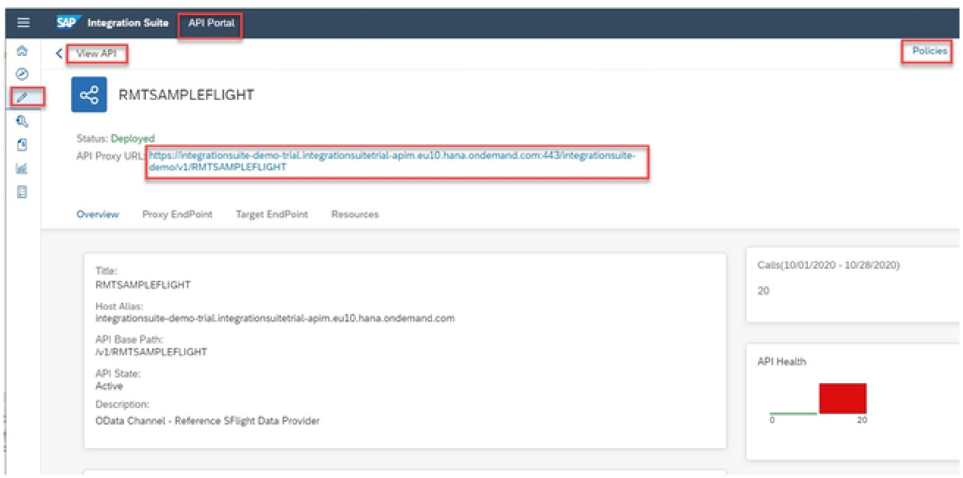


Figure 65: Second Step

3: Configure your API Proxy via policies

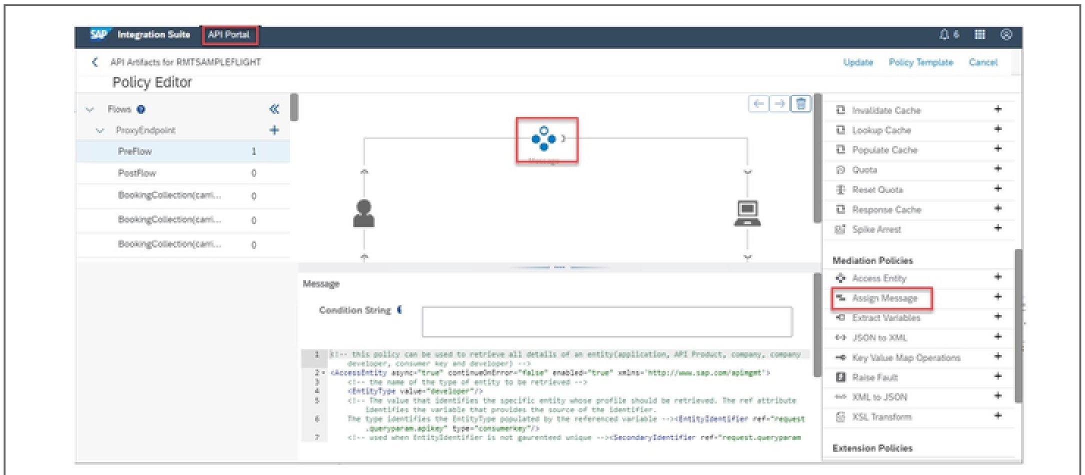


Figure 66: Third Step

4: Create a product/application and sell it. Manage within the Developer Portal

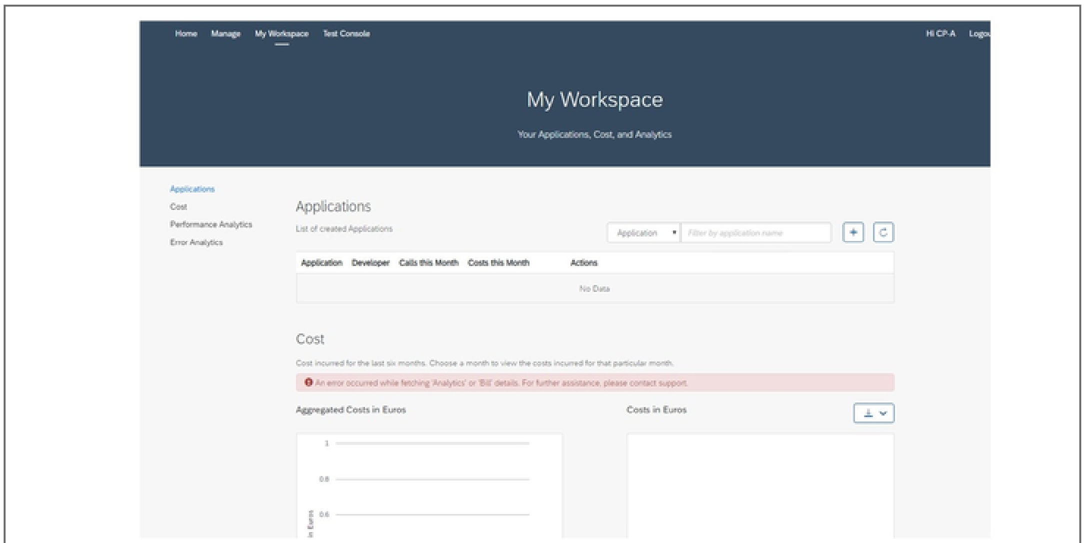


Figure 67: Fourth Step

The figure illustrates the screen, in which the settings for the fourth step are performed.

To the point

Focused statements about API management:

Configuration instead of coding.

Efficient API Publishing.

Broad analytics.

Whats New: API management is now a part of the integration suite.

Learn more:

https://help.sap.com/viewer/66d066d903c2473f81ec33acfe2ccdb4/Cloud/en-US/7d8514b4ab46455e8416723003b414d7.html

LESSON SUMMARY

You should now be able to:

- Explain the use of API Management

Unit 3
Lesson 6

Explaining SAP Graph

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain the use of SAP Graph

SAP Graph

In this lesson we will cover the following topics:

- What can I do with it?
- How does it work?

What can I do with it?

What is SAP Graph?

- SAP Graph is the **easy-to-use** API for the data of the Intelligent Enterprise from SAP.
- It provides an **intuitive programming** model that you can use to easily build new extensions and applications using SAP data.
- SAP Graph **wraps** the APIs of **existing products** into a single **harmonized API** layer across the existing source systems.
- This allows you to easily **build applications** and **extensions** for the SAP Intelligent Enterprise. SAP Graph **is based on OData**

SAP Graph technology is initially used in the four main processes of the intelligent enterprise.

SAP Graph focuses on the Lead-to-Cash (specifically on the SAP S/4HANA and SAP C/4HANA components and their products), Source-to-Pay and Recruit to Retire processes. Other processes and corresponding object types will be added in the future.

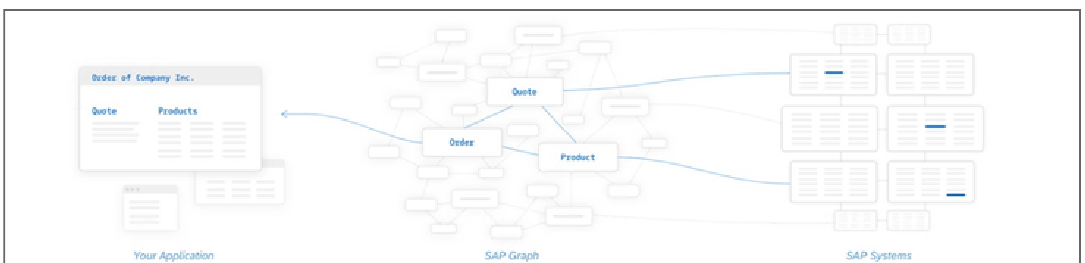


Figure 68: SAP Graph Overview

In this figure you can see that different professional domains, quota, product and order are summarized in one API.

How does it work ?- What do I do for it?

A graph describes a structure of nodes connected by edges, which we use to represent any kind of more or less complex networks. Some examples of graphs are computer networks with machines as nodes and network links as edges, social networks with users as nodes, and social relationships as edges. Graphs are helpful to visualize relationships as a formal representation of complex networks.

SAP Graph schema

In the Intelligent Enterprise we consider business entities, such as sales quotes, products, materials, workforce persons, skills, or cost centers, as nodes.

You can follow the edges between the nodes as links. Edges in the SAP Graph schema have a direction which makes them directed edges, other types of graphs, like road networks, often have undirected edges.

The directed edges in SAP Graph point from one node to another node, creating links (connections) between them. The links between the directed edges are labelled. However, the labels are often the same as the destination.

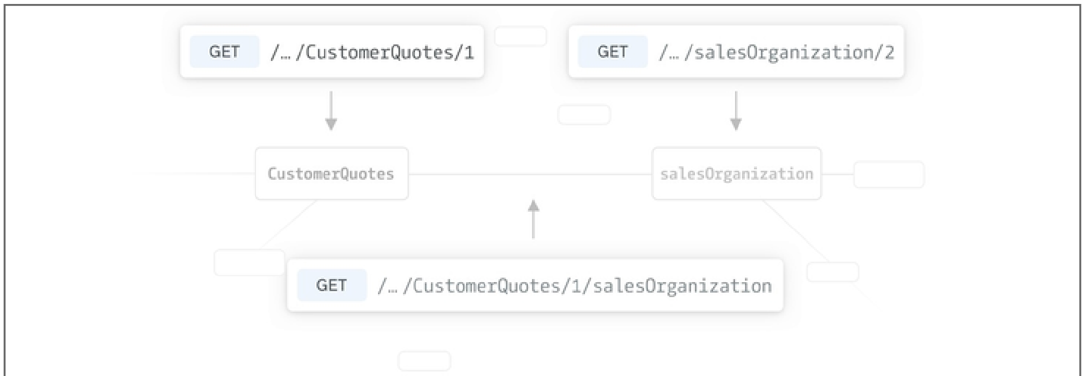


Figure 69: Example Schema: CustomerQuotes salesOrganization

The CustomerQuotes is a node connected to salesOrganization nodes by the link with the salesOrganization label. The SAP Graph API helps you explore these types of relationships by exposing the business entities as resources in its API.

To the point

Focused statement: What is SAP Graph?

SAP Graph is the easy-to-use API for the data of the Intelligent Enterprise from SAP.

It provides an intuitive programming model that you can use to easily build new extensions and applications using SAP data.

SAP Graph wraps the APIs of existing products into a single harmonized API layer across the existing source systems.

This allows you to easily build applications and extensions for the SAP Intelligent Enterprise. SAP Graph is based on OData.

Learn more

https://explore.graph.sap/

LESSON SUMMARY

You should now be able to:

- Explain the use of SAP Graph

Unit 3
Lesson 7

Explaining the Integration Advisor

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Create B2B mappings

Integration Advisor

In this lesson we will cover the following topics:

- Introduction
- How does it work ?- What do I do for it?
- Library of Type Systems
- Sample Scenario
- Build a MIG (Message implementation guideline) for source and target•Mapping Guidelines (MAGs).
- Mapping Guidelines (MAGs).
- Use the proposal service.
- Generate different runtime artifacts.
- Use the MAG artifacts within your integration Solution.

Introduction

What is Integration Advisor?

- SAP Integration Advisor service is a cloud application.
- It uses a crowd-based machine learning approach to help you create integration content(Mappings) very easily.
- This mapping artifacts can be used within the PI or the CPI.

SAP Integration Advisor service is a cloud application that helps you to simplify and streamline the implementation flow of your B2B/A2A and B2G integration process. It uses a crowd-based machine learning approach to help you create integration content very easily. Tests indicate that you can speed up the content creation to deployment process by almost 60% using SAP Integration Advisor service. You can also manage and share your content, and leverage the content shared by other users of the application with similar business needs

The SAP Integration Advisor service solves the biggest problem that you face in B2B/A2A/B2G integration: multiple business partners who use different industry standards like UN/EDIFACT, SAP IDoc and ASC X12, to name a few. Each new standard will create a need for a new interface that facilitates this integration and doing this manually is extremely time-consuming.

With IA, you have a library of type systems that you can use a starting point, use the messages in this type system library to:

Create a new message implementation guideline (MIG).

Create mapping guidelines (MAG) to map the standard you use in your business system to that of your business partner's.

Generate runtime artifacts (XSLT Mappings) that you can use in different integration solutions like SAP Integration Suite and SAP Process Orchestration.

How does it work ?- What do I do for it?

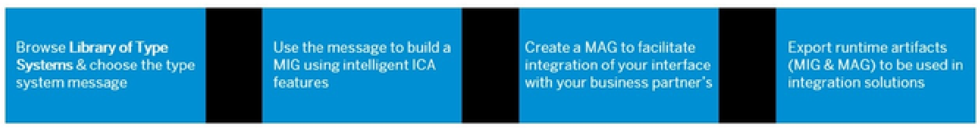
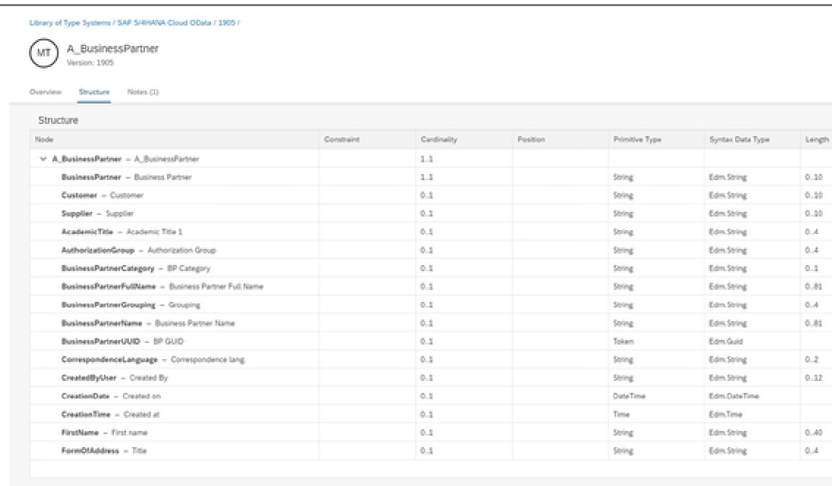


Figure 70: Step-by-Step Overview

We will examine the individual steps in more detail below.

Library of Type Systems

The library of type systems is a collection of message templates that are provided by agencies that maintain the B2B/A2A/B2G standards. Each of the type system is developed and maintained by the agency that owns it


Library of Type Systems / SAP S/4HANA Cloud OData / 1905 /

MT A_BusinessPartner

Version: 1905

Overview Structure Notes (1)

Structure

Node	Constraint	Cardinality	Position	Primitive Type	Syntax Data Type	Length
A_BusinessPartner – A_BusinessPartner		1..1				
BusinessPartner – Business Partner		1..1		String	Edm.String	0..10
Customer – Customer		0..1		String	Edm.String	0..10
Supplier – Supplier		0..1		String	Edm.String	0..10
AcademicTitle – Academic Title 1		0..1		String	Edm.String	0..4
AuthorizationGroup – Authorization Group		0..1		String	Edm.String	0..4
BusinessPartnerCategory – BP Category		0..1		String	Edm.String	0..1
BusinessPartnerFullName – Business Partner Full Name		0..1		String	Edm.String	0..81
BusinessPartnerGrouping – Grouping		0..1		String	Edm.String	0..4
BusinessPartnerName – Business Partner Name		0..1		String	Edm.String	0..81
BusinessPartnerUUID – BP GUID		0..1		Token	Edm.Guid	
CorrespondenceLanguage – Correspondence lang.		0..1		String	Edm.String	0..2
CreatedByUser – Created By		0..1		String	Edm.String	0..12
CreationDate – Created on		0..1		DateTime	Edm.DateTime	
CreationTime – Created at		0..1		Time	Edm.Time	
FirstName – First name		0..1		String	Edm.String	0..40
FormOfAddress – Title		0..1		String	Edm.String	0..4

Figure 71: Sample Type System From SAP S/4HANA Cloud OData

On the screenshot you can see the Business Partner OData interface.

Type Systems are:

SAP IDOCS

ASC X12

cXML

ICA Test

ISO

Odette

SAP S/4HANA Cloud OData

SAP S/4HANA Cloud SOAP

SAP S/4HANA On Premise IDoc

UN/CEFACT

UN/EDIFACT

Sample Scenario

A continuous example shows the individual steps

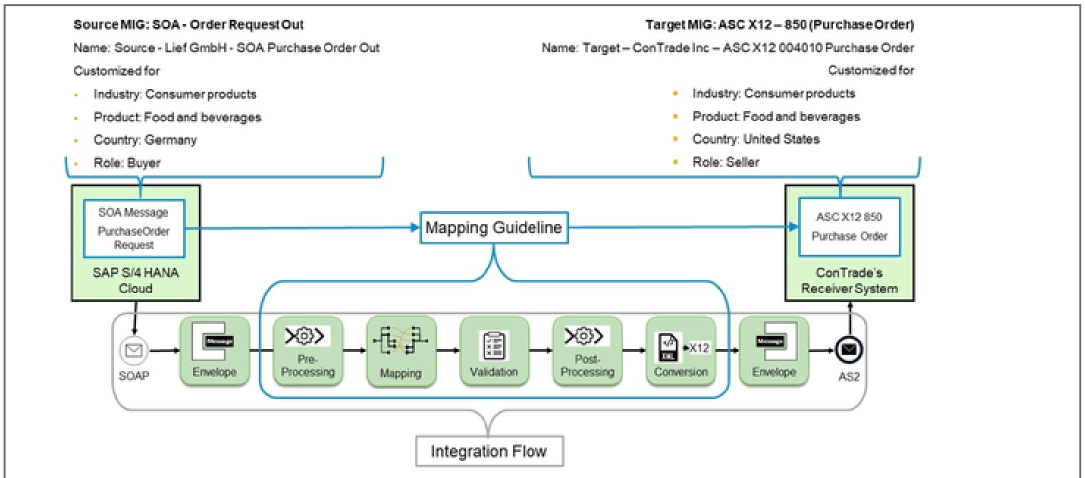


Figure 72: Overview of the Sample

The source is a Mig OrderRequest based on a SOAP interface. On the target side, an ASC x12 Purchase Order interface is created as MIG.

A mapping is then created between the source and the destination. The mapping artifacts can then be used -for example- in the CPI.

Build a MIG (Message implementation guideline) for source and target

Message implementation guideline, also referred to as MIG, is used as the source or target in a mapping guideline (MAG) of SAP Integration Advisor service. This is created using one of the messages in the type system that is relevant to your scenario as the template.

The MIG contains all the information for implementing a customized message interface. SAP Integration Advisor service uses the message in a type system as a starting point to ensure that you do not have to refer to any additional documents ti implement the interface. By providing a MIG, you ensure that all users who are involved in the process of implementing the interface have a clear understanding of the guidelines.

Target MIG ASC X12 MIG: Structure View

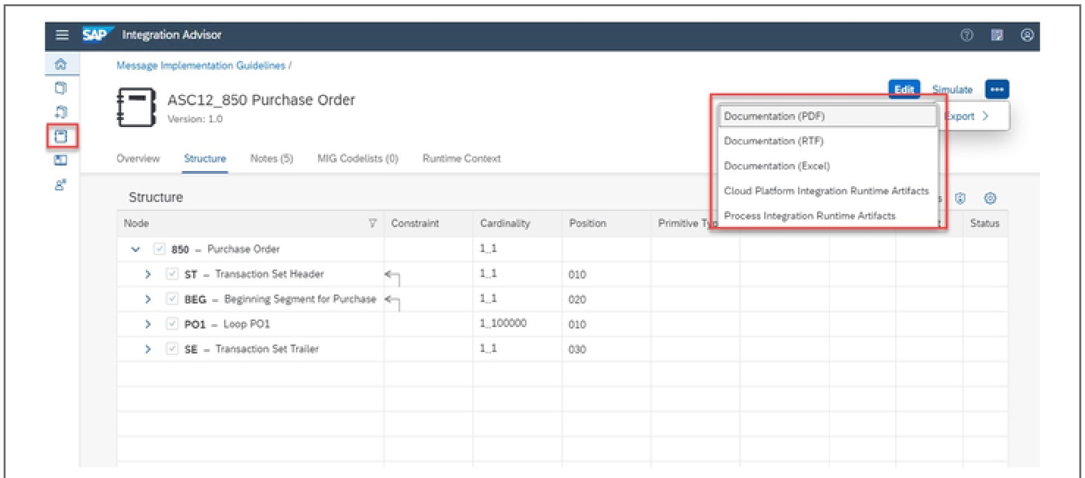


Figure 73: MIG for Target

You see the Structure of the target MIG. You can download a documentation as .pdf and also .xsd files to use within the cloud integration.

Source MIG OrderRequestOut - Structure view

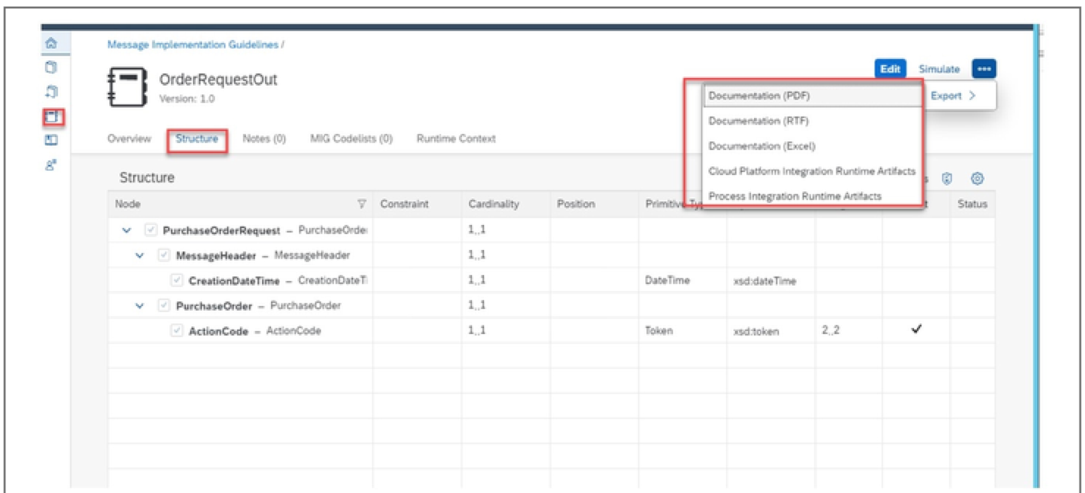


Figure 74: MIG for the Source

You see the Structure of the source MIG. You can download a documentation as p.df and also .xsd files to use within the cloud integration

Mapping Guidelines (MAGs)

A mapping guideline is the runtime artifact that you use in the mapping step of an integration application like SAP Integration Suite. This is used as a reference or guidance for implementing mapping in the integration application. By providing such an artifact, you simplify the mapping task for users who use messages that adhere to the A2A/B2B type system standards.

A Mapping Guideline (MAG) is based on a source and a target message implementation guideline. It demonstrates how the defined nodes at each side are mapped, describing all mapping elements in detail with definitions or notes and providing further instructions for the transformation, such as functions or code value mappings.

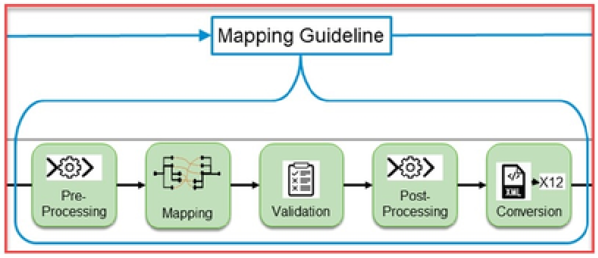


Figure 75: Mapping Guideline

In addition to the actual mapping, further steps are carried out in the MAG, pre- and post processing, validation and conversion.

Use the proposal service

All proposals are from a centrally provided and trained knowledge graph. The graph is trained by users; once they activate a MAG, their updates are anonymously pushed into the graph. A number of diverse machine-learning (ML)-based algorithms then calculate the best fit proposal according your business context and based on the selected source and target structure. The main intention is that all proposals are semantically correct, independent of the complexity of the source and target structures.

Proposal service in action

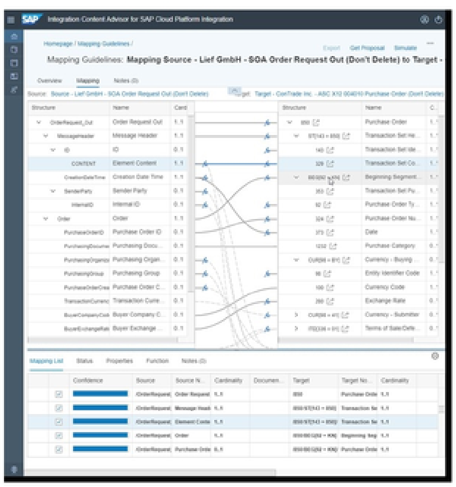


Figure 76: Proposal Service

The result of "Select Best Proposal" is shown in form of mapping lines from the source elements to the corresponding target elements

You can now also manually change or delete mapping entities as well as create additional ones.

Of course, this only works if the proposal service has been trained accordingly beforehand. This works in the background by using manual mappings: Export the runtime artifacts .

Generate different runtime artifacts

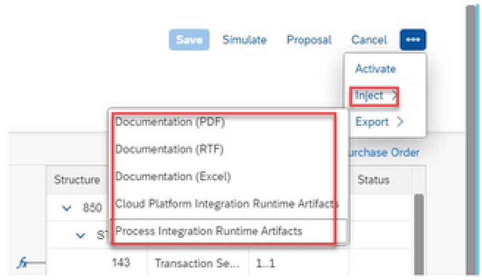


Figure 77: Runtime Artifacts

Generate different runtime artifacts :

XSLT Mappings for CPI

XSLT Mappings for PI

Documentation

In addition to a documentation of the mapping, various runtime artifacts (XSLT) are created. It is even possible to import the XSLT mapping directly into an iFlow via Inject. Use the XSLT Mappings within your CPI.

Use the MAG artifacts within your integration Solution

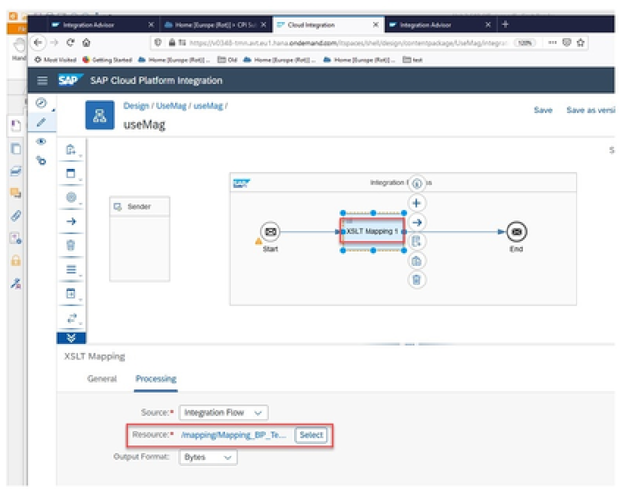


Figure 78: IFlow with xslt Mapping Created from Content Advisor

The figure shows a sample XSLT mapping script.

To the point

Focused statement: What is Integration Advisor?

SAP Integration Advisor service is a cloud application.

It uses a crowd-based machine learning approach to help you create integration content (Mappings) very easily.

This mapping artifacts can be used within the PI or the CPI.

Learn more:

https://open.sap.com/courses/s4h20

https://help.sap.com/viewer/368c481cd6954bdfa5d0435479fd4eaf/Cloud/en-US/6b9fe2d753534bebadcfa9080228bd94.html

LESSON SUMMARY
You should now be able to:

- Create B2B mappings

Unit 3
Lesson 8

Explaining Cloud Integration

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain the use and the main components of Cloud Integration

Cloud Integration

In this lesson we will cover the following topics:

- What can I do with it?
- Sample of a out-of-the box scenario.
- How does it work and what does it do for me.
- What are the advantages?

What can I do with it?

- Support end-to-end process integration through the exchange of messages.
- The SAP Integration Suite helps you to connect cloud and on-premise applications with other SAP and non-SAP cloud and on-premise applications.
- This service has the capabilities to process messages in real-time scenarios spanning different companies, organizations, or departments within one organization.
- It runs as an SaaS App on the SAP BTP.

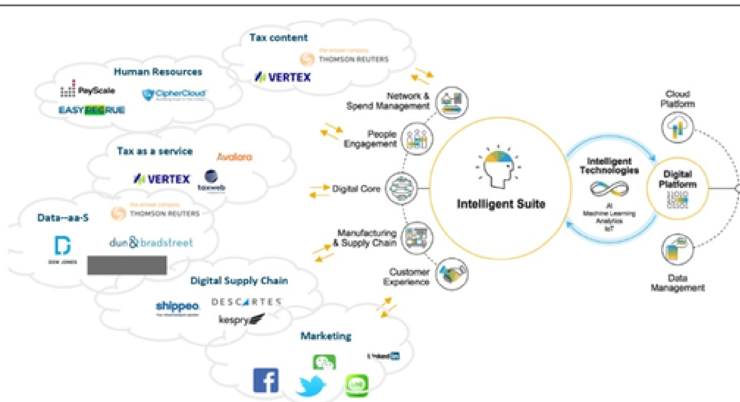


Figure 79: Out-of-the-box Business Process Integrations

On this figure the concept and the idea of out of the box integration is summarized again. On the left are indicated many common NO SAP partners, all of whom have to exchange data with the various components of the Intelligent Suite.

The Features are:

Implement diverse scenarios.

Connect to multiple endpoints.

Customize SAP integration scenarios.

Develop custom adapters.

Access public APIs.

Set up secure and reliable communication.

Implement various communication modes.

Integrate with SAP Process Orchestration.

That means in detail:

Implement diverse scenarios

Integrate processes and data in application-to-application (A2A) and business-to-business (B2B) scenarios.

Connect to multiple endpoints

Integrate various applications and data sources from SAP and non-SAP, on premise, as well as the cloud. The SAP Integration Suite comes with a set of pre-built adapters.

Customize SAP integration scenarios

Benefit from prepackaged integration content to jump-start integration projects and to set up productive scenarios with only minimum effort. You can extend predefined integration flows according to your requirements.

Develop custom adapters

Use the adapter SDK to build your own custom adapters for additional connectivity needs.

Access public APIs

Customize the access to the SAP Integration Suite with our public OData APIs.

Set up secure and reliable communication

Use our core integration and security capabilities for the safe and reliable processing of messages. Configure the way how messages are exchanged within an integration scenario so that the data involved is protected according to the newest security standards.

Implement various communication modes

Orchestrate business processes and integrate data in synchronous as well as in asynchronous scenarios. The SAP Integration Suite also supports reliable messaging processes based on asynchronous decoupling implemented by using queuing mechanisms.

Integrate with SAP Process Orchestration

Use SAP Integration Suite and SAP's on-premise integration Platform, SAP Process Orchestration, seamlessly integrated.

Sample of a out-of-the box scenario

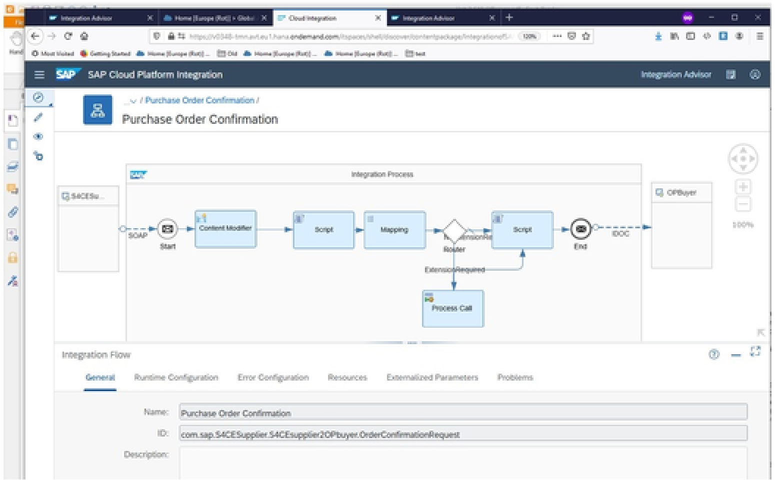


Figure 80: Sample of an Predefined iFlow

The figure shows the Cloud Integration Cockpit - Discover view. A predefined process is selected. The individual tiles mean predefined functionality for processing the data.

How does it work and what does it do for me

The cockpit is the central point for managing all activities associated with your subaccount and for accessing key information about your applications.

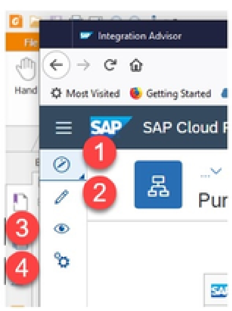


Figure 81: CPI Cockpit

Explanations about the bubbles:

1. Discover view
2. Design view
3. Operations view
4. Tenant settings

Discover view

Here you will find all available pre-defined iFlows that you can use directly.

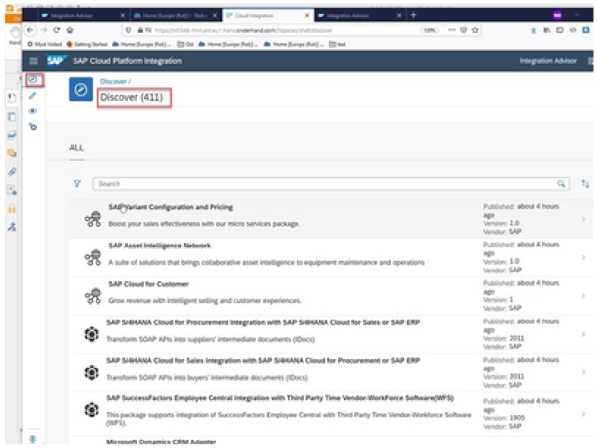


Figure 82: Discover View

Here you can explore and copy a lot of predefined iFlows.

Design view

Create your own iFlow or use an preconfigured iFlow.

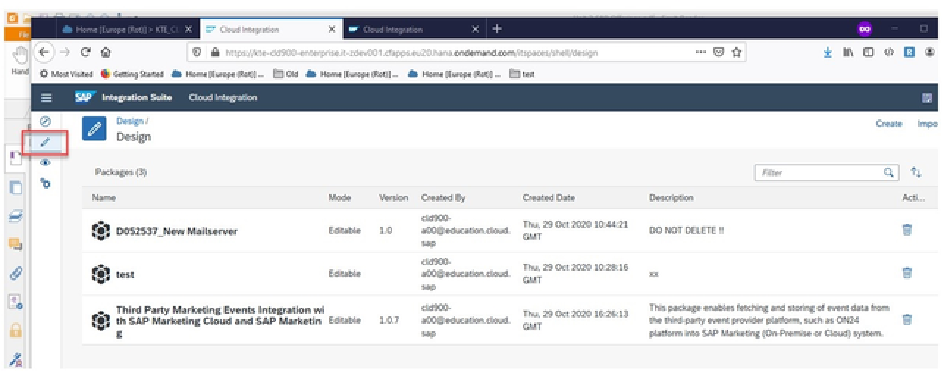


Figure 83: Design View

You will see two own and one pre-configured Scenario. Here you can create your own iFlow, configure an pre-configured or extend an already existing iFlow.

Operations view

Here you can :

- Monitor Message Processing
- Manage Integration Content
- Manage Security
- Manage Stores

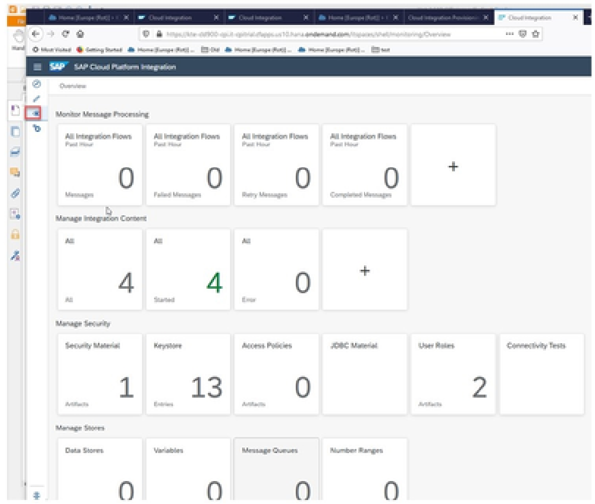


Figure 84: Operations View

In this view you can see the different usable functionalities. The implementation follows the DevOps approach.

Tenant settings view

Here you have several options to configure your tenant.

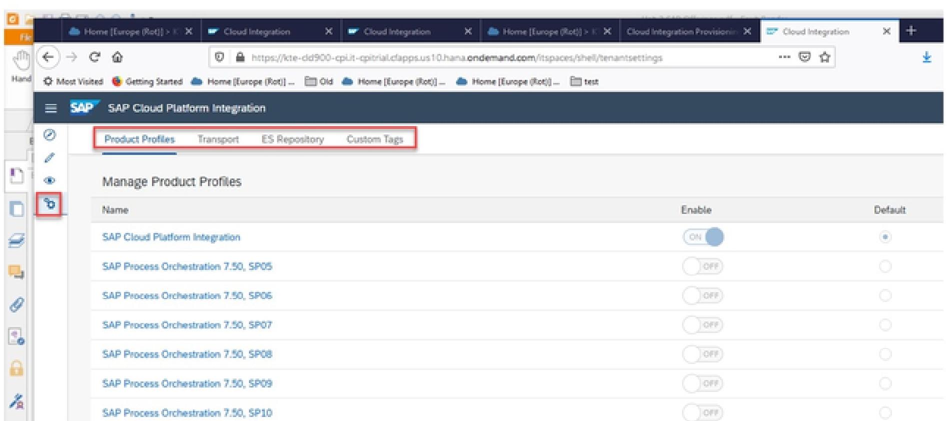


Figure 85: Tenant Settings View

Only administrative tasks are performed in this area. Here you can see the possibility to develop iFlows for the on-premise Process Integration and to process there as well.

What are the advantages?

Deployment as an integration as a service does **not require installation** or configuration.

All actions take place in the browser. This is in line with the **DevOps** approach.

Due to the **large amount of predefined technical integration scenarios**, you are **quickly productive**.

LESSON SUMMARY

You should now be able to:

- Explain the use and the main components of Cloud Integration

Unit 3
Lesson 9

Explaining SAP Enterprise Messaging

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain the features of SAP Enterprise Messaging

SAP Enterprise Messaging

In this lesson we will cover the following topics:

- What can I do with it?
- Features
- How does it work ?- What do I do for it?

What can I do with it?

Uses cases:

- SAP Enterprise Messaging is a fully-managed cloud service that allows applications to communicate through asynchronous events and seamlessly extend your digital core.
- Create responsive applications that work independently and participate in event-driven business processes inside your company and across your business ecosystem for greater agility and scalability.

Features

The service provides the following key features:

- Decouple communication between applications, services, and systems using asynchronous messaging patterns.
- Publish business events from SAP and non-SAP sources across hybrid landscapes from the digital core to extension applications.
- Build event-driven extensions.
- Subscribe to business events from core SAP systems like SAP S/4HANA, SAP SuccessFactors, SAP Concur, and SAP Cloud for Customer or third-party sources to enable cloud-native extensions to respond to the latest business developments.
- Seamless connectivity Implement extension and integration scenarios through decoupled communication between applications, services, and systems as part of the SAP Integration Suite.

How does it work ?- What do I do for it?

Development

Enable your application to:

Connect to a queue or a queue subscription in the enterprise messaging service.

Send messages to a queue or queue subscription

Receive messages from a queue or queue subscription.

Develop messaging applications using Java and Node.js and deploy them to the service.

Look up events from an event source such as SAP S/4HANA.

Use REST APIs for messaging.

Administration

Create and delete queues.

Create and delete queue subscriptions.

Generate application confiJurDtLons that facilitate the use of protocol agnostic libraries.

Check your application confiJurDtLons for missing properties.

Create, edit, and delete event channel groups.

Use REST APIs to manage queues and queue subscriptions.

Create message clients with unique credentials.

Provide access rules for each message client.

Create a message client with an event catalog.

Events

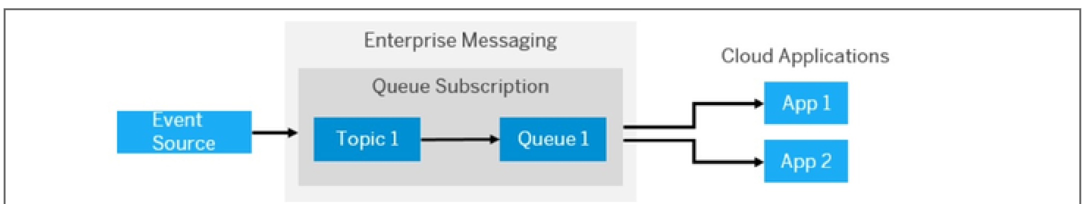


Figure 86: Concept of Events

An event notifies a receiving application that an object at the event source has changed. An event source is the system or application from which the event originates. A receiving application needs to create a connection to the event source to facilitate the flow of events. The service can receive events from an event source, lookup events, and discover events. Different event sources can publish events to the service.

Asynchronous Messaging

Asynchronous messaging facilitates communication between a sending and receiving application or system. Messages are sent to a queue where they're stored until the receiving application acknowledges them. The service provides capabilities for storing and transmitting messages through queues and queue subscriptions.

Queues

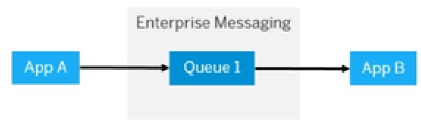


Figure 87: Concept of Queues

The service enables applications to communicate with each other through message queues. A sending application sends a message to a specific named queue. There's a one on one correspondence between a receiving application and its queue. The message queue retains the messages until the receiving application consumes them. You can manage these queues using the service.

Queue Subscriptions

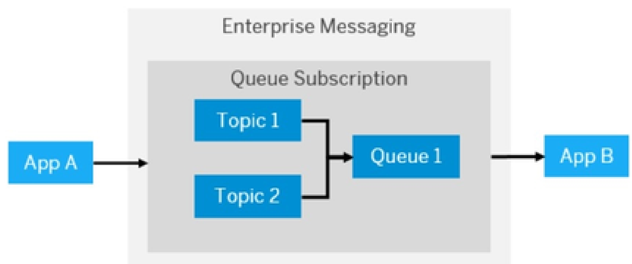


Figure 88: Concept of Queue Subscriptions

The service enables a sending application to publish messages and events to a topic. Topics don't retain messages but, it can be used when each message needs to be consumed by a number of receiving applications. Topics must be managed through queue subscriptions. In queue subscriptions, the service enables a sending application to publish messages to a topic that directly sends the message to the queue to which it's bound. For example, events from an SAP S/4HANA system (event source) can only be sent to a topic. A queue subscription ensures that the message is retained until it's consumed by the receiving application.

Message Client

A message client allows you to connect to the service through its unique credentials to send and receive messages. The message client can run within SAP BTP or outside it. You can create multiple message clients that can be distinguished with a set of credentials that consist of a namespace and connection rules. The credentials must also define the list of queues or topics to which the message client can send and receive messages. The credentials are stored in the service descriptor that you define while creating a service instance.

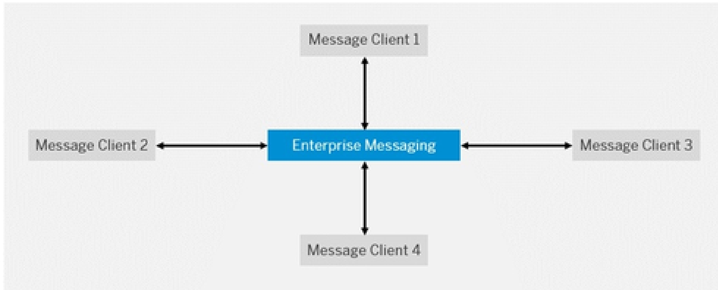


Figure 89: Concept of Message Client

The namespace is a unique prefix that defines all the queues or topics that have been created in the context of a particular message client. When you manage queues or topics in the service, providing the namespace allows message clients to identify the queues or topics to which communication must be made. The connection rules specify the queue or topic to which a message client must publish or consume messages.

The default service plan facilitates a connection between different message clients in a subaccount through its unique credentials. Message clients can communicate with each other using the service. Each message client has a set of queues and topics associated with it. All these queues and topics belonging to one message client are exposed to other message clients through the unique credentials in the service descriptor.

Event Catalog

An event source can have a list of events that can be published or consumed by the service. This list of events is known as event catalog. The catalog can contain events that are to be published or consumed. Each event in a catalog has a payload schema. An event source can provide an event catalog end point that gives the list of events that are published or consumed by the message client.

Webhooks

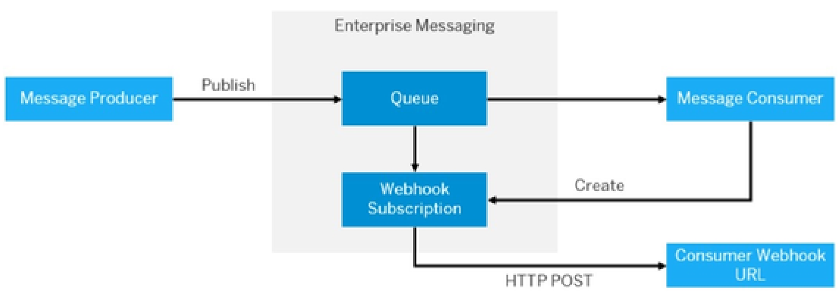


Figure 90: Concept of Webhooks

Webhooks allow you to set up an integration to send real-time data from one application to another when an event occurs. A webhook subscription notifies the receiving application when an event is triggered. The data is sent through HTTP POST to the receiving application that handles the data. The exchange of data is done through a webhook URL. A webhook URL is configured by the receiving application. When an event is triggered, an HTTP POST payload is sent to the webhook URL. The data sent to the webhook URL is known as payload.

To the point

Focused statements:

SAP Enterprise Messaging is a fully-managed cloud service that allows applications to communicate through asynchronous events and seamlessly extend your digital core.

Create responsive applications that work independently and participate in event-driven business processes inside your company and across your business ecosystem for greater agility and scalability.

Whats New: The enterprise messaging service is now called Event Mesh.

Learn more: https://discovery-center.cloud.sap/serviceCatalog/event-mesh?region=all

LESSON SUMMARY
You should now be able to:

Explain the features of SAP Enterprise Messaging

Unit 3
Lesson 10

Explaining SAP Open Connectors

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain the use of SAP Open Connectors

SAP Open Connectors

In this lesson we will cover the following topics:

- What can I do with it?
- Use Cases
- Main Features
- Connectors
- How does it work ?- What do I do for it?

What can I do with it?

Our platform - combined with our unique Connectors (more then 160) - is designed to unify the developer experience across all kinds of applications and services. Regardless of the application's back end - REST, SOAP, Proprietary SDK, Database, etc., - SAP Open Connectors creates a unified API layer and standards-based implementation across every environment. This ensures that developers, integration users, and their use cases are decoupled from the back end services on which they rely.

Use Cases

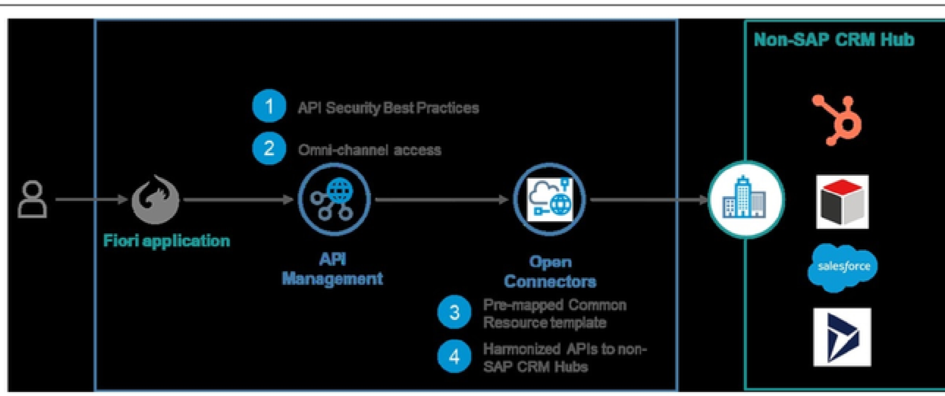


Figure 91: Use Case 1

We are using SAP Open Connectors to access data from non-SAP application using harmonized APIs from CRM Hubs. This harmonized APIs would be managed by SAP API Management where all the security best practices and governance policies would be applied. Finally we will be building a SAP Fiori application to display this harmonized data from any of the non-SAP CRM applications of your choice. A high level solution blue-print of this sample application is captured in the below figure.

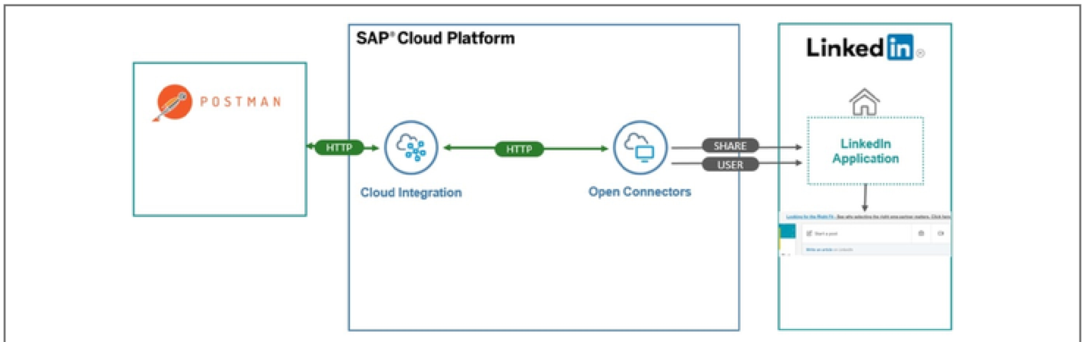


Figure 92: Use Case 2: Create a New Post in LinkedIn by Sending the Message from POSTMAN

Open Connectors is connected to the Linkedin Connector. The provided harmonised API is used via Cloud Integration to send a message to LinkedIn.

Main Features are:

Use our prebuild connectors in your integration scenarios.

Design and implement complex integration workflow templates.

Define your own resource and create mappings to your cloud apps.

Extend a connector or build your own.

Querying using the Open Connectors Query Language.

Use the SAP Open Connectors bulk services to upload and download data in bulk from an endpoint in a normalized way.

Configure security settings and set up your users and accounts.

Use the SAP Open Connectors events framework to receive notifications about changes to resources.

Use OAuth Proxy to have multiple environments, such as development, QA, etc, with one endpoint application.

Connectors

Connectors are pre-built API integration that enables a connection into a specific API Provider endpoint (for example: Salesforce, Quickbooks, or Marketo). All connectors start with a normalized set of features, including authentication, resources, paging, errors, events and search. At the hub level, we also seek to support the richer set of APIs that an application provides, even when not all the connectors in that category share that resource. For example, Salesforce Sales Cloud has APIs that many other CRM services do not support. You can find these APIs that are specific to just Salesforce in the documentation for that connector.

Further facts:

Connectors share common services including discovery, search query, pagination, bulk uploading and downloading, logging, and interactive documentation.

Methods are normalized and accessible through RESTful APIs.

Complete data payloads are returned in JSON and available to transform and normalize to a common resource.

SAP keeps each connector up-to-date with changes at the endpoint.

Each connector is a multi-tenant connector supporting an unlimited number of authenticated accounts with no additional code required.

How does it work ?- What do I do for it?

Sample :

Simplify connectivity to third-party document storage application from SAP Open Connectors.

1. Log in to open Connectors

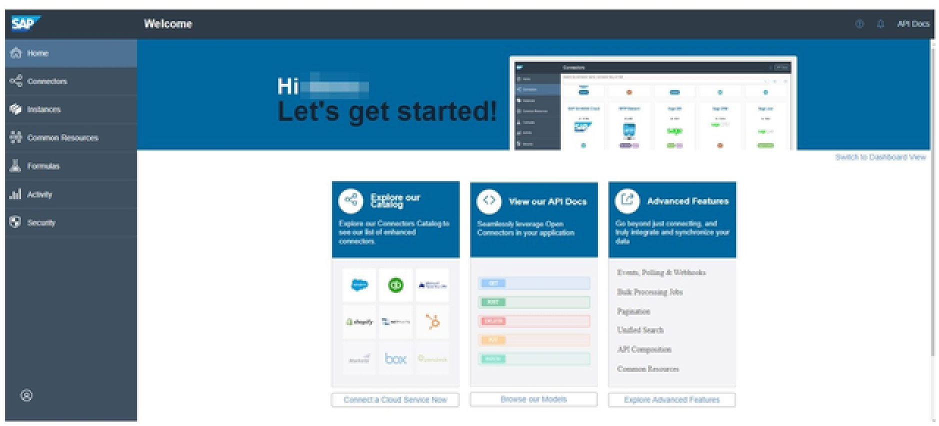


Figure 93: Step 1

1. Log in to open Connectors

Step 2

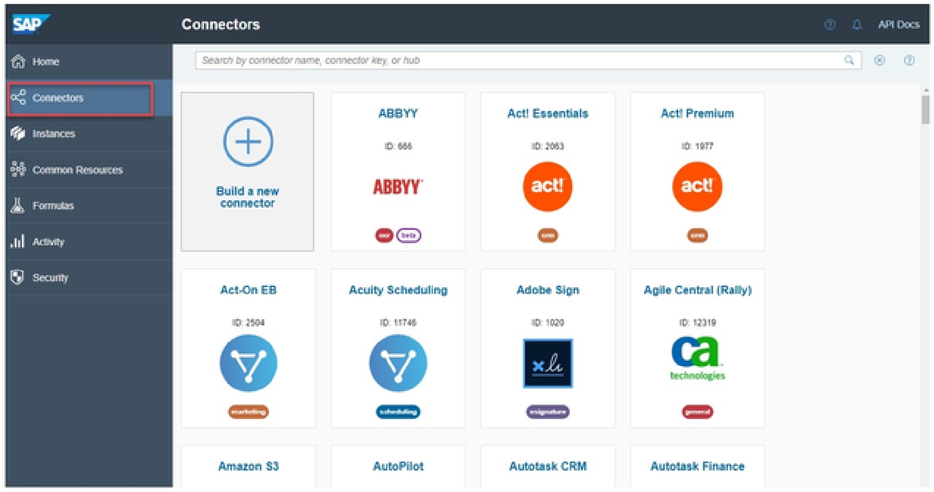


Figure 94: Step 2

2. Click on the Connectors tab to view all the available pre-built, feature rich connectors.

Step 3

3. Choose Dropbox.

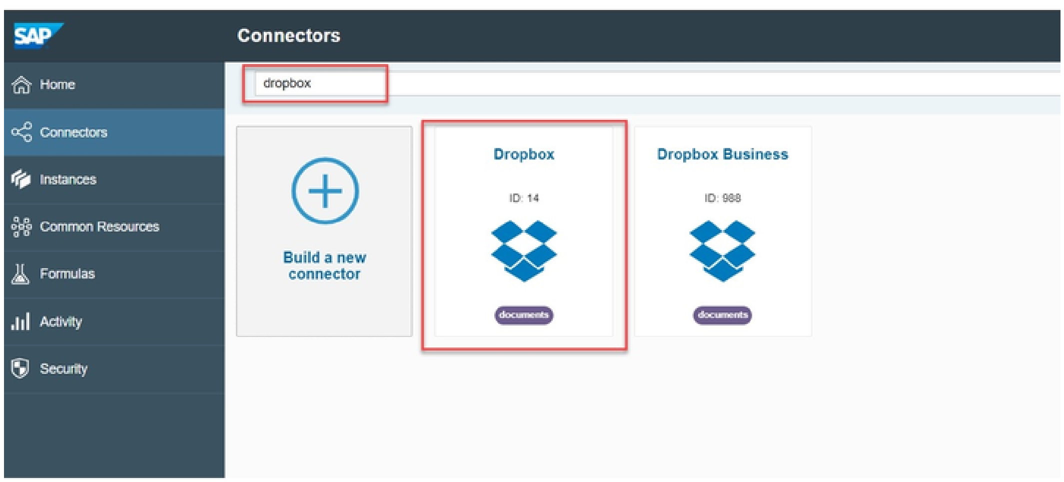


Figure 95: Step 3

3. Choose Dropbox.

Step 4

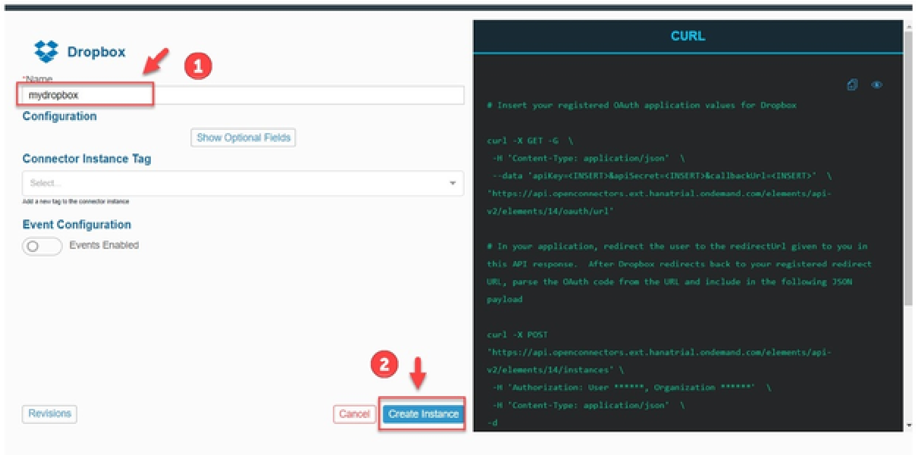


Figure 96: Step 4

Hover over the Dropbox connector and select the option Authenticate to connect to your own Dropbox tenant.

You would be navigated to the Create Authenticated Connector page, enter a name for your authenticated connector say mydropbox and click on the Create Instance button

Step 5

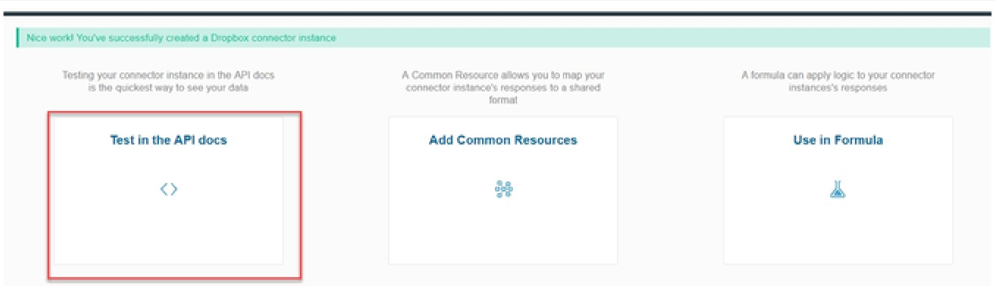


Figure 97: Step 5

After the authenticated connection to your third-party application has been created successfully you would be able to test out the connection from SAP Open Connectors. To try out the RESTFul APIs for the specific third-party application, click on the option Test in the API Docs.

To the point

Focused statements about open Connectors:

Use our pre-build connectors for one API in your integration scenarios.

Design and implement complex integration workflow templates.

Define your own resource and create mappings to your cloud apps.

Extend a connector or build your own.

- Querying using the Open Connectors Query Language.
- Use the SAP Open Connectors bulk services to upload and download data in bulk from an endpoint in a normalized way.
- Configure security settings and set up your users and accounts.
- Use the SAP Open Connectors events framework to receive notifications about changes to resources.
- Use OAuth Proxy to have multiple environments, such as development, QA, etc, with one endpoint application.

Learn more: https://help.openconnectors.ext.hana.ondemand.com/home/take-a-tour

LESSON SUMMARY
You should now be able to:

- Explain the use of SAP Open Connectors

Learning Assessment

1. Which of the folowing belongs is part of the Integration Suite?

   Choose the correct answers.

   - [ ] **A** Cloud Integration
   - [ ] **B** Data Intelligence
   - [ ] **C** Open Connectors
   - [ ] **D** SAP Graph

2. ISA-M can be used for the following:

   Choose the correct answers.

   - [ ] **A** As Governance Blueprint to provide integration architecture guidance.
   - [ ] **B** As Enabler for an integrative use of different user groups.
   - [ ] **C** As Assessment Tool to define and update your integration strategy.
   - [ ] **D** As Accelerator for Pattern-Based Interface Development (Reuse).

3. What are the four main use cases of SAP?

   Choose the correct answers.

   - [ ] **A** Design to Operate.
   - [ ] **B** Build to Run.
   - [ ] **C** Recruit to Retire.
   - [ ] **D** Source to Pay.
   - [ ] **E** Lead to Cash.

4. In the SAP API Business Hub, only APIs from SAP can be accessed. To reach APIs from partners, you must use the SAP Partner API Business hub.

   Determine whether this statement is true or false.

   - [ ] True
   - [ ] False

5. API management is in the middle between APIs delivery and APIs consumers.

   Determine whether this statement is true or false.

   - [ ] True
   - [ ] False

6. SAP Graph offers an intuitive programming model.

   Determine whether this statement is true or false.

   - [ ] True
   - [ ] False

7. The SAP Integration Advisor provides the following elements:

   Choose the correct answers.

   - [ ] **A** Mapping Guideline Editor
   - [ ] **B** Documentation
   - [ ] **C** Testing Scripts
   - [ ] **D** Runtime Artifacts

8. Features of Cloud Integration are:

   Choose the correct answers.

   - [ ] **A** Connect to multiple endpoints.
   - [ ] **B** Customize SAP integration scenarios.
   - [ ] **C** Access public APIs
   - [ ] **D** Implement various communication modes

9. SAP Enterprise Messaging is a fully-managed cloud service that allows applications to communicate through asynchronous events and seamlessly extend your digital core.

   Determine whether this statement is true or false.

   - [ ] True
   - [ ] False

10. It is possible, to use the SAP Open Connectors events framework to receive notifications about changes to resources.

    Determine whether this statement is true or false.

    - [ ] True
    - [ ] False

Learning Assessment - Answers

1. Which of the folowing belongs is part of the Integration Suite?

   Choose the correct answers.

   - [x] **A** Cloud Integration
   - [x] **B** Data Intelligence
   - [x] **C** Open Connectors
   - [x] **D** SAP Graph

   This is correct.

2. ISA-M can be used for the following:

   Choose the correct answers.

   - [x] **A** As Governance Blueprint to provide integration architecture guidance.
   - [ ] **B** As Enabler for an integrative use of different user groups.
   - [x] **C** As Assessment Tool to define and update your integration strategy.
   - [x] **D** As Accelerator for Pattern-Based Interface Development (Reuse).

   This is correct.

3. What are the four main use cases of SAP?

   Choose the correct answers.

   - [x] **A** Design to Operate.
   - [ ] **B** Build to Run.
   - [x] **C** Recruit to Retire.
   - [x] **D** Source to Pay.
   - [x] **E** Lead to Cash.

   This is correct. The four main use cases are: Lead to Cash, Source to Pay, Recruit to Retire, Design to Operate.

4. In the SAP API Business Hub, only APIs from SAP can be accessed. To reach APIs from partners, you must use the SAP Partner API Business hub.

   Determine whether this statement is true or false.

   - [ ] True
   - [x] False

   This is correct. The SAP API Business Hub offers all available APIs.

5. API management is in the middle between APIs delivery and APIs consumers.

   Determine whether this statement is true or false.

   - [x] True
   - [ ] False

   This is correct.

6. SAP Graph offers an intuitive programming model.

   Determine whether this statement is true or false.

   - [x] True
   - [ ] False

   This is correct.

7. The SAP Integration Advisor provides the following elements:

   Choose the correct answers.

   - [x] **A** Mapping Guideline Editor
   - [x] **B** Documentation
   - [ ] **C** Testing Scripts
   - [x] **D** Runtime Artifacts

   This is correct. It does not contains testing scripts.

8. Features of Cloud Integration are:

   Choose the correct answers.

   - [X] **A** Connect to multiple endpoints.
   - [X] **B** Customize SAP integration scenarios.
   - [X] **C** Access public APIs
   - [X] **D** Implement various communication modes

   This is correct.

9. SAP Enterprise Messaging is a fully-managed cloud service that allows applications to communicate through asynchronous events and seamlessly extend your digital core.

   Determine whether this statement is true or false.

   - [X] True
   - [ ] False

   This is correct.

10. It is possible, to use the SAP Open Connectors events framework to receive notifications about changes to resources.

    Determine whether this statement is true or false.

    - [X] True
    - [ ] False

    This is correct.

UNIT 4 SAP Extension Suite

UNIT OBJECTIVES

- Explain, why extensibility is so important
- Explain the tools used for creating an extension
- Discover the place and the features of the Cloud Connetor in connectivity
- Explore the area of endpoint building

Explain which channels are available within the SAP Extension Suite

Identify which tools for the deployment and operate area exist

Undergo a birds view In-App extensions

Undergo a birds view Side-by-Side extensibility

Unit 4 Lesson 1

Explaining the Extension Strategy

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain, why extensibility is so important

Extensibility Suite

In this lesson we will cover the following topics:

- What is the SAP Extension Suite?
- Short repetition
- Categorization into three areas - first look
- Going deeper in the categories
- What are extensions ?
- Where can I find the information if and how i can extend an SAP SaaS app via side-by-side?

What is the SAP Extension Suite?

- With the Integration Suite, you can make sure that you have the **right data** at the **right time** in the **right system** in the correct format.
- The **Extension Suite** now leverages the data **provided by the Integration Suite** and adds UX, processes and development capabilities on top.

First of all, the Extension Suite comes was best run methodologies and guidance and guidelines for the customers on how to implement a good user experience, a good process integration and with very efficient, cloud based developer tools. In this chapter we will only take care of the Extension Suite.

Short repetition

Remember the Intelligent Enterprise strategy chapter. The following picture was shown there.

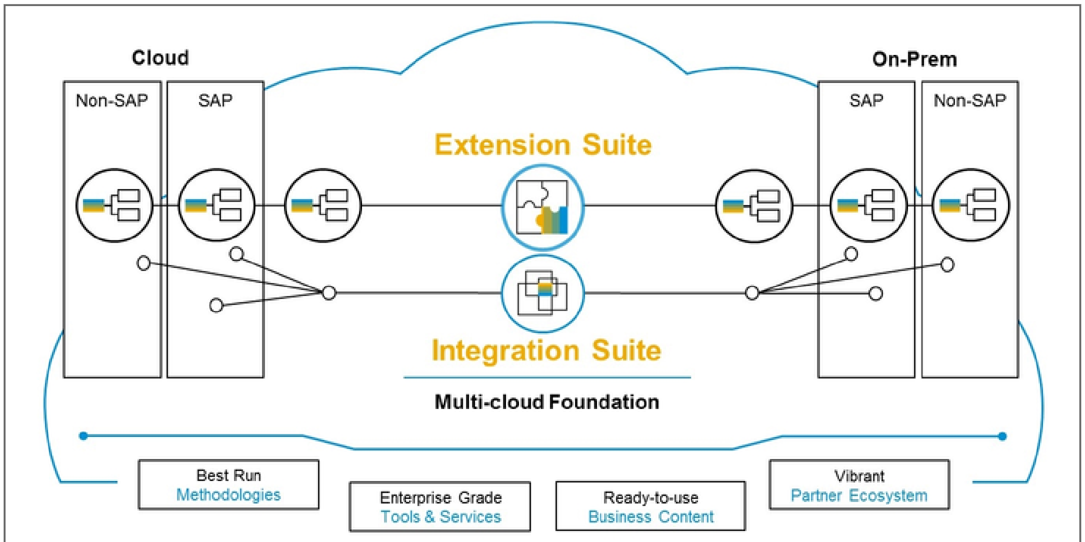


Figure 98: Position of the SAP Integration Suite and SAP Extension Suite within SAP BTP

On the left-hand side, you see all the cloud solutions, may it be an SAP system or a non-SAP system.

And on the right-hand side, you see systems on-premises. So here, SAP systems with SAP S/4HANA or an SAP Enterprise Suite, for instance, but it could also be non-SAP systems.

Categorization into three areas - first look

To easier understand the offered tools and services we have introduced three categories in the Extension Suite as seen here.

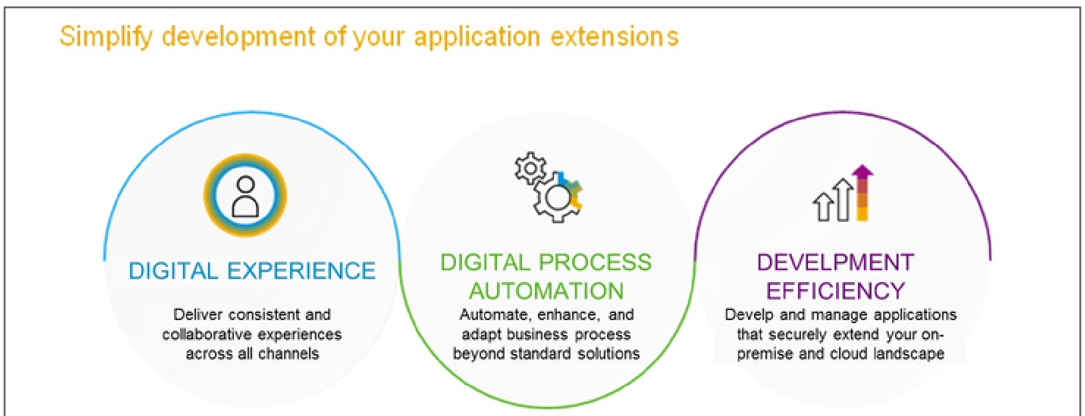


Figure 99: The Three Categories of the SAP Extension Suite

It has no technical impact to have this separation as all these services work hand in hand. You have development tools from "Development Efficiency" to build a modern, Digital Experience exposing digital, automated processes to end users.

Going deeper in the Categories

If we take a closer look at the Extension Suite, we see three main pillars.

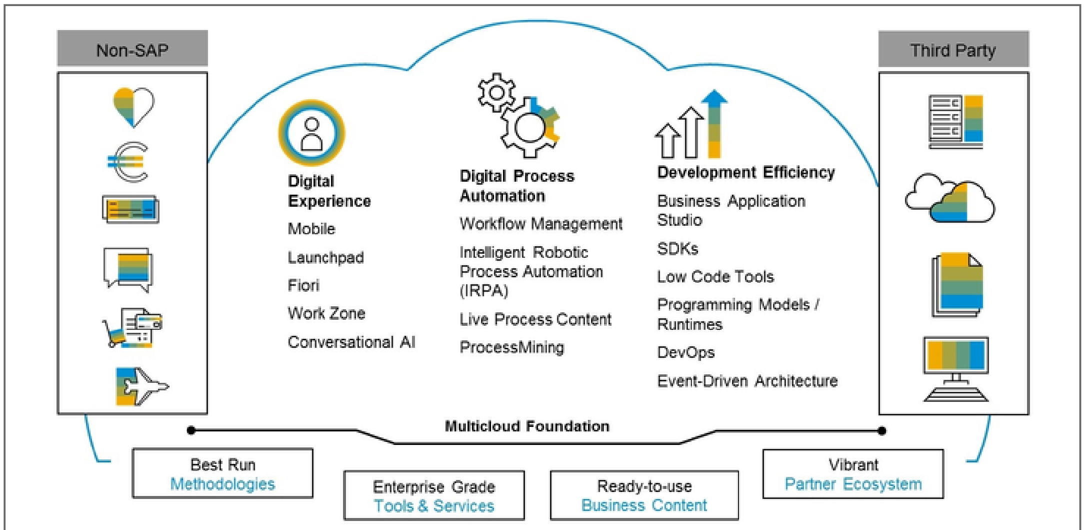


Figure 100: The 3 Main Pillars of the Extension Suite

On the left-hand side, you see the digital experience. This contains all the tools and services that helps you to provide an engaging user experience for your customers as well as your employees.

So, from a technology point of view, it's all the technologies, how you build Web applications, mobile applications, and conversational bots, for instance.

On the right-hand side then, you see the digital process automation area. These are the tools and services that help you to build processes, adapt processes, and also get an overview of the current state of all the processes, and gain insights into your processes.

The third pillar is the development efficiency. So this is the toolset and the area where you see the development tools and also the low-code tools, we later come to that in more detail, that help you to build the digital experience on the one hand and the digital process automation on the other.

Digital experience - tools and services are:

- Mobile
- Launchpad
- SAP Fiori technology
- Work zone
- Conversational AI

Digital process automation - tools and services are:

- Workflow Management
- Intelligent Robotic Process Automation (iRPA)
- Live Process Content
- Process Mining

Development efficiency - tools and services are:

- Business Application Studio

SDKs

Low Code Tools

Programming Models / Runtimes

DevOps

Event-Driven Architecture

What are extensions?

SAP Extension Suite Extensions: The Engineering View

Software engineering principle that allows for **enhancements** without impairing existing system functions.

Extensions can be **addition of new functionality** or through **modification of existing functionality**.

Extensibility is measure of **ability to extend** and **effort required** to implement extension.

Extension Types - 3 basic types:

Classic

Web GUI

Web Dynpro

BAPIs

Function modules

In-App

Custom SAP Fiori app deployed to SAP S/4HANA.

Custom Business Objects deployed to SAP S/4HANA.

Other SaaS apps.

Side-by-Side

Custom development outside of the application to be extended Cloud services

3rd party services

Cloud services from SAP

The SAP Extension Suite wants to cover the green areas.

What can be extended with the side-by-side Approach?

Basically, all SAP software products can be extended. The prerequisites are:

- The application to be expanded provides a usable API.
- The application to be expanded can send events.
- It doesn't matter which SaaS app is to be extended.

Examples are:

- SAP Concur
- SAP Fieldglass
- SAP S/4HANA essential
- SAP SuccessFactors

Where can I find the information if and how i can extend an SAP SaaS app via side-by-side?

In the SAP API business hub I find this information. We took a closer look at this in a previous chapter.

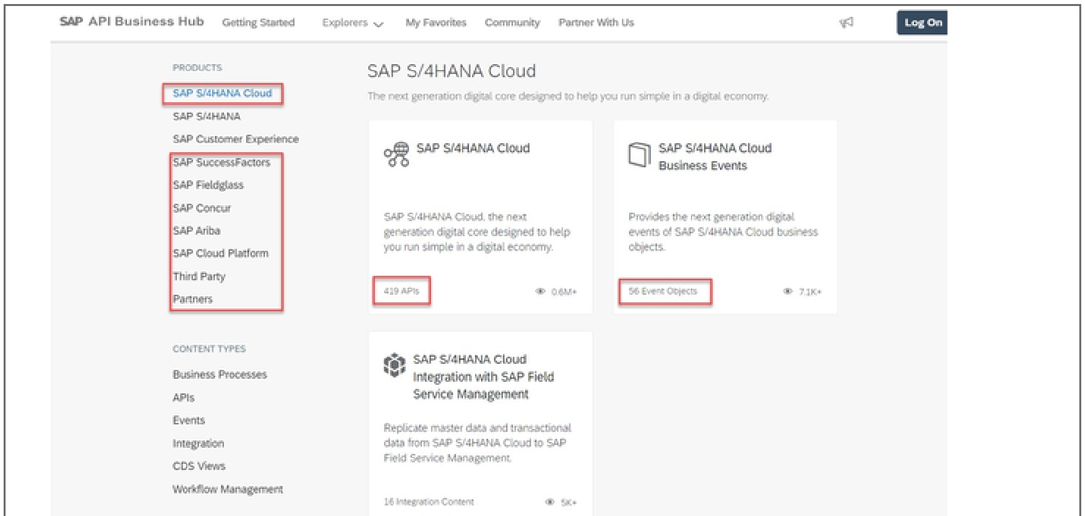


Figure 101: Screenshot from SAP API Business Hub

In the SAP API Business HUB you can see the available API es and/or events for the respective products, which can be used for a side-by-side extension. In addition to this white listed information, there are others that are only available in cooperation with SAP.

To the point

Focused statement: The three categories of the Extension Suite are:

Digital Experience

Digital Process Automation

Development Efficiency

Focused statement:

You can add additional technical functionality only side-by-side extensibility. To do this, use Extensibility Services at SAP BTP.

Learn more:

https://discovery-center.cloud.sap/serviceCatalog/?category=extensionsuite-digitalprocessautomation

https://www.sap.com/products/extension-suite.html

LESSON SUMMARY

You should now be able to:

- Explain, why extensibility is so important

Unit 4
Lesson 2

Building side-by-side Extensions

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explain the tools used for creating an extension

Tools for Extension Building

In this lesson we will cover the following topics:

- Extensions step-by-step
- SAP Extension Suite at a Glance
- Use cases

Extensions step-by-step

The general approach to the creation and operation of an extension is followed.

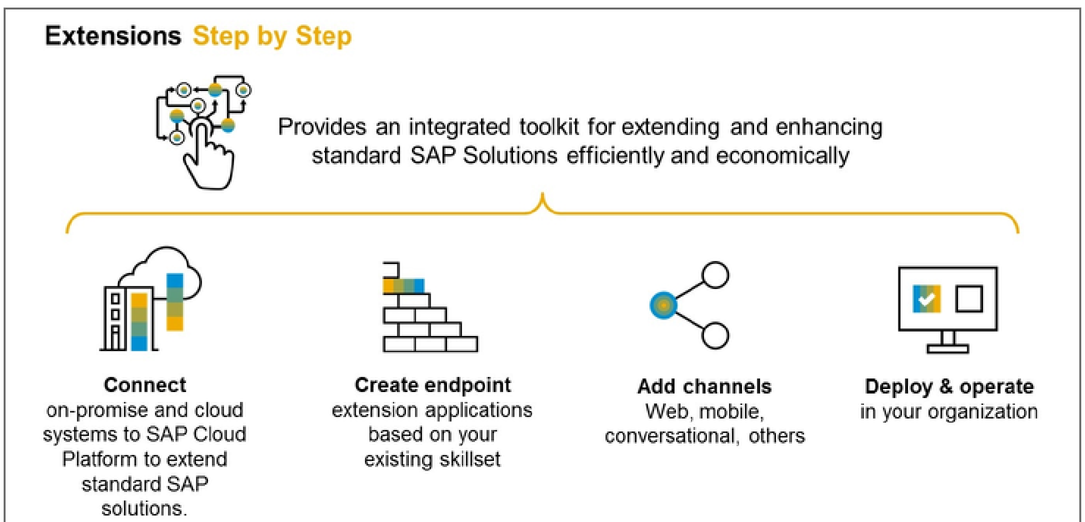


Figure 102: Extensions, Step-by-Step

Explanations about the figure:

1. Connect

   Connect on-premise and cloud systems to SAP BTP to extend standard SAP solutions.

2. Create endpoint

   Create endpoint extension applications based on your existing skill set.

3. Add channels

   Web, mobile, conversational, others.

SAP Extension Suite at a Glance

If you combine the general procedure with the tools and concepts of the Extension Suite, you will come to the following overview.

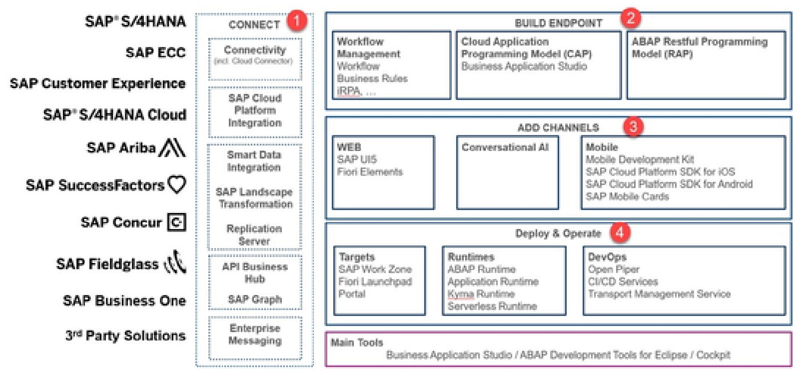


Figure 103: Extension Suite at a Glance

On the left side are the SaaS applications to be expanded. The following tools and concepts are available.

Explanations:

1: Connect

- Cloud Connector
- Cloud Integration (Integration Suite)
- API Business HUB
- SAP Graph
- Enterprise Messaging (Integration Suite)
- Smart Data Integration

2: Build Endpoint

- Workflow Management
- Cloud Programming Model (CAP)
- Micro services
- ABAP Restful Programming Model (RAP)

3: Add Channels

- Mobile Services

4: Deploy and Operate

- Portal
- SAP Fiori Launchpad

- Workzone
- Runtimes (ABAP, Cloud Founddry, Kyma, Serverless)
- DevOps (Ci/CD services, Piper, Transport)

In the following we will refer to this picture again and again so that you know where you are.

Use cases

Real-time Pricing Freight Platform -Transportation / Supply Chain.

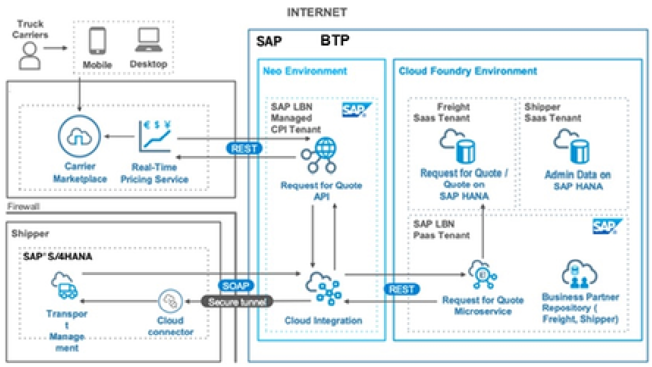


Figure 104: Customer Business Scenario

Use case:

Logistic customer brings SAP Logistic Business Network customers transparent pricing, a dense carrier network, and always-on availability from the nationwide, digitally enabled carriers using customers Freight's platform. In addition to unlocking this carrier capacity to SAP's LBN shippers, the customers Freight and SAP LBN integration also delivers continued innovation from customer' platform to SAP LBN customers via new product enhancements as well as the ability to further automate the supply chain within SAP's suite of products to better control transportation management. Utilizing real-time rates directly in workflows in SAP software enables customers to save time and money Guaranteed capacity directly into the software infrastructure of SAP customers

Components:

- SAP HANA
- SAP Integration Suite
- SAP S/4HANA, TM

Value Driver:

- Enables customers to save time and money on freight and operational costs
- Automate workflows to meet goals
- Make sourcing capacity more efficient
- Reduce underutilized assets
- Increase transparency and streamlined logistics communication
- Cut trucking costs

Lower freight costs through real-time visibility of market prices versus carrier contracted prices

Saves time by meeting dynamic logistics needs with real-time rates pricing & guaranteed capacity

Provides actionable insights and transparent market conditions which will help shippers estimate freight cost savings versus contracted rates and take actions.

Business Pattern:

Creating true business innovation wit new business models

Applying real time business processes

Multi-channel, custom UX

Time Recording - Professional Services Industry / Finance and Accounting: Customer Business Scenario
title

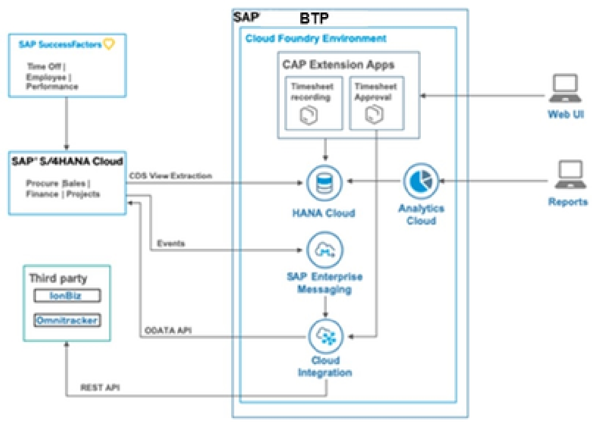


Figure 105: Customer Business Scenario 1/4

Use case:

Customer offers consulting services and wants to provide its employees or service consultants a light-weight application which can allow them to record their time worked on a project.

Customer runs lot of customer projects with different personas who would like to enter the time worked for a specific project.

This time is then approved by a 'Project Manager' persona who can then understand and plan resources working for a project.

Components:

SAP HANA

SAP Analytics Cloud

SAP Enterprise Messaging

SAP Integration Suite

Cloud Application Programming Model (CAP)

SAP SuccessFactors

SAP S/4HANA Cloud

IonBiz (3rd Party)

Omnitracker (3rd Party)

Value Driver:

Light weight UI to enter and approve Timesheet Application.

Extending the standard business process for managing projects.

Customer has 'HANA Cloud' as one source of truth from different systems.

Cloud application allows easy integration of customer's third-party systems.

Business Pattern:

Custom UI with Logic and Persistence.

Extend the Digital Core with new processes.

Data Integration Hub Pattern.

Replication of data from SAP S/4HANA Cloud to SAP BTP.

Equipment Management and Maintenance - Constructions and Operations / Asset Management

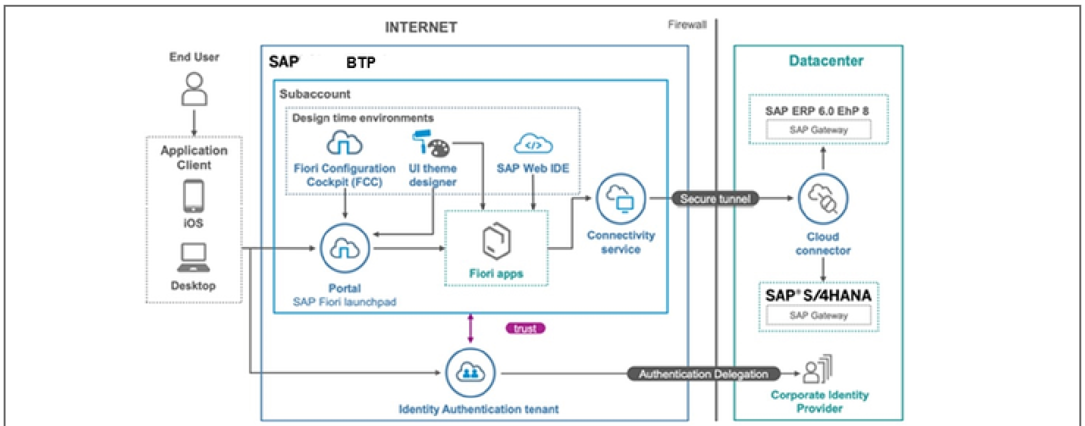


Figure 106: Customer Business Scenario 2/4

Use case:

Customer needs to digitize paper-based processes within equipment management & maintenance in order to meet the needs of steadily increasing volume of business.

Data had to be manually entered, using paper, into SAP ETM, PM, & BI. Data was consistently obsolete due to time-delayed posting. Big opportunity for improving User Experience. There were no existing apps from SAP Fiori, so the SAP ETM & SAP PM were adapted & subsequently developed in-house.

Customer needs to Digitize Paper-Based Processes within Equipment Management & Maintenance. Existing SAP Processes have been combined with SAP Fiori Apps. Using SAP BTP, the apps are readily available to technicians on iOS & Windows Mobile devices. SAP Fiori gives customer the opportunity to access our ERP directly through apps and to receive equipment data such as scheduling, profitability, analysis, repair orders, or equipment documents and perform in real time.

Customer requirements included to make data available in Real-Time & On-Site in order to maximize Equipment Management efficiency. Data has to available via Mobile Devices. Customer also wanted to merge related processes that are spread across ERP into Applications to remove duplicate entries and to validate process for Entry Accuracy.

Components:

SAP Cloud Portal service

SAP Identity and Authentication Services

SAP Integration Suite

SAP ERP 6.0 EhP 8

SAP S/4HANA

Value Driver:

Digitizing processes & providing employees with access to real-time Information - resulting in increased productivity & customer satisfaction.

Eliminated annual need of 200,000 paper pages.

Real-time bookings = 10-fold increase in efficiency.

Business Pattern:

Digitizing existing business processes.

SAP Fiori applications and mobile applications accessing back end data in real-time.

Custom UX with reading data from SAP back end.

Real-time Industry Management Oil & Gas / Operations

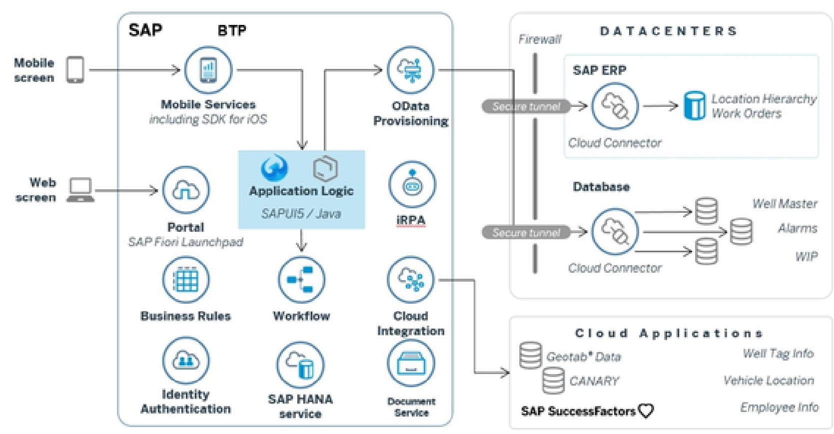


Figure 107: Customer Business Scenario 3/4

Use case:

Customer was looking to reduce the non-productive time during well operations. Non-productive time caused by mechanical failures, weather, and logistical problems add substantial overhead to operations. Reductions in these overheads can result in lower costs and improved health, safety, and environmental performance.

Components:

Mobile Services, OData Provisioning, Portal, iRPA, Workflow Management, Integration Suite, HANA DB, Document Service

SAP ERP

SAP SuccessFactors

Legacy DB, Geotab Data (3rd party)

Value Driver:

Faster decision making ability for increased efficiency.

Reducing operational cost by providing remote assistance, 12% reduction in cycle times.

Leveraging AI to identify data anomalies for faster reaction.

Improved employee health and safety by 10%.

Business Pattern:

Applying real time business processes.

Applying innovative AI-based analytical capabilities.

Providing multi-channel UX.

Bringing together data from various systems.

Innovating Customer Service Automotive / Customer Experience

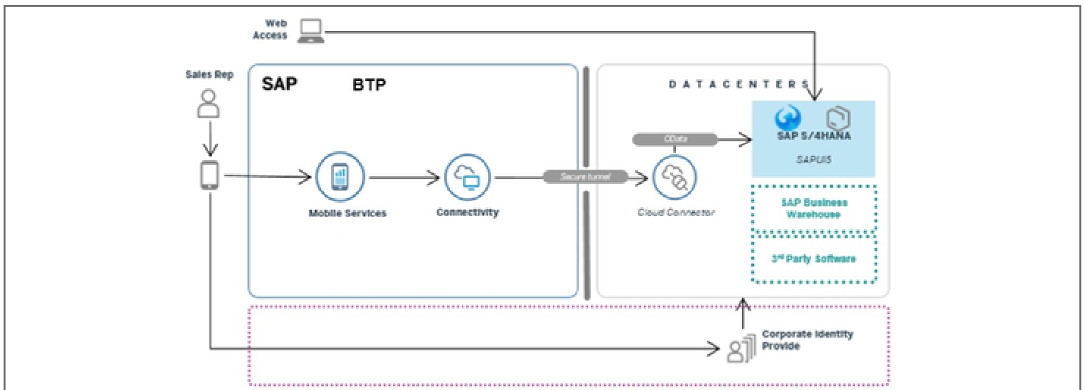


Figure 108: Customer Business Scenario 4/4

Use case:

Sales agents that are handling car delivery at the dealership did not have a consolidated insight into customer orders and production status as it was a paper intensive process to coordinate across multiple departments. To ensure they can provide a premium customer experience in new car delivery and react to changes on the fly, the company turned to experts from SAP and Apple to create a "car delivery" application.

Components:

Cloud Connector

Mobile Services

SAP S/4HANA

SAP Business Warehouse

Various 3rd party tools

Value Driver:

Deliver consistent premium experience.

Standardize processes across dealers.

Reduce underutilized assets.

Increase sales rep insight into order status for better change management.

Digitize custom car delivery process.

Business Pattern:

Digitizing existing business processes.

Applying real time business processes.

Multi-channel, custom UX.

LESSON SUMMARY
You should now be able to:

Explain the tools used for creating an extension

Unit 4
Lesson 3

Building Connections

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Discover the place and the features of the Cloud Connetor in connectivity

Cloud Connector Features

In this lesson we will cover the following topics:

- Classification in the SAP Extension Suite.
- Connectivity Service at SAP BTP.
- Cloud Connector - What can I do with it?
- Connecting cloud applications to on-premise systems.
- Connecting on-premise database tools to SAP HANA databases.
- Features

Classification in the SAP Extension Suite

In this lesson we will look at the tools highlighted in the figure below:

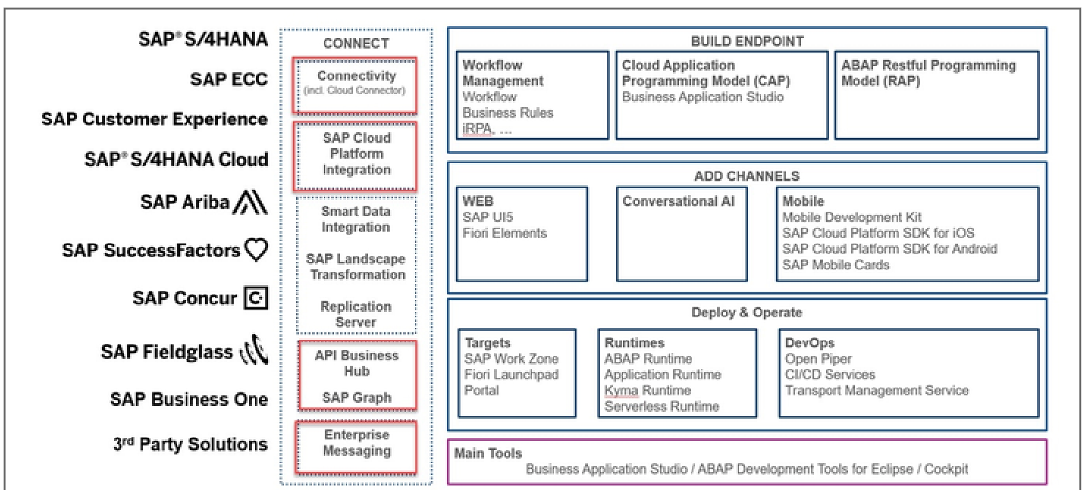


Figure 109: Overview of the Tools Under Consideration

In this lesson we will look at the following tools:

1: Cloud Connector

2: SAP Integration Suite

3: API Business HUB

4: SAP Graph

5: Enterprise Messaging

We have already discussed the API Business HUB, SAP Graph, SAP Integration Suite and SAP Enterprise Messaging as part of the Integration Suite. Therefore, we will only look at the SAP Cloud Connector below.

Connectivity Service at SAP BTP

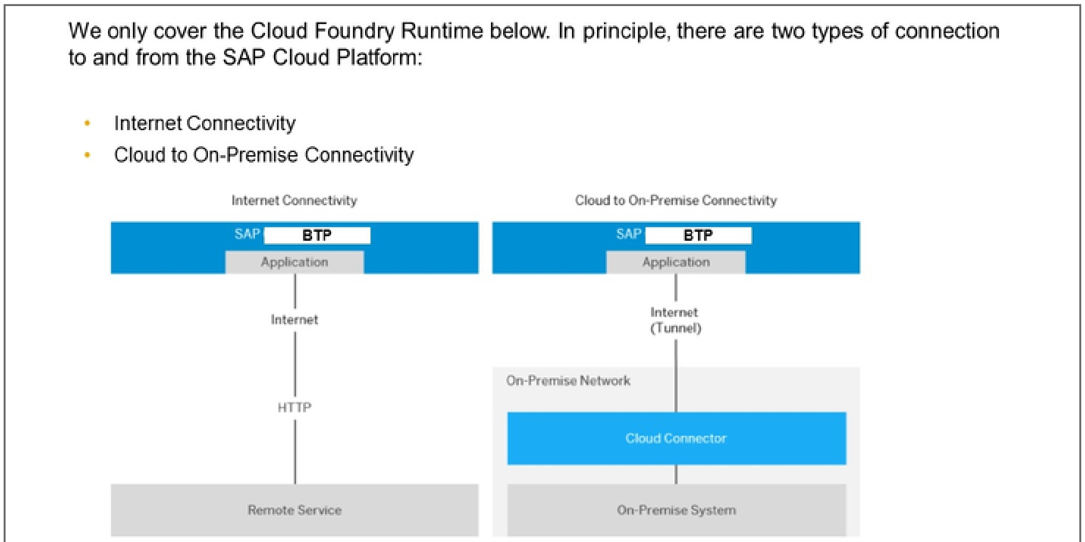


Figure 110: Cloud Connector Features

In the upper part you can see the SAP BTP Platform. In the lower part the two connected landscapes. One is a direct connection that can be reached via HTTPs, and an on-premise system connected via the Cloud Connector. Communication is bi-directional

The Connectivity service allows SAP BTP applications to securely access remote services that run on the Internet or on-premise. This service:

This service:

Allows subaccount-specific configuration of application connections via destinations.

Provides a Java API that application developers can use to consume remote services.

Allows you to make connections to on-premise systems, using the Cloud Connector.

Lets you establish a secure tunnel from your on-premise network to applications on SAP BTP, while you keep full control and auditability of what is exposed to the cloud.

Supports both the Neo and the Cloud Foundry environment for application development on SAP BTP.

The Connectivity service supports the following protocols and scenarios:

Cloud to on-premise

Http: HTTPS is not needed, since the tunnel used by the Cloud Connector is TLS-encrypted.

RFC: You can communicate with SAP systems down to SAP R/3 release 4.6C. Supported runtime environment is SAP Java Buildpack with a minimal version of 1.8.0.

TCP: You can use TCP-based communication for any client that supports SOCKS5 proxies.

Internet

HTTPS: Exchange data between your cloud application and Internet.

Cloud Connector - What can I do with it?

Serves as a link between SAP BTP applications and on-premise systems.

Runs as on-premise agent in a secured network.

Provides fine-grained control over the connectivity.

Lets you use the features that are required for business-critical enterprise scenarios.

Get more details:

Serves as a link between SAP BTP applications and on-premise systems:

Combines an easy setup with a clear configuration of the systems that are exposed to the SAP BTP.

Lets you use existing on-premise assets without exposing the entire internal landscape.

Runs as on-premise agent in a secured network:

Acts as a reverse invoke proxy between the on-premise network and SAP BTP.

Provides fine-grained control over:

On-premise systems and resources that can be accessed by cloud applications.

Cloud applications using the Cloud Connector.

Lets you use the features that are required for business-critical enterprise scenarios:

Recovers broken connections automatically.

Provides audit logging of inbound traffic and configuration changes.

Can be run in a high-availability setup.

Cloud Connector can be used for:

Table 3: Usage of Cloud Connector

Enviroments	Examples
Customer on-premise backend systems	SAP ERP, SAP S/4HANA
SAP Hosting	SAP HANA Enterprise Cloud (HEC)
Third-party IaaS providers	AWS, Azure, GCP

Cloud Connector can not be used for:

Table 4: No Usage of Cloud Connector for

Enviroments	Examples
SAP SaaS solutions	SAP SuccessFactors, SAP Concur, SAP Ariba and more.
SAP cloud-based enterprise solutions	SAP S/4HANA Cloud, SAP C/4HANA
Third-party PaaS providers	AWS, Azure, GCP
Third-party SaaS providers	

Connecting cloud applications to on-premise systems

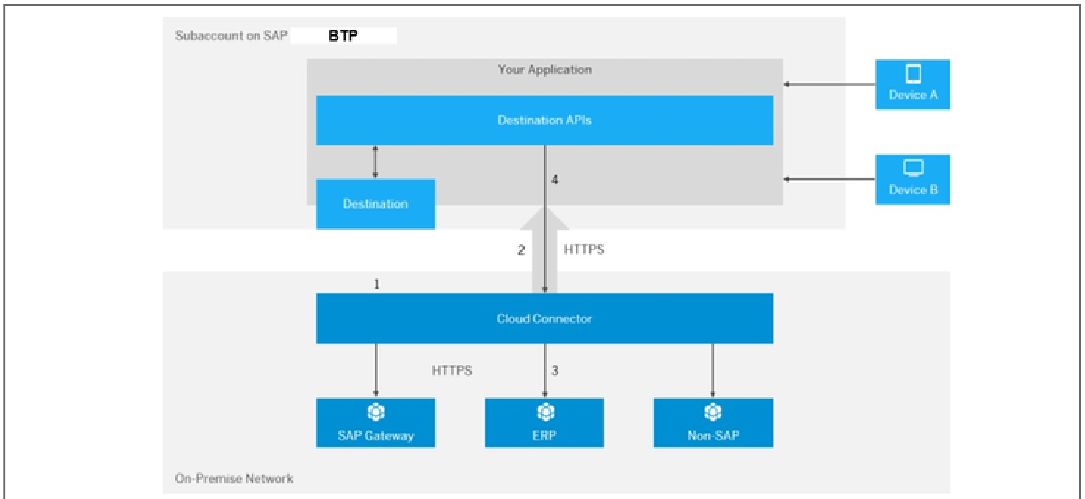


Figure 111: Schematic Representation of the Connection to On-Premise Systems

Data access is based on a subaccount. Here is the example of the Destination API's. The tunnel is TSL encrypted. The Cloud Connector then distributes the calls to various connected back end connections. The idea is to provide data from the on-premise network on the SAP BTP.

Connecting on-premise database tools to SAP HANA databases

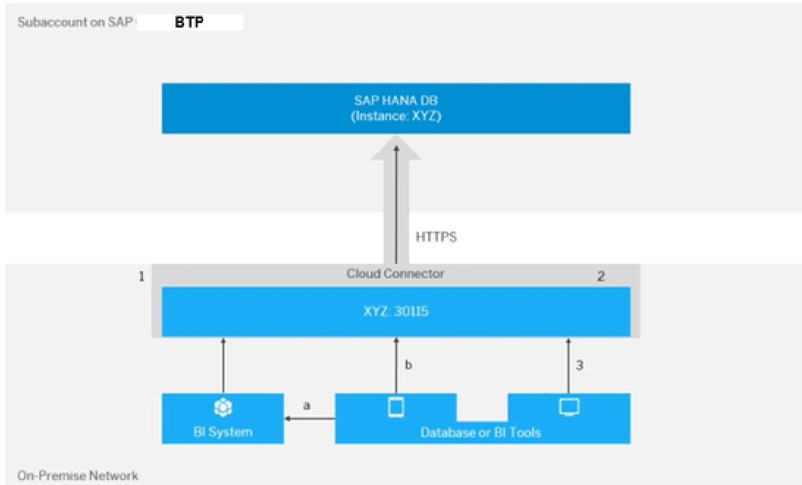


Figure 112: Schematic Representation of the Connection to SAP HANA Databases

In this case, the data access starts from the on-premise database. On the subaccount you need an SAP HANA service instance. The idea is to make data available from the cloud in the on-premise network.

In addition to the pure connection, other features are available:

- High Availability Setup
- Secure the Activation of Traffic Traces
- Monitoring
- Alerting
- Audit Logging

Get more in detail:

High Availability Setup

The Cloud Connector lets you install a redundant (shadow) instance, which monitors the main instance.

Secure the Activation of Traffic Traces

Tracing of network traffic data may contain business critical information or security sensitive data. You can implement a "four-eyes" (double check) principle to protect your traces.

Monitoring

Use various views to monitor the activities and state of the Cloud Connector.

Alerting

Configure the Cloud Connector to send email alerts whenever critical situations occur that may prevent it from operating.

Audit Logging

Use the auditor tool to view and manage audit log information.

To the point

Focused statement: The Connectivity service supports the following protocols and scenarios:

Cloud to on-premise:

- Http: HTTPS is not needed, since the tunnel used by the Cloud Connector is TLS-encrypted.
- RFC: You can communicate with SAP systems down to SAP R/3 release 4.6C. Supported runtime environment is SAP Java Buildpack with a minimal version of 1.8.0.
- TCP: You can use TCP-based communication for any client that supports SOCKS5 proxies.

Internet:

- HTTPS: Exchange data between your cloud application and Internet.

Cloud Connector - What can I do with it?

- Serves as a link between SAP BTP applications and on-premise systems.
- Runs as on-premise agent in a secured network. Use a secure tunnel between the on-premise system and SAP BTP
- Provides fine-grained control over the connectivity.

Learn more

https://help.sap.com/viewer/cca91383641e40ffbe03bdc78f00f681/Cloud/en-US/e54cc8fbbb571014beb5caaf6aa31280.html

https://tools.hana.ondemand.com/#cloud

LESSON SUMMARY
You should now be able to:

Discover the place and the features of the Cloud Connetor in connectivity

Unit 4
Lesson 4

Building Endpoints

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Explore the area of endpoint building

Endpoint Building

In this lesson we will cover the following topics:

- Classification in the SAP Extension Suite.
- What Is Workflow Management?
- Cloud Application Model (CAP)
- ABAP Restful Programming Model (RAP)
- SAP SDK.

Classification in the Extension Suite

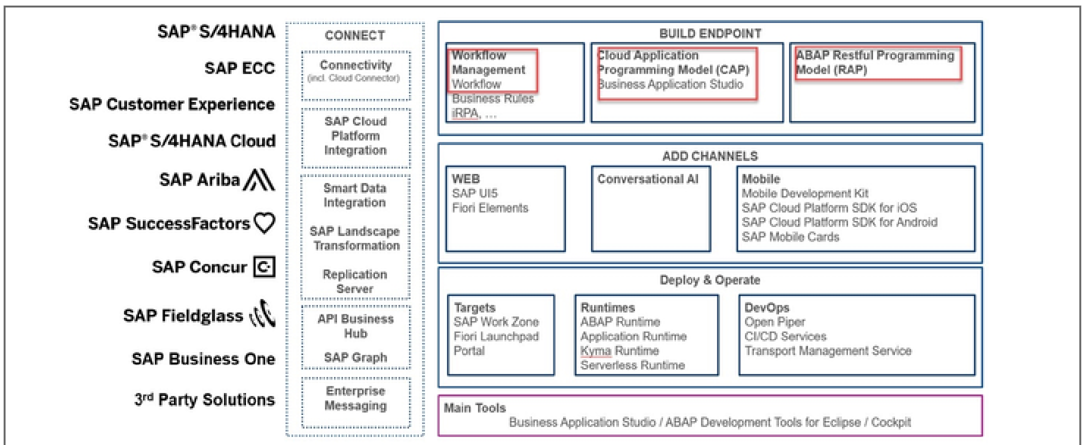


Figure 113: Overview of the Tools Under Consideration

In this lesson we will look at the following tools:

1: Workflow Management

2: Cloud Application Model (CAP)

3: ABAP Restful Programming Model (RAP)

What Is Workflow Management?

Digitize workflows, manage decisions, and gain end-to-end process visibility:

SAP Workflow Management allows you to digitize workflows, manage decisions, gain end-to-end process visibility, and configure processes in a low-code approach.

It allows users to build, run, and manage workflows.

It allows users to digitize and automate decision making.

It enables process excellence, process transparency, process transformation by providing one view of the process.

Users can configure the process flow without the involvement of IT to improve process efficiency.

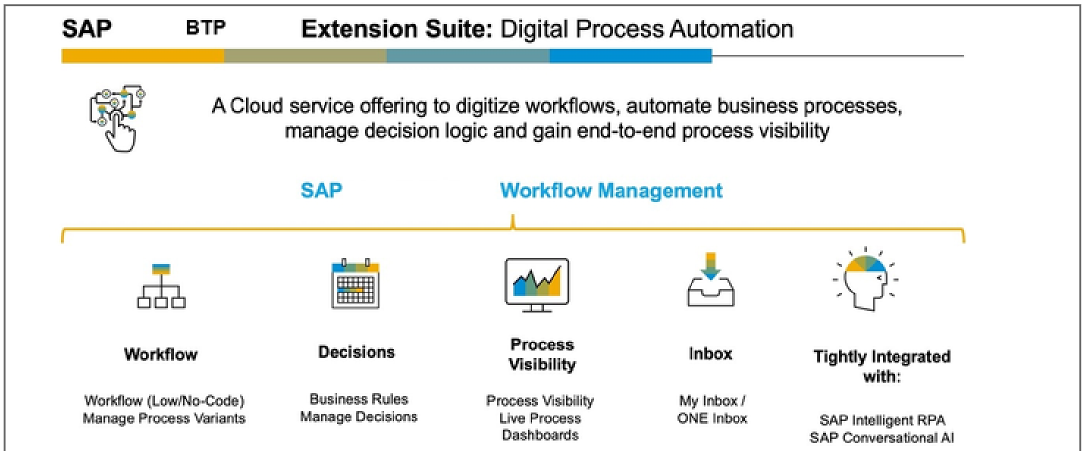


Figure 114: Digital Process Automation with SAP Workflow Management

SAP provides a comprehensive workflow platform that is heavily configuration (no-code)-based but also with textual code support.

Drag and drop is supported for process, integration and UX, including agent conversations and integration with RPA bots. With the release of SAP Workflow Management the following is combined:

Workflow

Business rules

Process visibility in one service.

With one user metric and service plan. We are also adding new features for business process experts to achieve process flexibility and also "live process content packages" as templates to accelerate your digital workflow automation projects.

There are several use cases where digital process automation with SAP Workflow Management will provide immediate value. You might want to group them like this:

Digital workflows and bots.

Process-driven application development.

Process excellence and optimization.

Facilitate process transformation journeys.

Immediate outcomes are:

Do more with less - empower employees through process automation, freeing up their time to do more meaningful work.

Ensure compliance - comply with policies and regulations via clear responsibilities and audit trails.

Flexibly adapt to new requirements - tailor business processes to the business needs and support new business model innovations.

Business process experts are now able to manage changing business needs directly on live, running processes. Customers can use dedicated business expert tooling to discover, configure, and run workflows, like capital expenditure approvals, business partner creation or procurement data collection. To speed up and simplify these changes without involvement of the IT department, customers will be able to leverage pre-defined live process content packages via SAP API Business Hub to manage workflows on top of SAP ERP Central Component (ECC), SAP Utilities and SAP S/4HANA (Procurement). This enhanced real-time process flexibility increases the level of automation, empowers business users and ensures compliance with business policies and regulations via clear responsibilities and audit trails.

Workflow management in your landscape

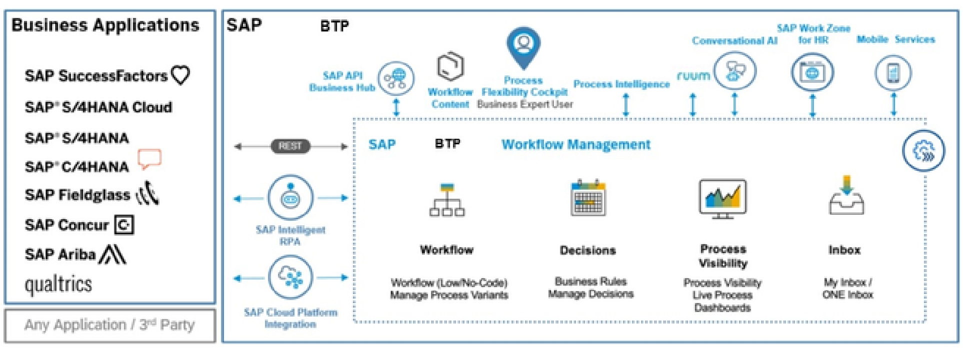


Figure 115: Workflow Management in Your Landscape

You can see on the left the possible data/event provider . These are connected via REST interfaces.

The saturation options are indicated in the upper part for example Mobile Services, SAP Work Zone and much more.

What is Cloud Application Model (CAP)?

The SAP Cloud Application Programming Model is a **framework of languages, libraries, and tools** for building enterprise-grade services and applications.

It guides developers along a ' **golden path'** of proven best practices and a great wealthof out-of-the-box solutions to recurring tasks.

The CAP framework features a mix of proven and broadly adopted open-source and SAP technologies, as highlighted in the figure below.

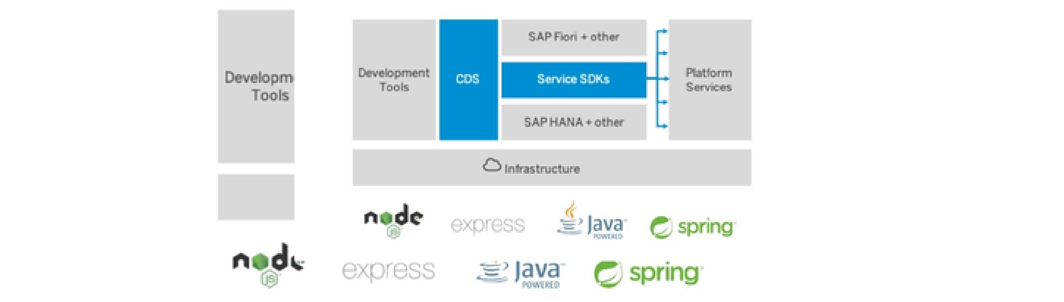


Figure 116: Schematic Overview

On top of open source technologies, CAP mainly adds:

Core Data Services (CDS) as our universal modeling language for both domain models and service definitions.

Service SDKs and runtimes for Node.js and Java, offering libraries to implement and consume services as well as generic provider implementations serving many requests automatically.

Focus on Domain, Powered by CDS

CAP places primary focus on domain, by capturing domain knowledge and intent instead of imperative coding - that means, **What, not How** - thereby promoting:

Close collaboration of developers and domain experts in domain modeling.

Out-of-the-box implementations for best practices and recurring tasks.

Platform-agnostic approach to avoid lock-ins, hence protecting investments.

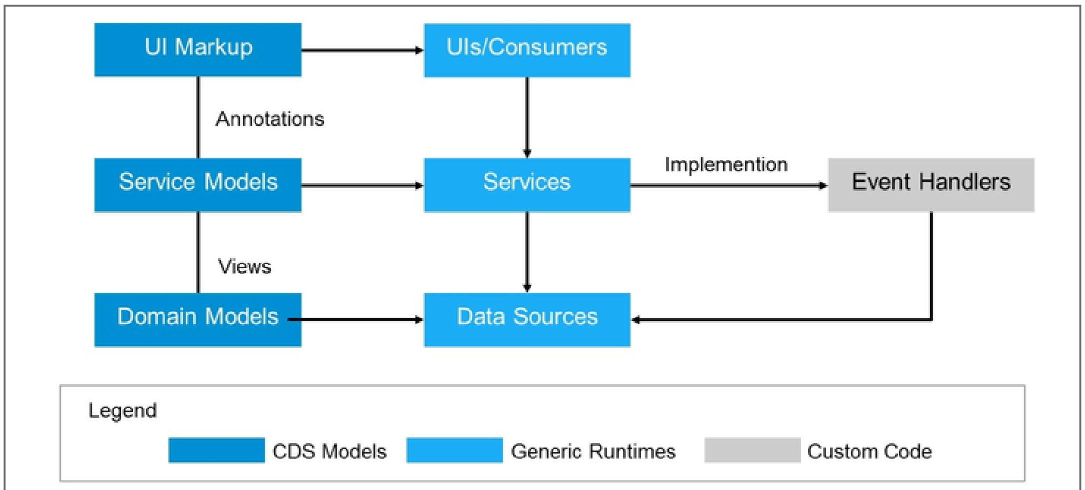


Figure 117: Connection of the Domain Models with the Service and the UI Model

The figure illustrates the prevalent use of CDS models (in the left column), which fuel generic runtimes, like the CAP service runtimes or databases.

Core Data Services (CDS)

CDS serves as our universal modeling language to capture static, as well as behavioral aspects of problem domains in conceptual, concise, and comprehensible ways, and hence is the very backbone of CAP.

```
entity Books : cuid {
  title : localized String;
  author : Association to Authors;
}

entity Orders : cuid, managed {
  descr : String;
  Items : Composition of many {
    book : Association to Books;
    quantity : Integer;
  }
}
```

Figure 118: Definition of the Domain Model with CDS

Two tables, books and orders with the corresponding attributes, are created. A link from orders to books is further created.

```
// Service Definition in CDS
service OrdersService {
  entity Orders as projection on my.Orders;
  action cancelOrder (ID:Orders.ID);
  event orderCanceled : { ID:Orders.ID }
}
```

Figure 119: Service Definitions in CDS

Complete UI's can also be defined via CDS.

In the service layer, any further implementations can be created, for example in Java with the cloud sdk and/or spring boot.

Golden Path

Following the golden path of the programming model helps you implement data models, services and UIs in order to develop stand-alone business applications or extend other cloud solutions, like SAP S/4HANA or SAP SuccessFactors.

Details about the "golden Path":

Table 5: The Golden Path

Your Task	Using	and let the framework take care of....
Define your Data Model	CDS	automatic deployment to **SAP HANA**

Your Task	Using	and let the framework take care of....
Define your Services	CDS	Translating to **OData** service definitions. Serving requests out of the box. Authorisation*) and standard validations. Filling in primary keys and audit information. Writing trace and audit logs. Health checks. etc.
Add Custom Logic	Java EE, Spring Boot, Node.js	Handling database connections Including tenant isolation Parsing input Serialising responses
Add SAP Fiori UIs	CDS Annotations	translation to **OData** vocabularies generating a **SAP Fiori elements** app skeleton
Reuse existing Serv- ices		

What is ABAP Restful Programming Model ?

The ABAP RESTful programming model defines the architecture:

- for efficient end-to-end development of intrinsically SAP HANA-optimized OData services (such as SAP Fiori apps) in SAP BTP, ABAP Environment or Application Server ABAP.

It supports the development of all types of SAP Fiori applications as well as publishing Web APIs.

It is based on technologies and frameworks such as Core Data Services (CDS) for defining semantically rich data models and a service model infrastructure for creating OData services with bindings to an OData protocol and ABAP-based application services for custom logic and SAPUI5-based user interfaces - as shown in the figure below.

Architecture Overview

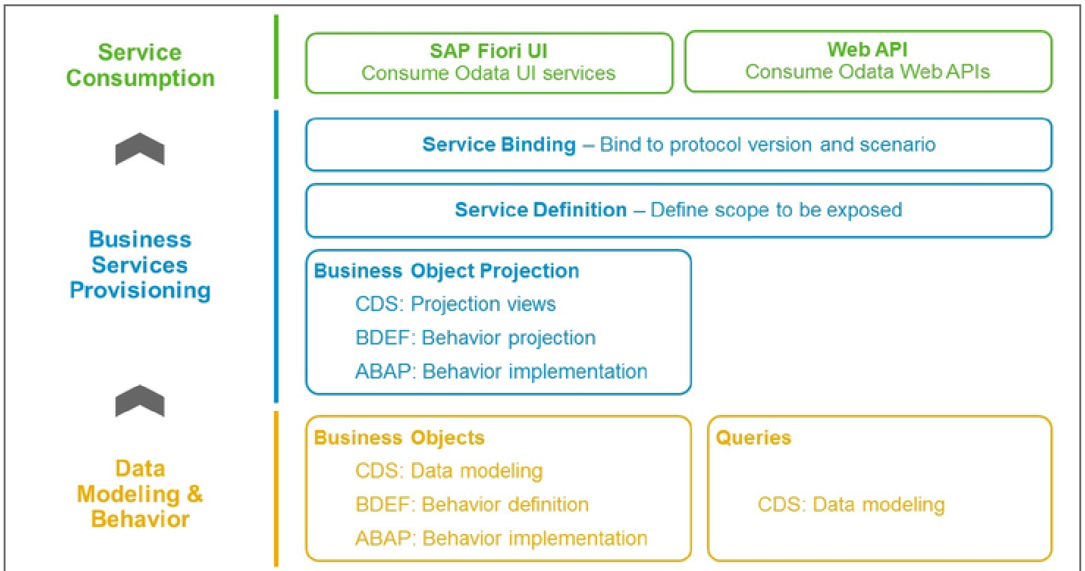


Figure 120: Architecture Overview

The architecture can be divided into:

- Data Modelling & Behavior
- Business service Provisioning
- Service Consumption

Data Modelling & Behavior

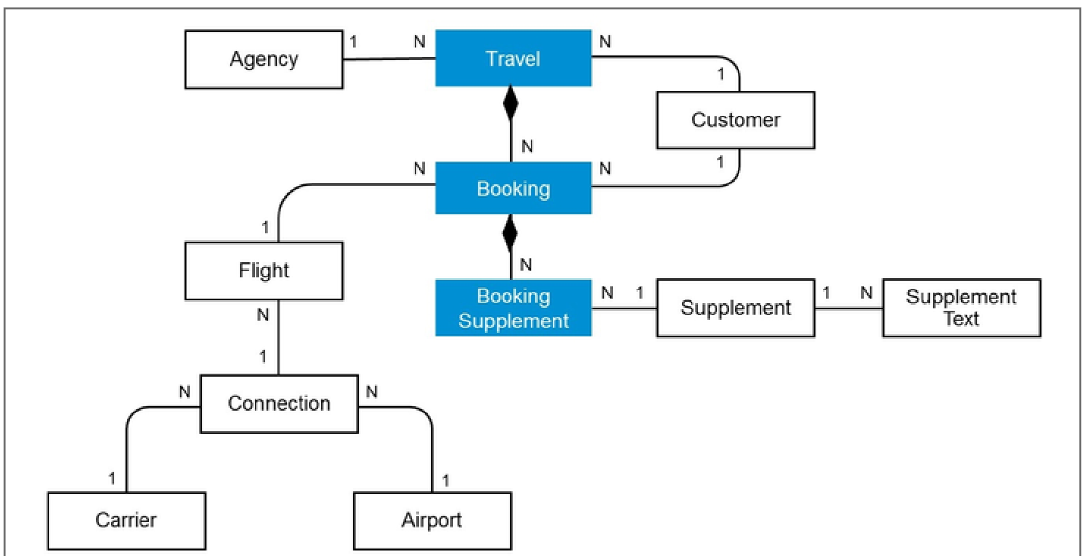


Figure 121: Sample of an Data Model

The layer of data modeling and behavior deals with data and the corresponding business logic.

The data model comprises the description of the different entities involved in a business scenario. We use CDS data modelling therefore.

The behavior describes what can be done with the data model, for example if the data can be updated. (ABAP or BDEF)

Business Service Provisioning

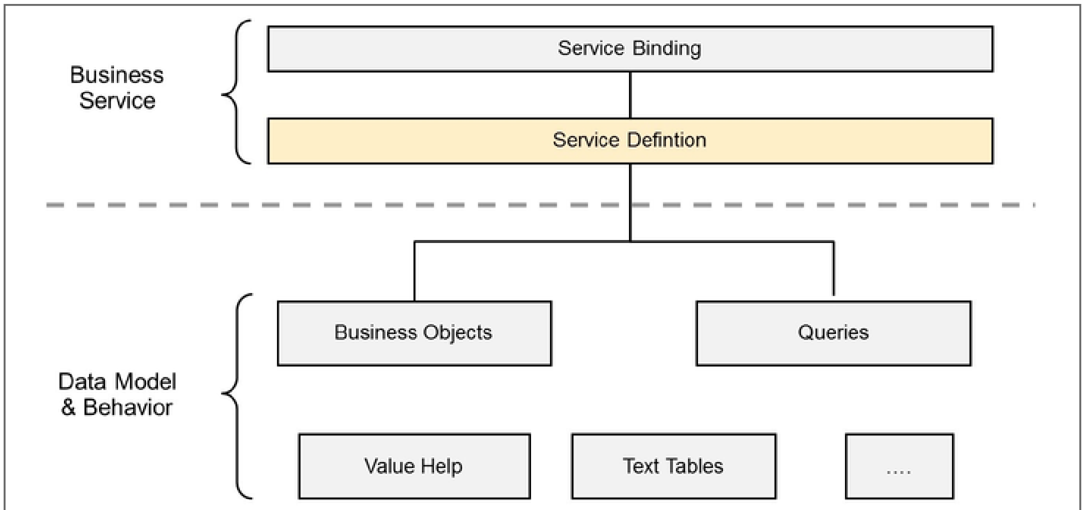


Figure 122: Relationship Between Data Model and Behavior and Exposed Business Service

In the context of the ABAP RESTful programming model, a business service is a RESTful service which can be called by a consumer. It is defined by exposing its data model together with the associated behavior. It consists of a service definition and a service binding.

Service Consumption

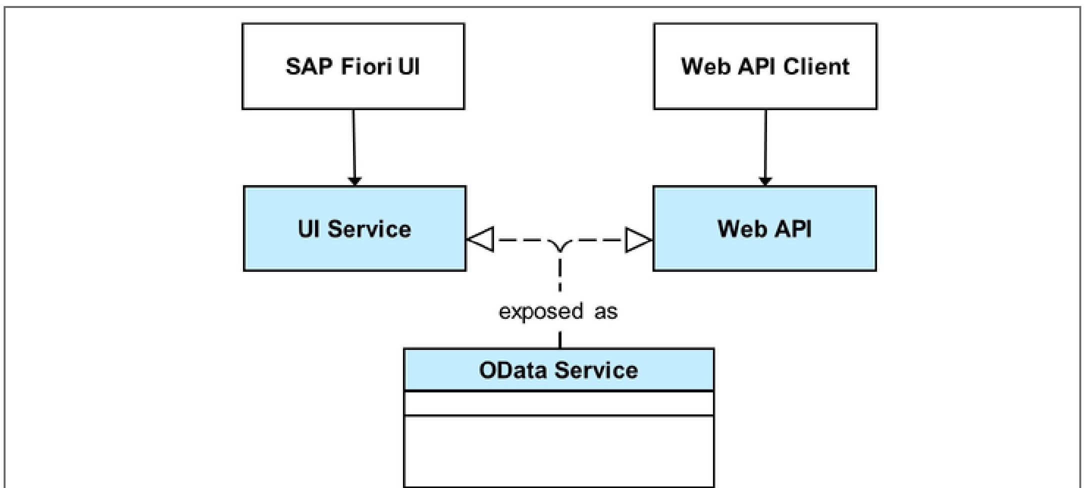


Figure 123: Overview Service Consumption

An OData service can be exposed as a UI service, that can be consumed by an SAP Fiori UI, or as a Web API that can be consumed by any OData client.

SAP SDK

The Cloud SDK is a set of tools and libraries for consuming, building or extending SAP services and applications in the cloud-native way and deploying them to SAP BTP.

SAP Cloud SDK was conceived as a collection of best practices for developing cloud-native applications withing SAP. Eventually it grew into a fully-fledged and feature rich development library consisting of three main components.

We will have a quick look to the parts of SAP Cloud SDK.

The SAP SDK consists of the following layers:

- The implementations of the SAP Cloud SDK.
- Pre-Generated Type-Safe API Client Libraries.
- Business Partner OData API within API Business Hub
- The code generator for OData and REST.
- Cloud abstractions for multiple runtimes
- Cloud native development best practices
- Valuable extras
- Documentation and support

We will have a quick look to the parts of SAP Cloud SDK.

The SAP Cloud SDK consists of three different implementations:

- Java
- Continuous Delivery Toolkit
- JavaScript Typescript

Use Java or java script for your cloud native development. Java has the largest range of functions:

- Caching
- Resilience
- Client for BAPI/RFC

Both implementations can also be used in the Cloud Application Programming Model (CAP).

Continuous Delivery belongs to the Deploy & Create section and is discussed there.

Pre-Generated Type-Safe API Client Libraries

For SAP services, applications, and platforms to simplify integration and extensibility:

- A set of REST client libraries
- SAP S/4HANA Cloud OData v2/v4
- SAP S/4HANAOn-premise OData v2/v4

For all API' for SAP S/4HANA listed in the SAP API Hub and others, there are predefined Java classes in the Cloud SDK. Access is via Typed Clients. This can be done with OData V2 , V4 and REST.

Business Partner OData API within API Business Hub

```
service.createBusinessPartner(partner);
service.getBusinessPartnerByKey("id");
service.getAllBusinessPartner();
service.updateBusinessPartner(partner);
service.deleteBusinessPartnerAddress(address);
```

```
service.getBusinessPartnerByKey("id")
    .select(BusinessPartner.TO_BUSINESS_PARTNER_ADDRESS
        .select(BusinessPartnerAddress.CITY_CODE, BusinessPartnerAddress.COUNTRY)
        .filter(BusinessPartnerAddress.CITY_CODE.notEqualTo("1234"))
        .orderBy(BusinessPartnerAddress.COUNTRY.desc())
    )
    .execute(destination);
```

Figure 124: Business Partner OData API Within API Business Hub

You have -for example- Basic CRUD Functions, Filter and much more to consume a predefined OData API.

About the code generator for OData and REST:

To cover your custom needs, you can use the SAP SDK code generator.

It offers the following functions: test an own service, ship a library, get a client for a configured system, consumes a service, which is not yet shipped prepackaged with Cloud SDK.

It works with: OData v2, OData v4 and OPEN API 3.X.

While a number of pre-generated API's are already provided in the cloud sdk, of course, the possibility to create Java classes based on Custom OData or REST interfaces with the Code Generator also includes the possibility.

The Cloud SDK supports multiple runtimes:

SAP Cloud Foundry

Kyma (Kubernetes)

Cloud native development best practices

The Cloud native development approach consists of:

Multitenancy

Authentication Flows

Caching (only Java)

Resilience (only Java)

Connectivity

Proxies

Etag & Headers

Automatic CSRF

These libraries are integrated to enable the implementation of frequently used development patterns in the cloud.

Additional functionality:

Client for BAPI/RFC (only Java)

Integration

Extensions

Works with CAP

Maximum flexibility with advanced un-typed API layer

Runs in a browser (only JS)

Supports Business Application Studio

About documentation and support:

There is a lot of documentation and support available for SAP Cloud SDK.

To the point

Focused statement: The ABAP RESTful programming model defines the architecture:

for efficient end-to-end development of intrinsically SAP HANA-optimized OData services (such as SAP Fiori apps) in SAP BTP, ABAP Environment or Application Server ABAP. An architecture for building OData services.

It supports the development of all types of SAP Fiori applications as well as publishing Web APIs.

It is based on technologies and frameworks such as Core Data Services (CDS) for defining semantically rich data models and a service model infrastructure for creating OData services with bindings to an OData protocol and ABAP-based application services for custom logic and SAPUI5-based user interfaces.

Focused statement: What is Cloud Application Model (CAP)?

The SAP Cloud Application Programming Model is a framework of languages, libraries, and tools for building enterprise-grade services and applications.

It guides developers along a 'golden path' of proven best practices and a great wealth of out-of-the-box solutions to recurring tasks.

Learn more

https://cap.cloud.sap/docs/

https://help.sap.com/viewer/923180ddb98240829d935862025004d6/Cloud/en-US/289477a81eec4d4e84c0302fb6835035.html

LESSON SUMMARY

You should now be able to:

- Explore the area of endpoint building

Unit 4
Lesson 5

Adding Channels

LESSON OBJECTIVES

After completing this lesson, you will be able to:

Explain which channels are available within the SAP Extension Suite

Channels

In this lesson we will cover the following topics:

Classification in the SAP Extension Suite.

What is SAP Conversational AI?

Bot-Building Platform.

What is SAP Ui5?

SAP Mobile Services.

Classification in the SAP Extension Suite

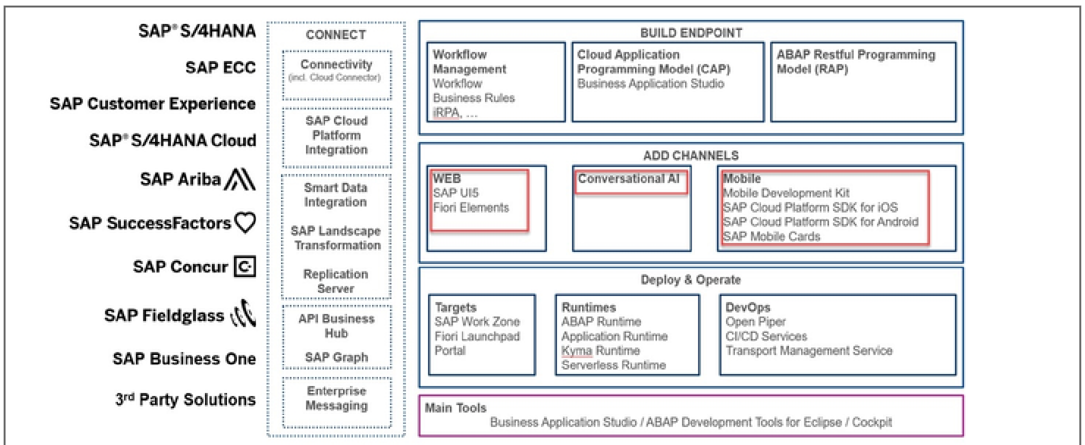


Figure 125: Overview of the Tools Under Consideration

In this lesson we will look at the following tools:

1: SAP UI5

2: Conversational AI

3: Mobile

The figure illustrates, to which area these tools belng.

What is SAP Conversational AI?

SAP Conversational AI helps users **efficiently manage business tasks**

As the conversational AI layer of SAP's Business Technology Platform, it enables users to **build** and **monitor** intelligent chatbots in one interface to automate tasks and workflows.

You can expect:

Bots in any language

Leverage the power of our highly performing NLP technology capable of building human-like AI chatbots in any language.

Low-code bot building.

Customize and expand ready-to-use chatbots.

Multilingual

Easy to use

Simple UX, collaboration, organizational accounts, versioning and environments: we've made a bot platform all teams can smoothly use.

Fastest time to market

Through an efficient bot building process and a full integration to the SAP software suite, you can launch your bot in days.

Bot-Building Platform

Your way to build bots.

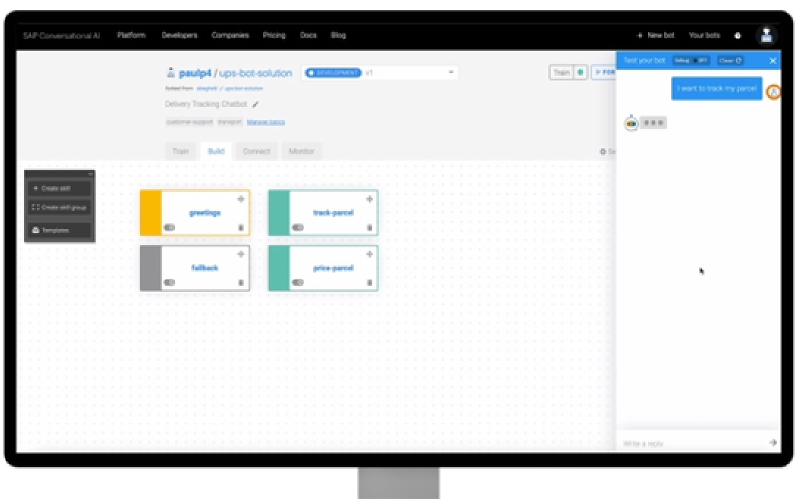


Figure 126: Bot Building Platform

Bot training

Analyze text inputs and enrich key data in any language with our world-class natural language processing (NLP) technology.

Bot building

Build adaptable conversational flows and skills through our powerful bot builder.

Bot connector

Connect chatbots to any SAP solution, external communication channel, or back-end system.

Bot analytics

Understand how customers and employees talk to your chatbot and improve the user experience based on usage and training data.

What is SAPUI5?

UI5 apps run on smartphones, tablets, and desktops. The UI controls automaticallyadapt themselves to the capabilities of each device and make the most of the availablereal estate.

It comes with built-in support for architectural concepts like MVC, two-way data binding, and routing.

It Includes standards like MVC and various data-binding types.

Further facts:

- Choose between different view formats (XML, HTML, JavaScript, JSON).
- Binding with OData, JSON, XML and other data formats.
- Built-in support tool for exploring the object tree and binding status.

Created by professionals, for modern developers building state of the art web applications.

It comes with all features needed to cover most current application requirements, with standards high enough to be delivered in standard SAP solutions:

- Translation and internationalization support.
- Extensibility concepts at code and application level.
- High Contrast theme to aid visually impaired users.

What is SAP Fiori?

SAP Fiori is the design language that brings great user experiences to enterprise applications. Based on user roles and business processes, SAP Fiori simplifies doing business.

SAP Fiori is a paradigm shift away from monolithic ERP solutions towards light-weight apps tailored to the users' tasks.

It is:

- Role-Based
- Adaptive
- Coherent
- Simple
- Delightful

SAP Mobile Services

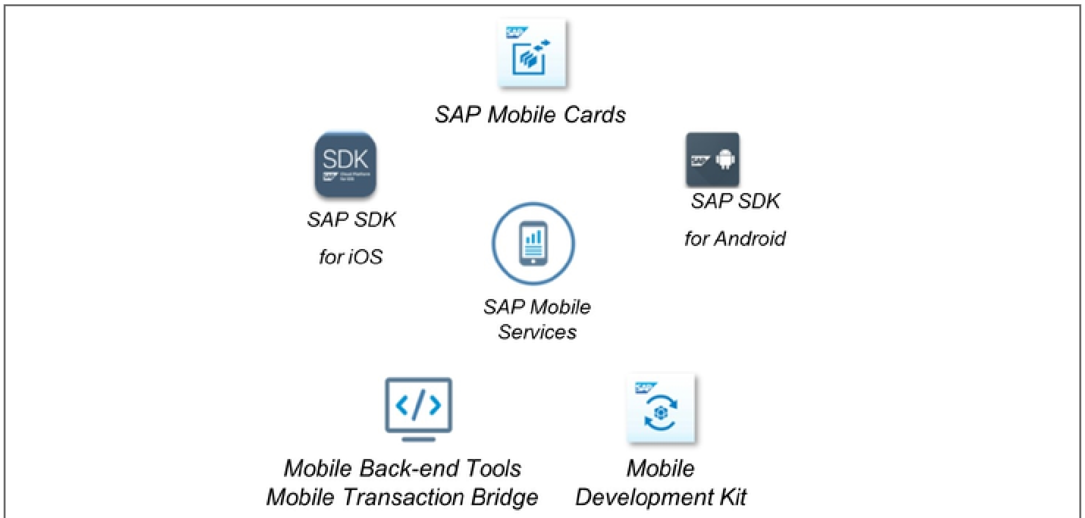


Figure 127: SAP Mobile Services, Overview

SAP Mobile Services:

1: Mobile Cards

2: Platform SDK (iOS, Android)

3: Mobile Development Kit

Key capabilities are:

Offline OData synch, push notification.

Enterprise security with encryption and authentication.

Reporting and Usage analytics.

Scalable supportability features.

Hybrid or full-cloud scenarios supported.

Back-end mobile integration and creation toolset.

Benefits

Increase developer productivity with support for a wide range of mobile app types to meet all use cases.

Engage a highly mobilized workforce, consumers, suppliers and partners with preferred channel.

Increase user productivity via anytime, anywhere connectivity with back end systems.

Scale to meet large enterprise app and user demands.

LESSON SUMMARY
You should now be able to:

Explain which channels are available within the SAP Extension Suite

Unit 4
Lesson 6

Explaining Deployment and Operations

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Identify which tools for the deployment and operate area exist

Deployment and Operations

In this lesson we will cover the following topics:

- Classification in the SAP Extension Suite.
- What is SAP Work Zone?
- What is SAP Fiori Launchpad?
- What is SAP Portal?
- What is Open Piper?
- What are CI/CD services?
- SAP Cloud Transport Management Services

Classification in the SAP Extension Suite

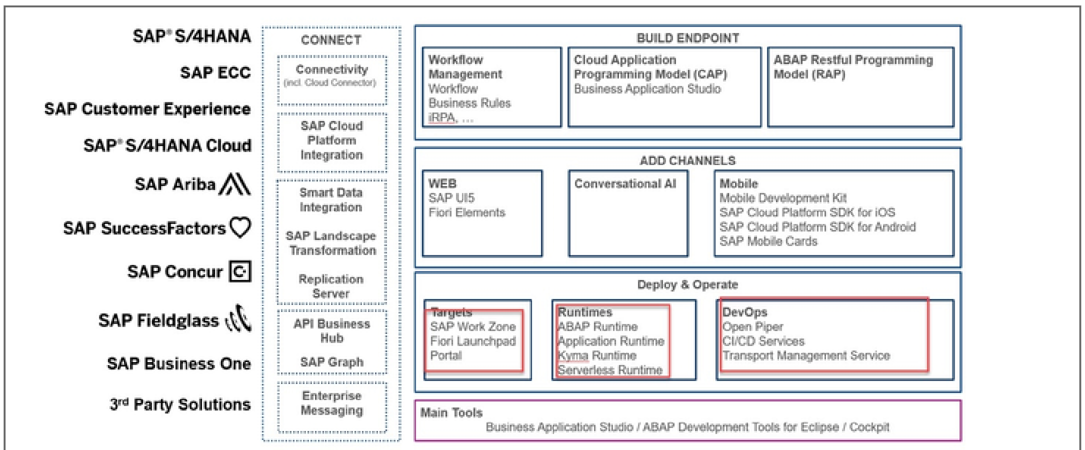


Figure 128: Overview of the Tools Under Consideration

In the following we will deal with the following components:

1: SAP Work Zone

2: SAP Fiori Launchpad

3: Portal

4: Open Piper

5: CI/CD services

6: Transport Management Services

The runtimes are discussed in another lesson.

What is SAP Work Zone?

SAP Work Zone provides an intuitive digital experience solution,

Which increases user engagement and productivity,

By providing a delightful and personalized working environment

To easily access relevant business applications, information and collaboration at the moment of truth.

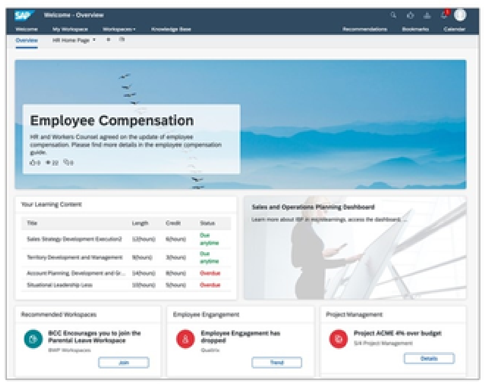


Figure 129: Sample of SAP Work Zone

It is based on Jam, and it is personalized.

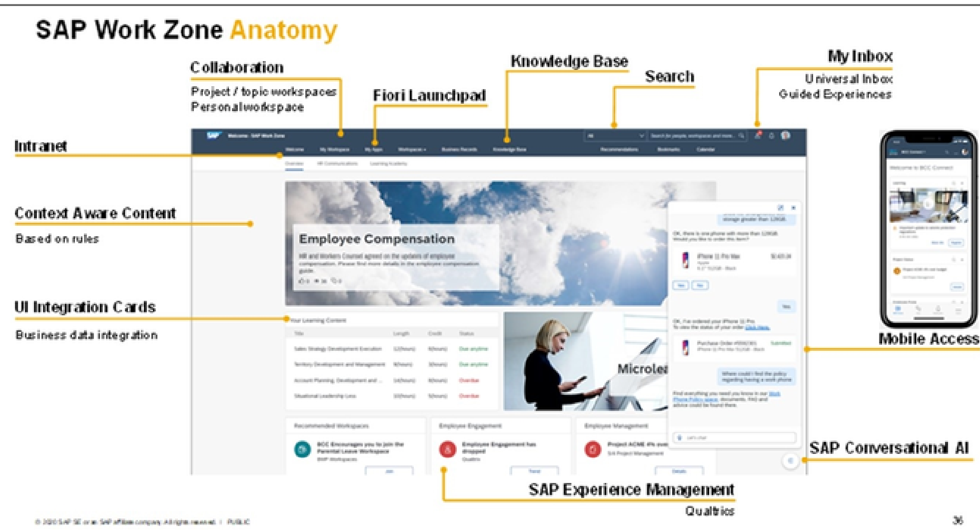


Figure 130: Anatomy of a Work Zone App

All apps, tasks, etc. that I need for my job every day should be on this workplace.

What is SAP Fiori Launchpad ?

SAP Launchpad services enables organizations to establish an intuitive, central entry point,

Unifying access to multiple SAP products and solutions (cloud and on-premise).

Key Capabilities and Benefits are:

Key Capabilities

Central home page providing harmonized user experience.

Role-based access to applications (fro example from multiple SAP S/4HANA systems).

Application integration for different SAP UI technologies and third-party web apps.

Configuration of launchpad services and content (for example custom roles and tiles).

Integration with central services such as inbox, notifications and SSO.

Benefits

Users quickly and easily find relevant apps and services to get their job done efficiently.

Streamlined business roles and end-to-end business processes with app-to-app navigation across products.

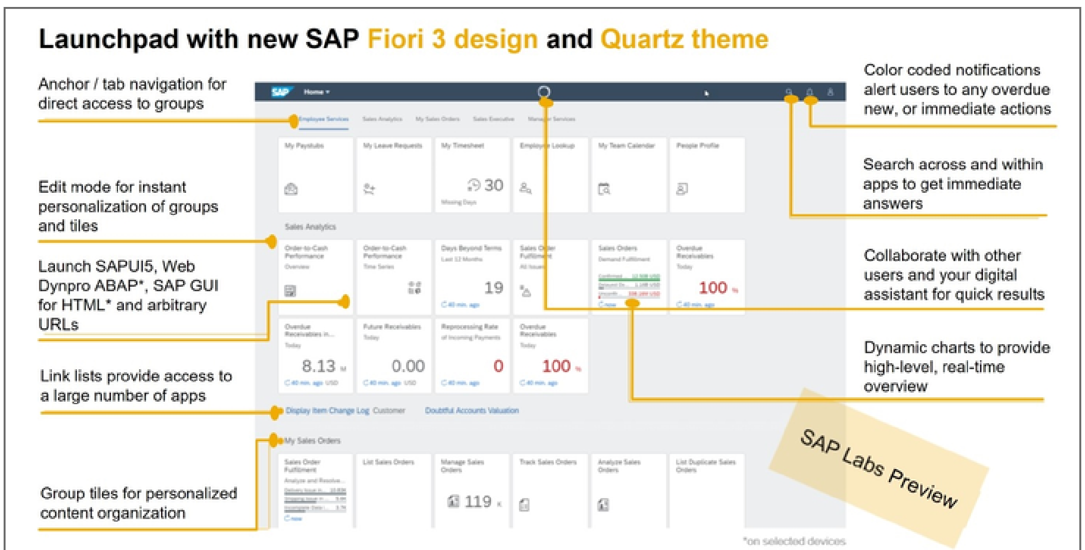


Figure 131: Sap Labs Preview Version of the Launchpad

The difference to SAP Work Zone is, among other things, that there is no collaboration here.

What is SAP Cloud Portal service?

SAP Cloud Portal service empowers organizations to build digital experiences through engaging **business sites**

Key capabilities:

- Build engaging user experiences with SAP Fiori 3 or custom site design.
- Provide **central access to applications** (cloud and on premise).
- Integrate apps based on SAPUI5/Fiori and classic SAP UI technologies.
- Use **public APIs to extend** and optimize site experience.

Benefits:

- Simplify and streamline **access to applications** and information.
- Increase user satisfaction and engagement.
- Drive **UX innovations** and gain agility using cloud services.
- Enhance administration efficiency via templates and open integration.

Use case of portal service

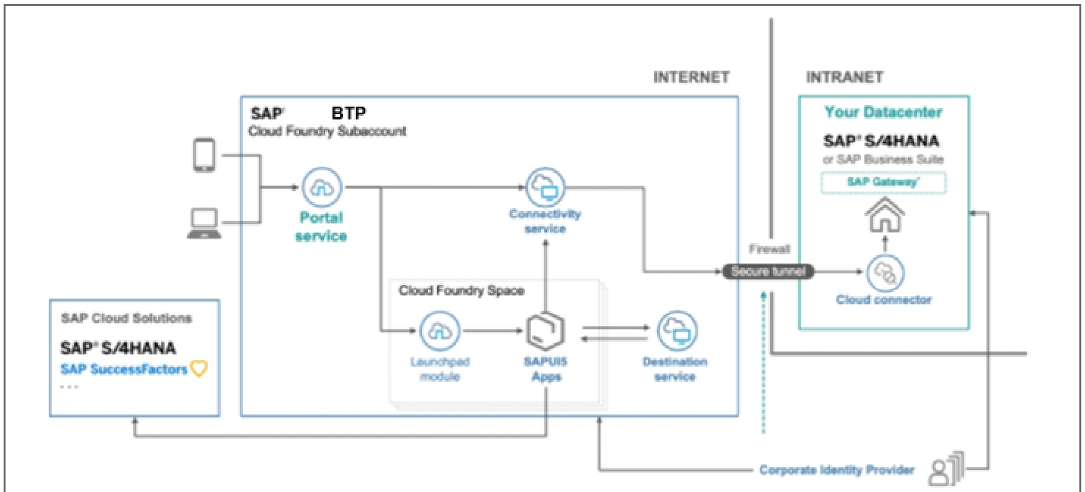


Figure 132: Use Case of Portal Service

The portal can be accessed directly from the mobile phone. The Launchpad modules and apps can be used on the portal.

Sample of an Portal App

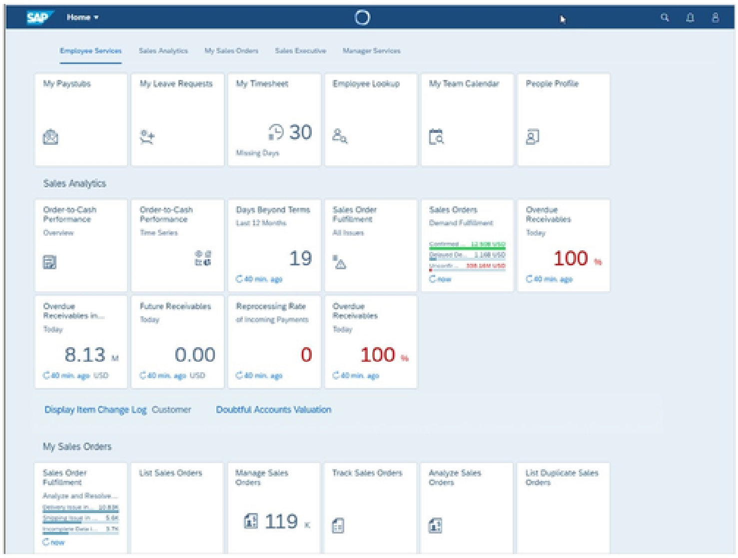


Figure 133: Sample of an Portal App

The figure shows a sample of an Portal App.

CI/CD services

What are CI/CD services?

At the moment, SAP offers three different solutions that help you apply CI/CD in your software development:

- SAP Continuous Integration and Delivery.
- Project "Piper".
- Continuous Integration and Delivery Best Practices Guide. The CI/CD Best Practices Guide provides best practice procedures to implement continuous delivery pipelines on any CI/CD stack and demonstrates how to apply the principles of CI/CD to SAP-specific technologies.

SAP Continuous Integration and Delivery is a service on SAP BTP, which lets you configure and run predefined continuous integration and delivery pipelines that test, build, and deploy your code changes. At the moment, it supports the development of SAP Cloud Application Programming Model (CAP) and SAPUI5/SAP Fiori applications.

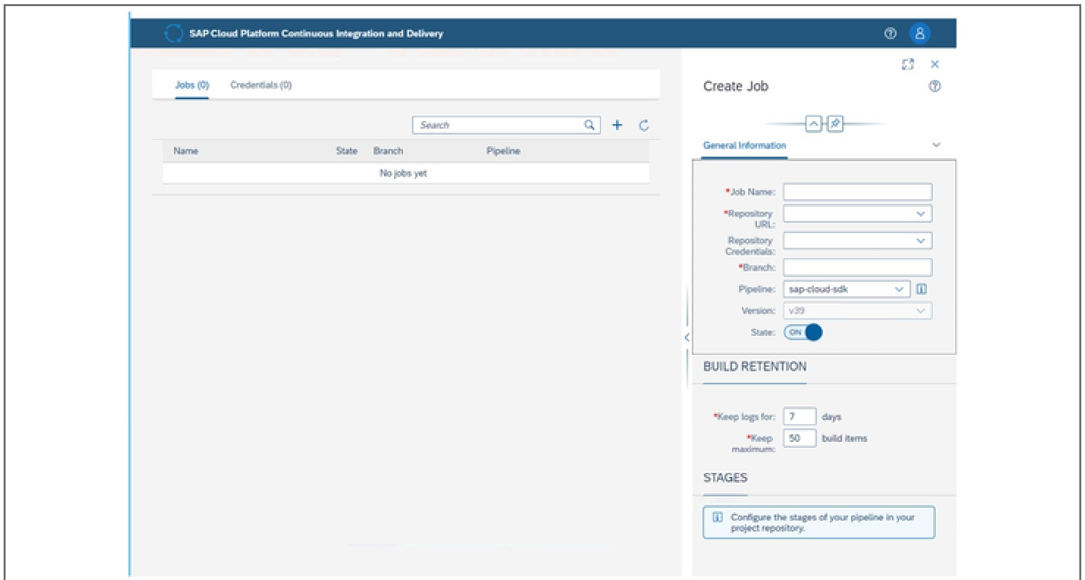


Figure 134: Cockpit of SAP Continuous Integration and Delivery

It is a service on the SAP BTP.

In the cockpit, the individual tasks/jobs are configured.

Open Piper

What is Open Piper?

Continuous delivery is a **method to develop** software with **short feedback cycles**

It is applicable to projects both for **SAP BTP** and SAP **on-premise platforms**

SAP implements tooling for continuous delivery in project "Piper". The goal of project"Piper" is to substantially **ease setting up continuous delivery** in your project using SAP technologies.

Project "Piper" offers you the following artifacts:

A set of ready-made Continuous Delivery pipelines for direct use in your project.

ABAP Environment Pipeline.

General Purpose Pipeline.

SAP Cloud SDK Pipeline.

A shared library that contains reusable step implementations, which enable you to customize our pre-configured pipelines, or to even build your own customized ones.

A standalone command line utility for Linux and a GitHub Action.

Note: This version is still in early development. Feel free to use it and provide feedback, but don't expect all the features of the Jenkins library.

A set of Docker images to setup a CI/CD environment in minutes using sophisticated life-cycle management.

Also used in CAP.

Project "Piper" is an open-source project that provides pre-configured Jenkins pipelines, which you can use in your own Jenkins main infrastructure and adapt according to your needs, if necessary. It consists of two components:

A shared library, which contains the description of steps, scenarios, and utilities required to use Jenkins pipelines.

A set of Docker images.

Sample Pipline in jenkins running at Docker

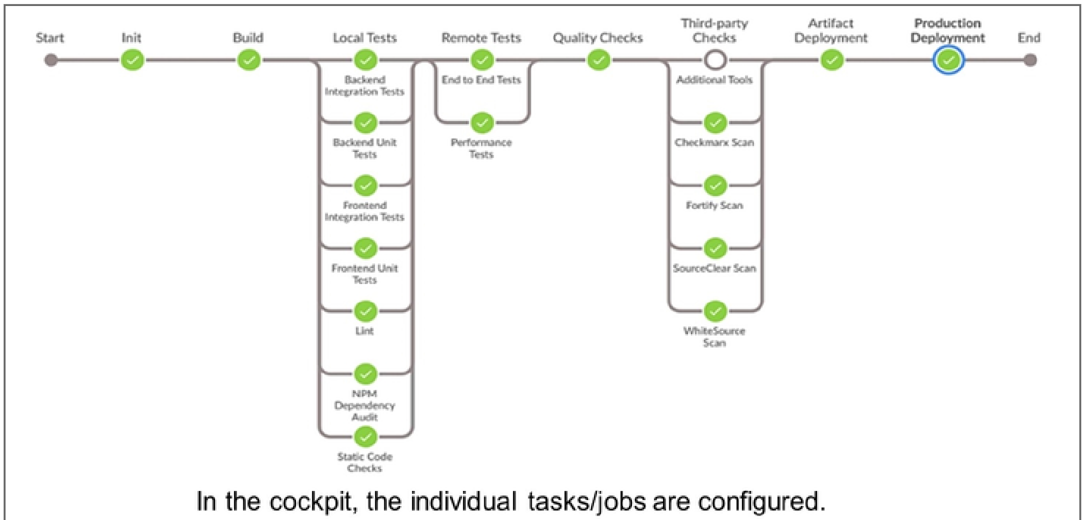


Figure 135: Sample Pipline in Jenkins Running at Docker

You will see several steps with configured functionality, Init, Build, Local Tests …

SAP Cloud Transport Management Services

What are Transport Management Services ?

Manage transports of development artifacts and application-specific content.

SAP Cloud Transport Management lets you manage transports between SAP BTP accounts in Neo and Cloud Foundry environments, such as from DEV to TST and PROD accounts.

It lets you transport development artifacts (in form of Multitarget Application archives) and application-specific content, such as SAP BTP Integration content.

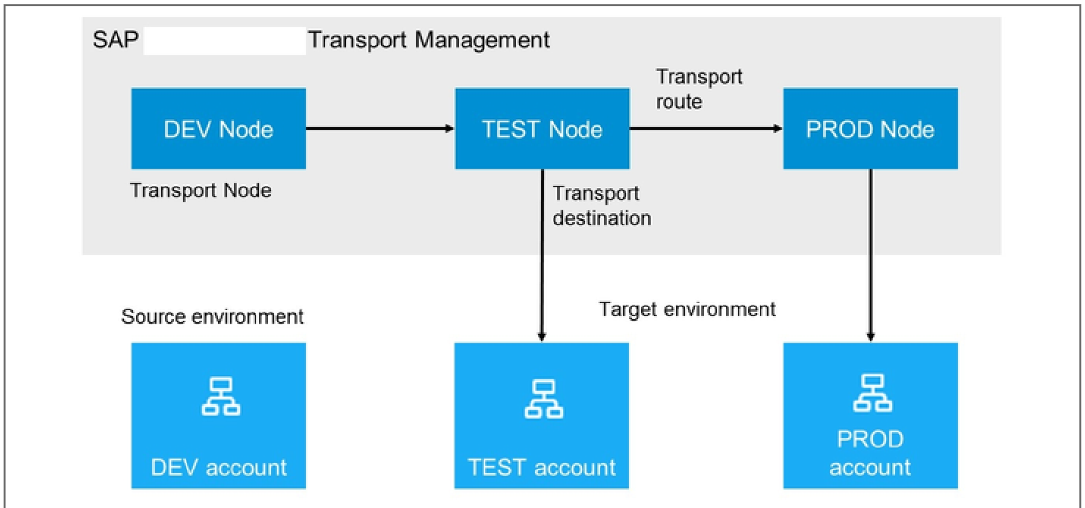


Figure 136: Overview Graphic

Here, the possible things are still shown schematically.

Cockpit of the SAP Cloud Transport Management

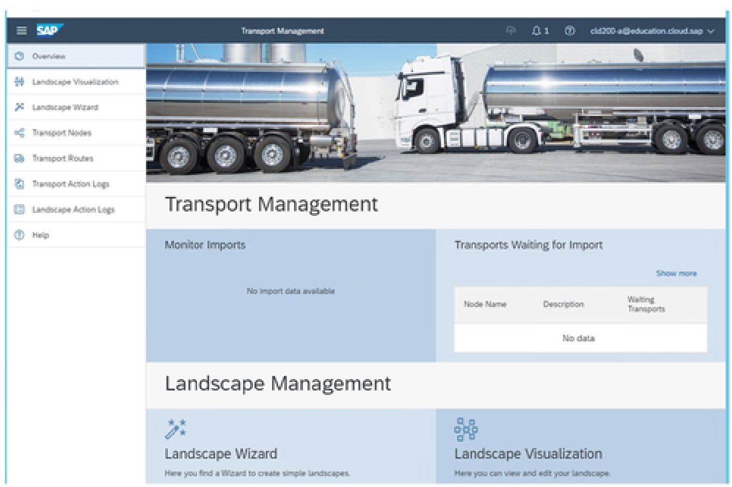


Figure 137: SAP Cloud Transport Management

On the left is the menu with which can configure and monitor transports.

To the point

Focused statement: What is SAP Fiori Launchpad?

SAP Launchpad services enables organizations to establish an intuitive, central entry point.

Unifying access to multiple SAP products and solutions (cloud and on-premise).

Learn more: https://discovery-center.cloud.sap/serviceCatalog/launchpad-service?service_plan=standard&region=all

Focused statement:

Cx Server is a life-cycle management tool to bootstrap a pre-configured Jenkins instance within minutes. All required plugins and shared libraries are included automatically. It is based on Docker images with Linux as OS provided by project "Piper".

Whats New:

There is now also a CI/CD Servive on the SAP BTP that uses this project as an implementation.

https://discovery-center.cloud.sap/serviceCatalog/continuous-integration-&-delivery?region=all

Learn more:

https://sap.github.io/cloud-sdk/

https://www.project-piper.io/

LESSON SUMMARY
You should now be able to:

- Identify which tools for the deployment and operate area exist

Unit 4
Lesson 7

Undergoing a Birds View: In-App Extensions

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Undergo a birds view In-App extensions

In-App Extensions, a Birds View

In this lesson we will cover the following topics:

- Birds view of possible extensions independent of the SAP S/4HANA version.

Birds view of possible extensions independent of the SAP S/4HANA version

All in-app extensions are technically implemented inside the core of SAP S/4HANA.

As a result, no remote communication between the extension and the extended app is required.

SAP S/4HANA offers the following in-app extension options depending on the version.

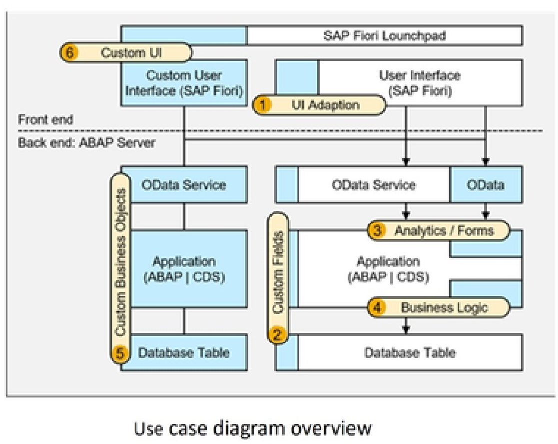


Figure 138: Possible Extensions, Independent of the SAP S/4HANA Version

The figure shows all possible options:

1 User Interface Adaptation

Key users may change the layout of tables and forms directly in the running user interface.

A special graphical user interface adaptation mode provides the mechanisms to hide fields in existing forms, tables, or filters; to rename labels; to add fields to the user interface from the field repository; and to move fields or entire blocks to other sections of the screen.

2 Custom Fields

In the Custom Fields and Logic SAP Fiori application, you can add and edit custom fields to extend SAP database tables, CDS views, and OData APIs.

Moreover, this extension application helps to define how custom fields are used in user interfaces, reports, or forms.

Furthermore, you can use custom fields along with predefined business scenarios so that all involved business objects are extended and values are passed along automatically.

3 Custom CDS Views, Analytics, Forms

The Custom CDS View application enables customers to create new views based on an existing view model supporting features like associations, joins, transformations, field relabeling, selections, etc.

With OData APIs on top of these views, customers may, in addition, expose the data for consumption outside the SAP S/ 4HANA core.

4 Business Logic

Many SAP S/ 4HANA applications have enhancement spots (also called Business Add-Ins [BAdIs]). With the ABAP web editor, customers may create custom logic for the BAdIs.

Typically, customers implement additional checks, set default values, or create mappings in combination with custom fields.

5 Custom Business Objects

Besides custom views, you can create completely new business objects using the Custom Business Objects application.

This application helps define the business object structure in the form of business object nodes, which automatically create the required database tables

6 Custom User Interfaces

You can use the SAP Web IDE to create your own SAP Fiori user interfaces either from scratch or based on templates. The SAP Fiori user interfaces consume SAP S/ 4HANA's RESTful OData interfaces.

To to point

Focused statement:

In an SAP R/3 or SAP S/4HANA on-premise system, additional Business Logic can be added to ABAP in an inApp expansion approach.

Learn more: https://www.sap.com/corporate/de/documents/2018/05/606d1ee8-037d-0010-87a3-c30de2ffd8ff.html

LESSON SUMMARY
You should now be able to:

Undergo a birds view In-App extensions

Unit 4
Lesson 8

Undergoing a Birds View: Side-by-Side Extensibility

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Undergo a birds view Side-by-Side extensibility

Side-By-Side Extensions in SAP S/4HANA, a Birds View

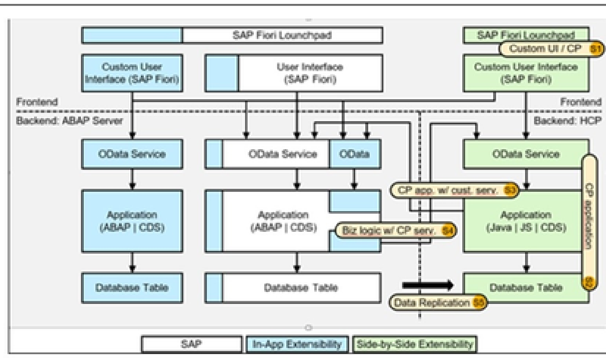
In this lesson we will cover the following topics:

- Birds view of possible extensions independent of the SAP S/4HANA version.

Extensions Independent of the SAP S/4HANA Version

Remember what side-by-side extension means:

- Extension and integration with custom or standard business applications of SAP BTP. Either via stable APIs or through business events from an SAP product, for example SAP S/4HANA.
- All side-by-side extensions are technically implemented outside the core of SAP S/4HANA.
- As a result, remote communication between the extension and the extended app is required.



Use case diagram overview

Figure 139: Possible Extensions Independent of the SAP S/4HANA Version

The picture shows the side-by-side extensibility use cases:

S1: Custom UI on SAP BTP

Custom SAP Fiori UI, build with SAP Web IDE, running on SAP BTP with SAP pre-defined released OData service.

Use UI5 technology, UI5 templates, editors, testing capabilities.

Build UI applications with offline support.

Deploy to your SAP BTP account (or to SAPUI5 ABAP repository).

S2: SAP BTP Application

SAP BTP application (Java, JavaScript, HANA) running on SAP BTP with SAP pre-defined released OData service.

Benefit from open standards and from a partner ecosystem that contribute value to existing solutions and services.

Integration Scenarios (Cloud Integration), benefit from SAP prepackaged integration content as reference templates to quickly realize new business scenarios.

S3: Custom UI on SAP BTP or SAP BTP app w/ custom OData service

Same as S1 and S2, but with custom OData service built with in-app extensibility tools.

S4: New SAP BTP Service called from SAP S/4 Extension

SAP S/4HANA, SAP Success Factors, SAP C/4HANA can fire business events to call apps on the SAP BTP.

S5: Analytics on SAP BTP (Replication)

Data replication to provide fast access to reporting data.

LESSON SUMMARY

You should now be able to:

Undergo a birds view Side-by-Side extensibility

Learning Assessment

1. Which of the following are extension types?

   Choose the correct answers.

   - [ ] **A** Top-down
   - [ ] **B** Classic
   - [ ] **C** In-App
   - [ ] **D** Side-by-Side
   - [ ] **E** Bottom-up

2. To create endpoints, which tools and concepts are available?

   Choose the correct answers.

   - [ ] **A** Workflow management
   - [ ] **B** Cloud Programming Model (CAP)
   - [ ] **C** Micro services
   - [ ] **D** ABAP Restful Programming Model (RAP)

3. The Connectivity service supports the Cloud to on-premise protocols and scenarios. Is HTTPS required?

   Choose the correct answer.

   - [ ] **A** Exchange data between your cloud application and Internet.
   - [ ] **B** HTTPS is not needed, since the tunnel used by the Cloud Connector is TLS-encrypted.
   - [ ] **C** You can use HTTPS-based communication for any client that supports SOCKS5 proxies.

4. The SAP Cloud Application Programming Model is a framework of languages, libraries, and tools for building enterprise-grade services and applications.

   Determine whether this statement is true or false.

   - [ ] True
   - [ ] False

5. Which view formats are available for SAPUI5?

   Choose the correct answers.

   - [ ] **A** XML
   - [ ] **B** HTML
   - [ ] **C** JavaScript
   - [ ] **D** JSON

6. CI/CD stands for Continuous Integration/Continuous Delivery

   Determine whether this statement is true or false.

   - [ ] True
   - [ ] False

7. Which options are available for In-App extensions?

   Choose the correct answers.

   - [ ] **A** Custom Fields
   - [ ] **B** Business Logic
   - [ ] **C** Custom Business Objects
   - [ ] **D** Custom UserInterfaces

8. Which statements are correct, concerning side-by-side extensibility?

   Choose the correct answers.

   - [ ] **A** All side-by-side extensions are technically implemented outside the core of SAP S/4HANA.
   - [ ] **B** Side-by-side extensibility means, that in an extensibility object multiple programming languages can be used.
   - [ ] **C** As a result, remote communication between the extension and the extended app is required.
   - [ ] **D** Extension and integration with custom or standard business applications of the SAP BTP. Either via stable APIs or through business events from an SAP product, for example SAP S/4HANA.

Learning Assessment - Answers

1. Which of the following are extension types?

   Choose the correct answers.

   - [ ] **A** Top-down
   - [x] **B** Classic
   - [x] **C** In-App
   - [x] **D** Side-by-Side
   - [ ] **E** Bottom-up

   This is correct.

2. To create endpoints, which tools and concepts are available?

   Choose the correct answers.

   - [x] **A** Workflow management
   - [x] **B** Cloud Programming Model (CAP)
   - [x] **C** Micro services
   - [x] **D** ABAP Restful Programming Model (RAP)

   This is correct.

3. The Connectivity service supports the Cloud to on-premise protocols and scenarios. Is HTTPS required?

   Choose the correct answer.

   - [ ] **A** Exchange data between your cloud application and Internet.
   - [x] **B** HTTPS is not needed, since the tunnel used by the Cloud Connector is TLS-encrypted.
   - [ ] **C** You can use HTTPS-based communication for any client that supports SOCKS5 proxies.

   This is correct.

4. The SAP Cloud Application Programming Model is a framework of languages, libraries, and tools for building enterprise-grade services and applications.

   Determine whether this statement is true or false.

   - [X] True
   - [ ] False

   This is correct.

5. Which view formats are available for SAPUI5?

   Choose the correct answers.

   - [X] **A** XML
   - [X] **B** HTML
   - [X] **C** JavaScript
   - [X] **D** JSON

   This is correct.

6. CI/CD stands for Continuous Integration/Continuous Delivery

   Determine whether this statement is true or false.

   - [X] True
   - [ ] False

   This is correct.

7. Which options are available for In-App extensions?

   Choose the correct answers.

   - [X] **A** Custom Fields
   - [X] **B** Business Logic
   - [X] **C** Custom Business Objects
   - [X] **D** Custom UserInterfaces

   This is correct.

8. Which statements are correct, concerning side-by-side extensibility?

   Choose the correct answers.

   - [x] **A** All side-by-side extensions are technically implemented outside the core of SAP S/4HANA.
   - [ ] **B** Side-by-side extensibility means, that in an extensibility object multiple programming languages can be used.
   - [x] **C** As a result, remote communication between the extension and the extended app is required.
   - [x] **D** Extension and integration with custom or standard business applications of the SAP BTP. Either via stable APIs or through business events from an SAP product, for example SAP S/4HANA.

   This is correct.

UNIT 5 Security

UNIT OBJECTIVES

- Figure out, how the domain model works and name the main services
- Differentiate between authentication and authorization and how both are used on SAP BTP
- Appreciate the secure implementation of a microapp

Unit 5
Lesson 1

Introducing Application Security

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Figure out, how the domain model works and name the main services

The Domain Model

In this lesson we will cover the following topics:

- Current Domain Model
- SAP Product Context

Current Domain Model

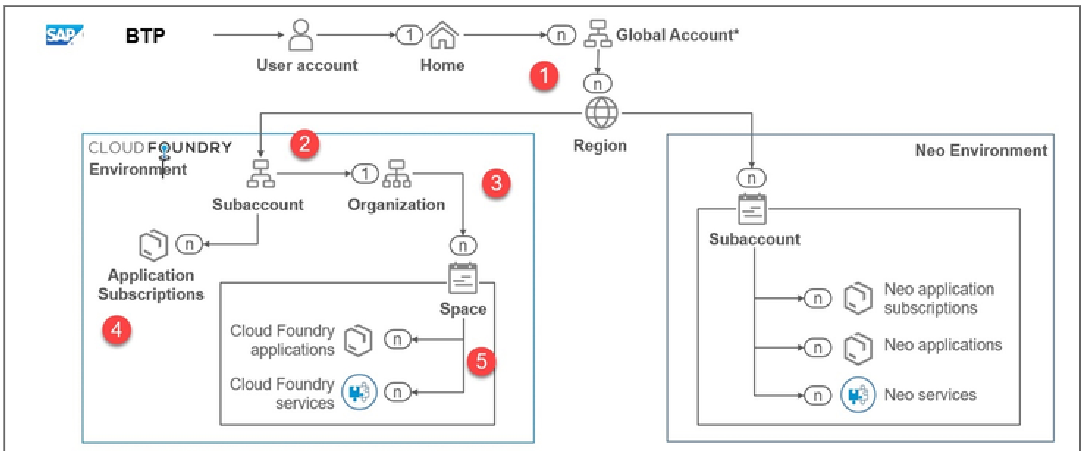


Figure 140: Current Domain Model

The figure above illustrates the current domain model.

Global Account

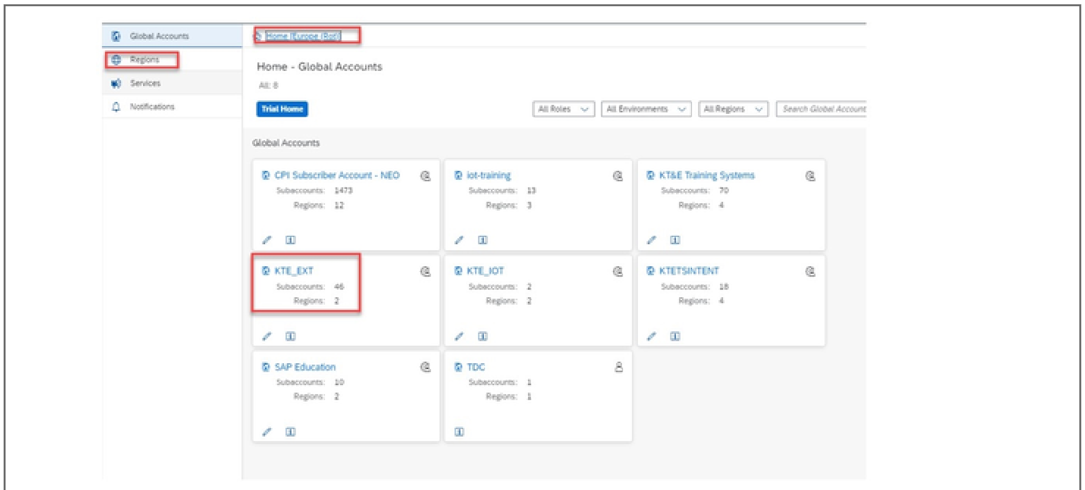


Figure 141: Global Account

A global account has a region, here red (1).

Subaccount

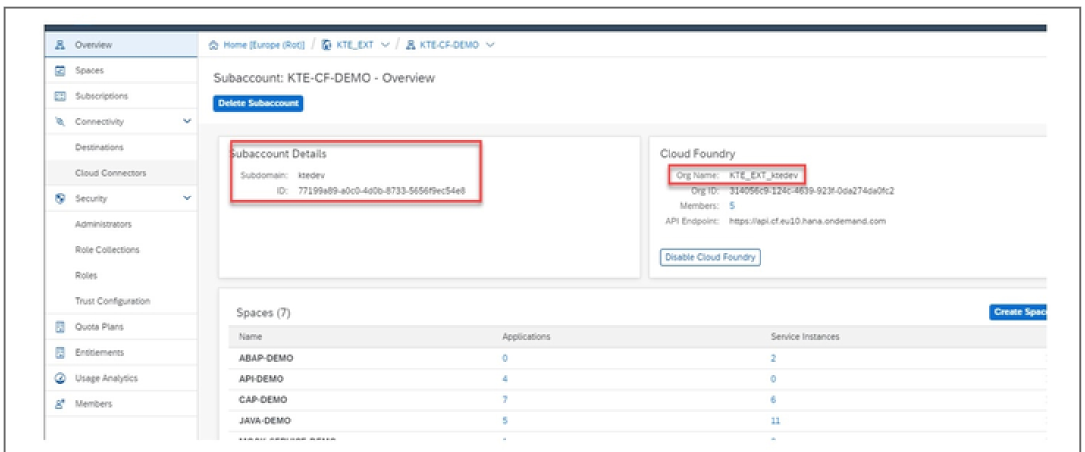


Figure 142: Subaccount

The figure illustratses, that a subaccount has a subaccount ID and exactly one organization (2) and (3).

Spaces

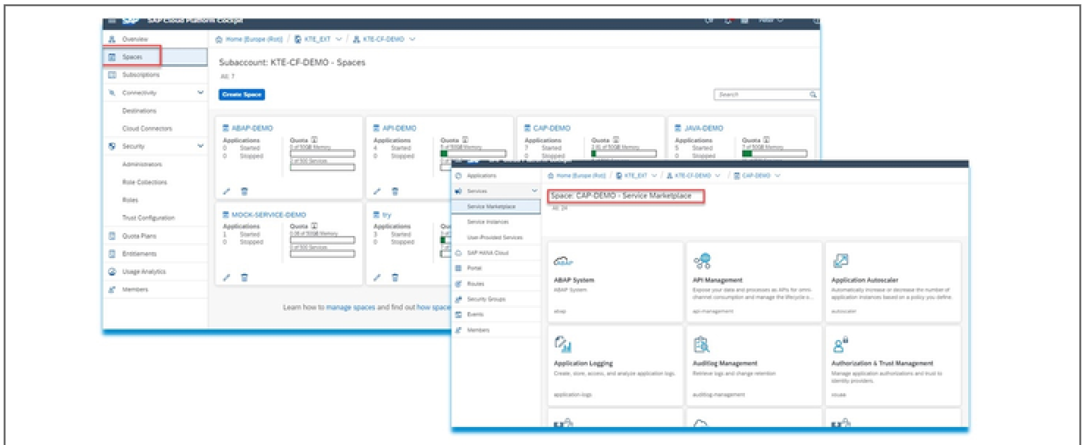


Figure 143: Spaces

The figure illustrates, that a subaccount has n Spaces (5) with services and applications in the Marketplace.

SAP Product Context

First, an overview of the services and tools in the area of security.

Figure 144: SAP Product Context

The figure illustrates the SAP product context. In detail:

Authentication & Single Sign-on

SAP Identity Authentication

SAP Single Sign-On

Identity Management

SAP BTP Identity Provisioning service

SAP Identity Management

Governance, Risk & Compliance

SAP Cloud Identity Access Governance

SAP Access Control

We focus on the services Identity Authentication and Identity Provisioning that help us to work safely.

Information about the Identity Authentication:

Simplify and secure cloud-based access to business processes, applications, and data with state-of-the-art authentication mechanisms, single sign-on, on-premise integration, and convenient self-service options

These are:

- Two-factor authentication
- Customer and partner onboarding
- Secure integration

SAP Cloud Identity Access Governance

Streamline identity and access management (IAM) in complex on-premise and cloud environments with SAP Cloud Identity Access Governance software. You can improve IAM practices with an intuitive, dashboard-driven interface and a simple single sign-on (SSO) experience in the cloud.

Information about the Identity Provisioning:

Automate identity lifecycle processes to provision identities and their authorizations to cloud and on-premise applications.

Define user access based on identity attributes such as current group or role assignment and location.

LESSON SUMMARY
You should now be able to:

Figure out, how the domain model works and name the main services

Unit 5
Lesson 2

Explaining Platform Security

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Differentiate between authentication and authorization and how both are used on SAP BTP

Platform Security

In this lesson we will cover the following topics:

- Identity Authentication: IDP.
- Identity Authentication: Service.
- User and Role Management on SAP BTP.
- Identity Provisioning Service.
- Bringing all together.

Identity Authentication: IDP

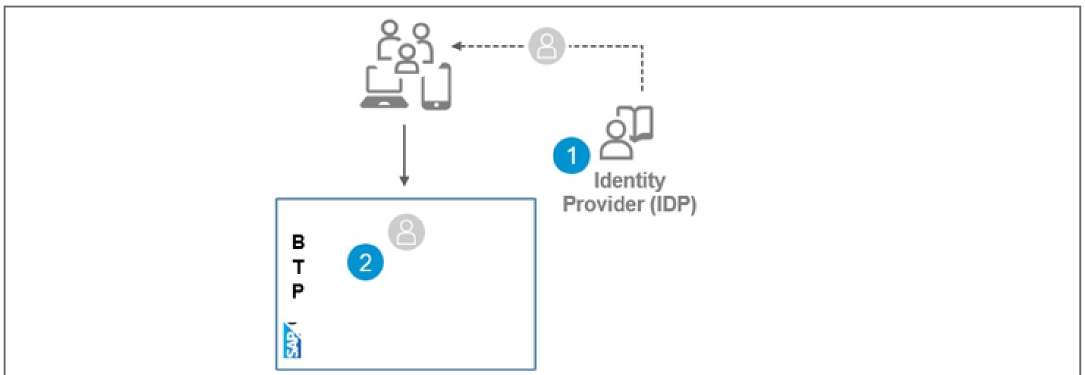


Figure 145: Identity Authentication: IDP

SAP BTP uses out of the box, the Identity Providers (IDP) for user authentication. He has the role of a user store:

(1) External Authentication providers (SAML).

(2) Authenticated identity is used inside SAP BTP.

Freedom of choice of IDP

SAML2 standard provides choice of authentication provider.

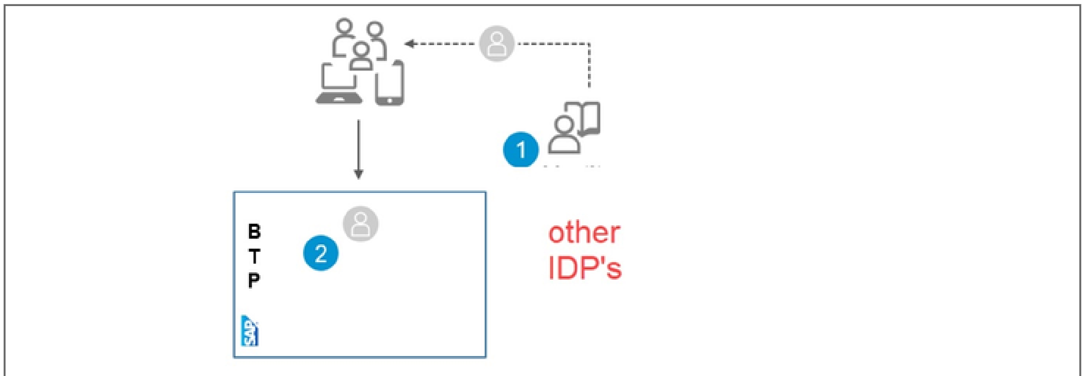


Figure 146: Freedom of Choice of IDP

SAP BTP Identity Authentication can use on-premise IDP's (AD, LDAP, SAP) for user authentication:

Re-use existing IDP, easy to implement.

Maintain central user repository (no user sync needed).

SAML2 capable IDP's can be integrated, for example:

Microsoft ADFS

Microsoft AZURE

Change the IDP on Subaccount Level

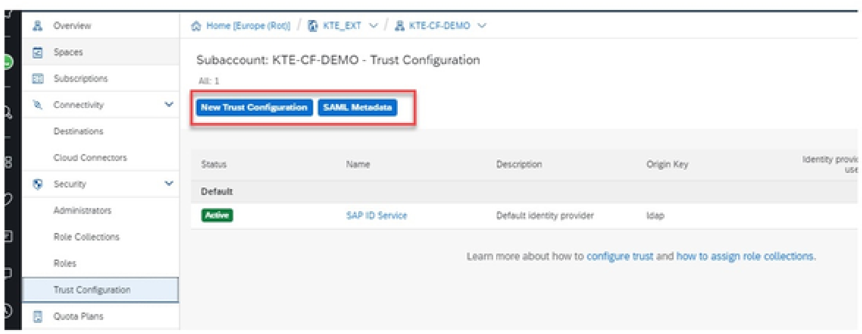


Figure 147: Change the IDP on Subaccount Level

You can change the SAML 2.0 IDP Provider on subaccount level.

SAML 2.0 Response after successful Login to IDP

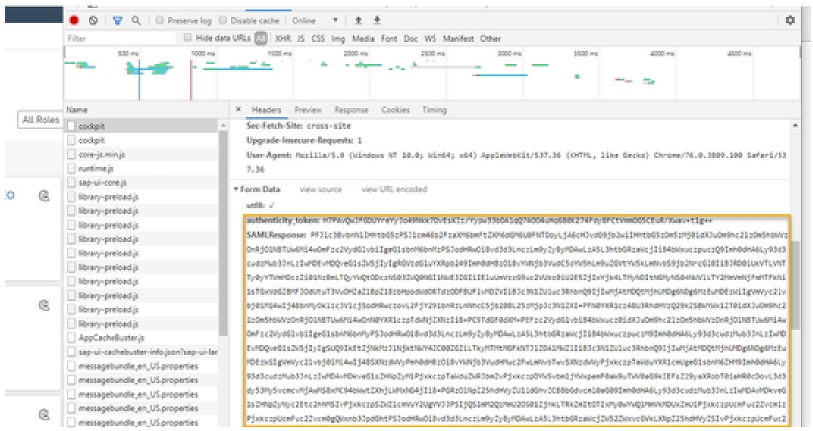


Figure 148: SAML 2.0 Response After Successful Login to IDP

Here is the SAML 2.0 Response from IDP to SAP BTP after successful login.

Identity Authentication: Service

By default, SAP always uses the IDP as the identity provider. However, it only offers basic functions like User Authentication as a user store.

To take advantage of all possibilities, the Identity Authentication Service can be licensed. It offers almost all conceivable options based on SAML 2.0 or OpenID Connect Standard. Additionally, you still need an Identity Provider, for example the IDP from SAP or from third-party companies.

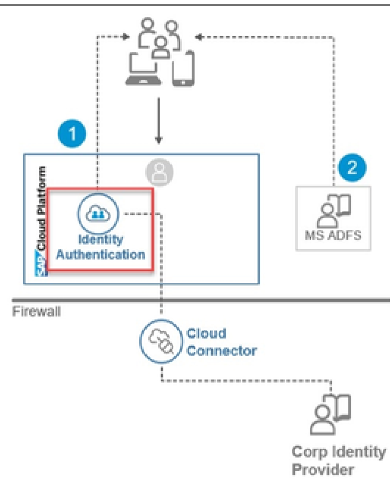


Figure 149: SAP Identity Authentication Service

The example above uses two IDPs.

The Features are:

Basic authentication

Re-use of Windows Domain logon

- Two-factor authentication
- Delegated logon

User and Role Management on SAP BTP

Table 6: User and Role Management on SAP BTP

Type	Description	Authentication Configuration
Platform Users	Member on Global - and sub-account, members on space level	Platform IDP, on Global Account Level
Application Developer/ Users	User that use Subscriptions and/or Market Place Services. Developers or Business developer.	IDP on subaccount Level
Business User	User that use business apps.	IDP on subaccount Level

No user identities are held on the SAP BTP. However, domain-dependent system and service roll and groups are used.

These roles and groups are either created directly on the SAP BTP, for example, or existing ones are imported and mapped to the Platform Roles or Groups. This is done with the SAP Cloud Identity Provisioning Service.

You can identify the following user types. A developer can also be a business.

Roles and Groups on the SAP BTP: Platform Users

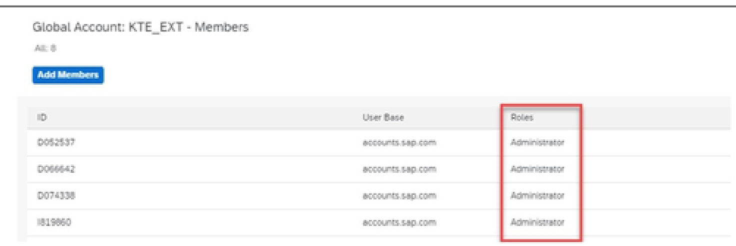


Figure 150: Platform Users: Global Account

On global level the Administrator role is assigned.

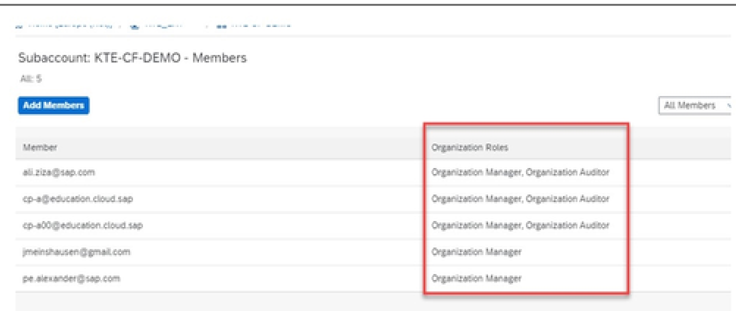


Figure 151: Platform Users: Subaccount

On subaccont level, the Organization Roles are assigned.

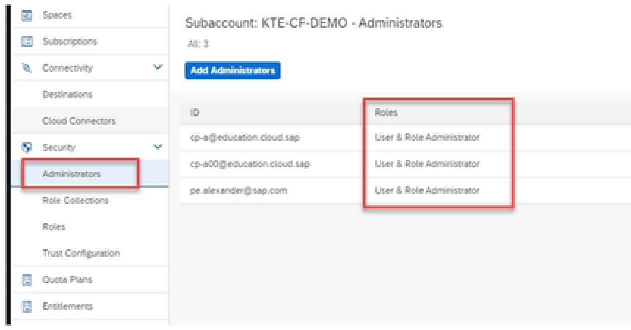


Figure 152: Platform Users, Subaccount: Security Administrator

On subaccont level also the Security Administrators are assigned.

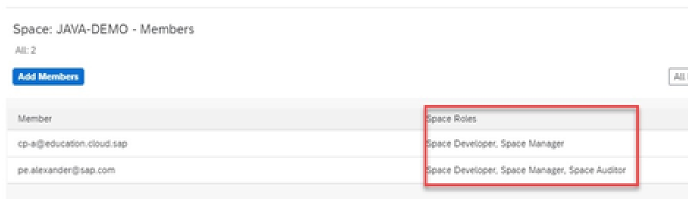


Figure 153: Platform Users: Space

On space level the Spaces Roles are assigned.

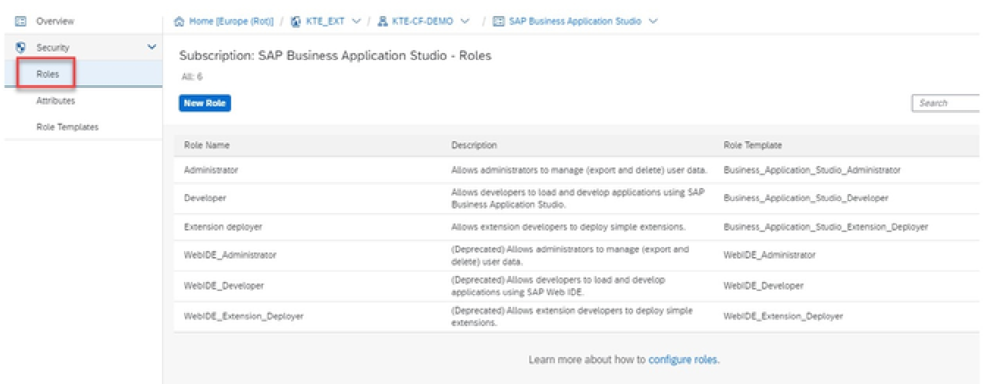


Figure 154: Application Users: Service Roles

However, the service roles still have to be assigned to the corresponding user. You can see how this can be done in the case of your own service in the next Lesson with a Business User.

SAP Provisioning Service

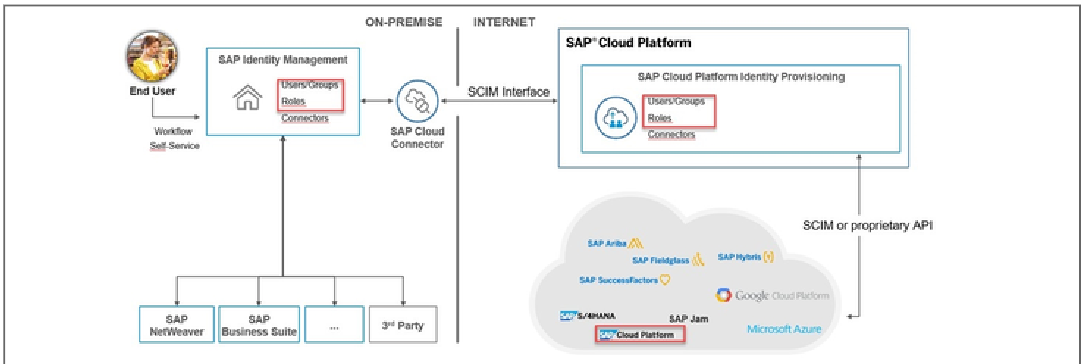


Figure 155: SAP Identity Provisioning Service

In order to use existing roles and groups, for example in the SAP BTP, these can be mapped manually via the SAP Identity Provisioning Service.

Bringing all together

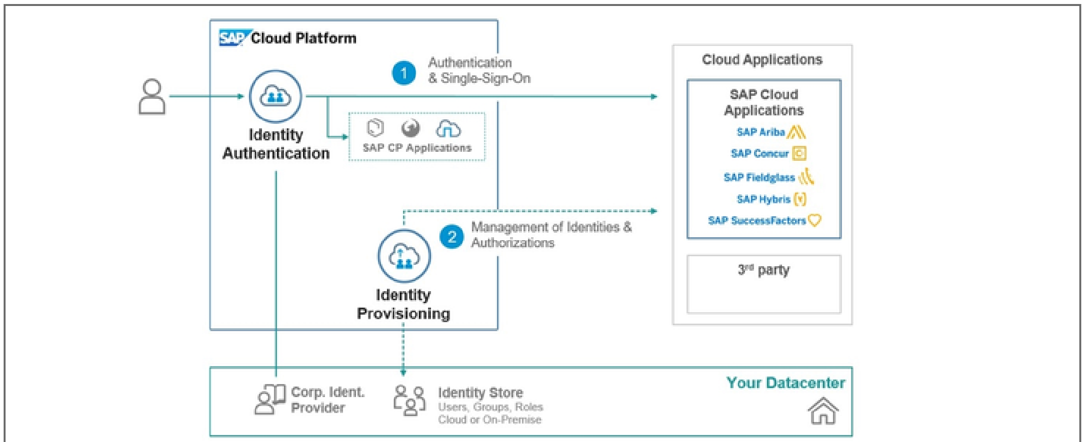


Figure 156: Bringing all Together

The combination of authentication and authorization looks like. This would also be the architecture to be aspired to. The SAP BTP Platform as a security hub.

LESSON SUMMARY

You should now be able to:

- Differentiate between authentication and authorization and how both are used on SAP BTP

Unit 5
Lesson 3

Explaining Application Security

LESSON OBJECTIVES

After completing this lesson, you will be able to:

- Appreciate the secure implementation of a microapp

Application Security

In this lesson we will cover the following topics:

- How Application Security works.
- The Approuter
- JSON Web Token (JWT)
- Extended Services for User Account and Authentication (XSUAA).
- Assigning Users to Application Roles.

How Application Security works

We limit ourselves to what we're doing in Cloud Foundry. The following example describes a microservices in Java.

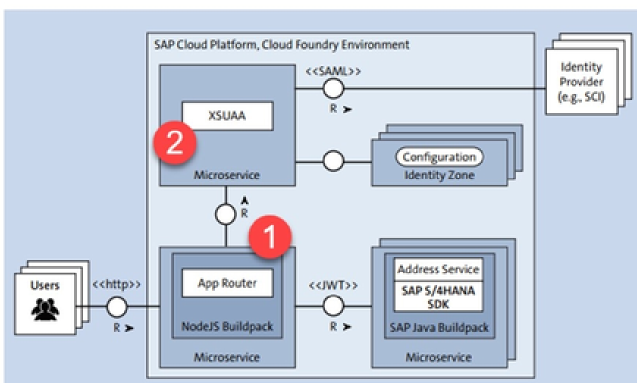


Figure 157: Main Blocks of Application Security

The figure above illustrates a high-level authentication setup.

Further details:

The App Router (1)

Its tasks are:

The App Router is the central entry point for our application it dispatches requests to our back end microservices, thus acting as a reverse proxy. The back end microservices should not be directly accessible by the client.

The App Router can serve static content such as web pages, SAPUI5, or another client-side code.

The App Router manages the authentication flows for our entire application

For authentication (who the user is) and authorization (what the user is allowed to do), the App Router takes all incoming, unauthenticated request and initiates an OAuth2 flow (authorization code grant) with the Extended Services for User Account and Authentication (XSUAA) service of the SAP BTP in the Cloud Foundry environment .

Install via Service Marketplace

The App Router is a Node.js component, distributed via the publically available SAP NPM registry.

The Application Router's Design-Time Descriptor: xs-app.json

```
{
  "welcomeFile": "index.html",
  "routes": [{
  "source": "^/api/(.*)",
    "target": "/api/$1",
    "destination": "app-destination"
  },{
  "source": "^/address-manager/(.*)",
  "target": "/address-manager/$1",
  "destination": "app-destination"
  }]
}
```

Figure 158: Router's Design-Time Descriptor: xs-app.json

The xs-app.json file can be used to configure many more aspects at design-time of the AppRouter like authentication types, cache control, compression threshold for outgoing traffic (by default, 1 KB), client-initiated logout, custom settings for CSRF protection, and much more.

The AppRouter's Runtime Configuration within the manifest.yaml

```
---
applications:
- name: approuter
  routes:
    - route: approuter- ↩
        <subdomain>.cfapps.eu10.hana.ondemand.com
  path: approuter
  memory: 128M
  env:
    TENANT_HOST_PATTERN: 'approuter- ↩
          (.*).cfapps.eu10.hana.ondemand.com'
    destinations: '[{"name":"app-destination", "url": ↩
          "https://address-manager-unsparse-subutopian ↩
          .cfapps.eu10.hana.ondemand.com", ↩
          "forwardAuthToken": true}]'
  services:
    - my-xsuaa
```

Figure 159: Runtime Configuration Within the manifest.yaml

Besides destinations and tenant host patterns, the AppRouter understands many different parameters that can be changed during runtime, without redeploying the entire AppRouter, which is consistent with the twelve-factor application principles. Recognized parameters include cross-origin policies, session timeouts, compression thresholds, JWT refresh times, etc.

The Application Security Descriptor: xs-security.json

```
{
  "xsappname": "extensibility-demo-advanced",
  "tenant-mode": "shared",
  "scopes": [
    {
      "name": "$XSAPPNAME.Display",
      "description": "display"
    }
  ],
  "role-templates": [
    {
      "name": "Viewer",
      "description": "Required to view things in our solution",
      "scope-references"     : [
        "$XSAPPNAME.Display"
      ]
    }
  ]
```

Figure 160: Security Descriptor: xs-security.json

The xs-security.json descriptor is an important design-time artifact to provide authorization scopes, role templates, and other attributes of the application itself as well as foreign applications. This file will be imported while the XSUAA Service will be create.

JSON Web Token (JWT)

JSON Web Token (JWT) is an open standard (RFC 7519) that defines a compact and self-contained way for securely transmitting information between parties as a JSON object. This information can be verified and trusted because it is digitally signed. JWTs can be signed using a secret (with the HMAC algorithm) or a public/private key pair using RSA or ECDSA.

Although JWTs can be encrypted to also provide secrecy between parties, we will focus on signed tokens. Signed tokens can verify the integrity of the claims contained within it, while encrypted tokens hide those claims from other parties. When tokens are signed using public/private key pairs, the signature also certifies that only the party holding the private key is the one that signed it.

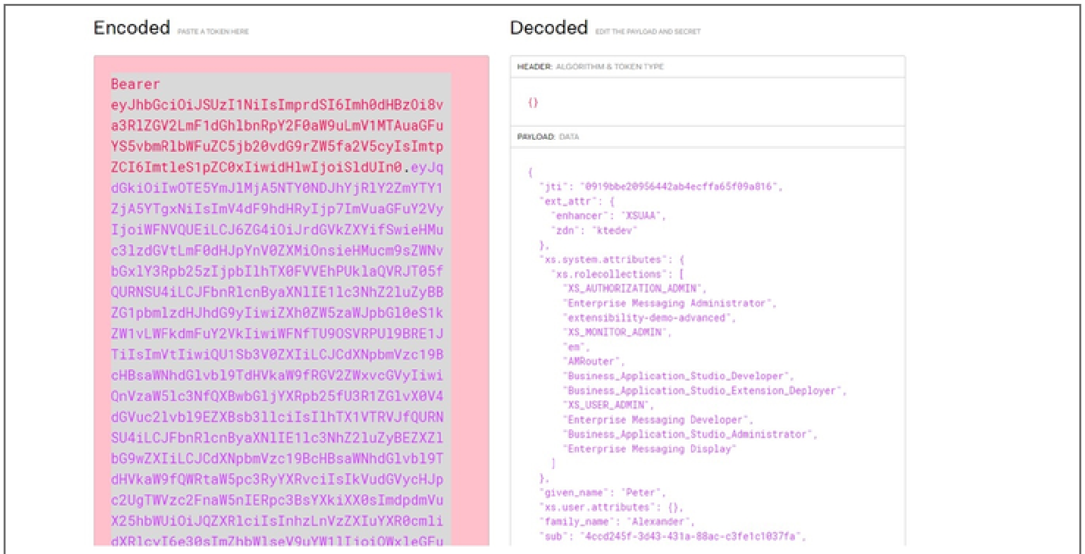


Figure 161: JWT Sample

The figure shows an example of a JWT.

When should you use JSON Web Tokens?

Here are some scenarios where JSON Web Tokens are useful:

Authorization

This is the most common scenario for using JWT. Once the user is logged in, each subsequent request will include the JWT, allowing the user to access routes, services, and resources that are permitted with that token. Single Sign On is a feature that widely uses JWT nowadays, because of its small overhead and its ability to be easily used across different domains.

Information Exchange

JSON Web Tokens are a good way of securely transmitting information between parties. Because JWTs can be signed-for example, using public/private key pairs-you can be sure the senders are who they say they are. Additionally, as the signature is calculated using the header and the payload, you can also verify that the content hasn't been tampered with.

User Assignment to Application Roles

Figure 162: Assigning Users to Application Roles

In the xs-security.json we define Role Templates, Roles and Scopes. We need to assign these to users who want to use our app in the subaccount under Trust.

To do this, we need to create a role collection in the subaccount and assign the role-template from the xs-security.json.

Procedure to assign users:

Create the Role Collection in the subaccount.

Assign the role template from the xs-security.json to the Role Collection.

Under Trust Configuration select the current SAP ID service and enter the corresponding user as email address.

Add the Role Collection to the user.

Protecting the back end application

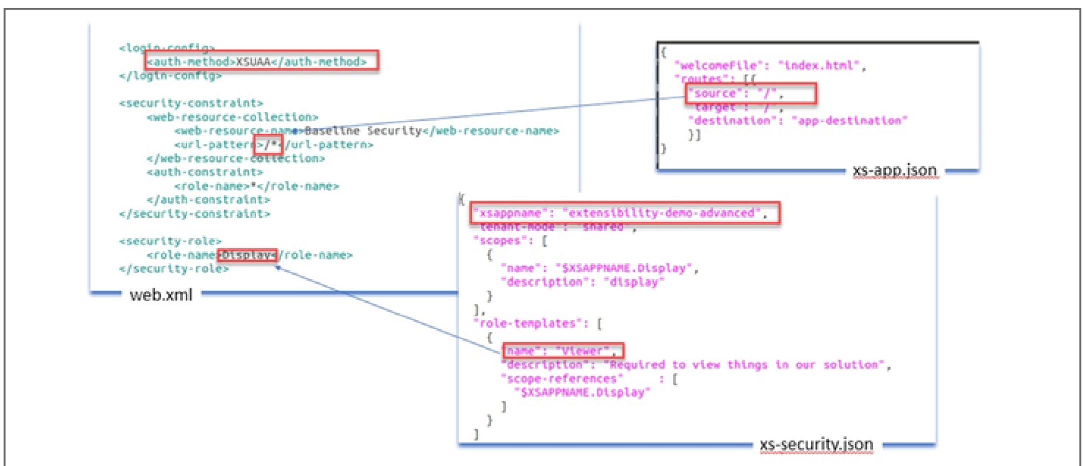


Figure 163: Use Scopes to Protect Your App

Now that we've introduced scopes and role templates to the XSUAA service instance, we'll need to protect our application accordingly. Basically, two places exist where you can check authorizations: inside the AppRouter and in the back end microservices.

Use Scopes to protect your app:

Within the xs-app.json you define the protected routes.

In the web.xml of your back end service you set the security requirements for Authentication and Authorization.

Protect the Implementation

Figure 164: Protect the Implementation

At the method level in my implementation you assign the scope.

To the point

Focused statement about the App Router (1). the tasks are:

The App Router is the central entry point for our application it dispatches requests to our back end microservices, thus acting as a reverse proxy. The back end microservices should not be directly accessible by the client.

The App Router can serve static content such as web pages, SAPUI5, or another client-side code.

The App Router manages the authentication flows for our entire application

LESSON SUMMARY

You should now be able to:

Appreciate the secure implementation of a microapp

Learning Assessment

1. The current domain model consists of:

   Choose the correct answer.

   ☐ **A** Space, Account, Subaccounts

   ☐ **B** Domain Host, Spaces, Accounts

   ☐ **C** Account, Subaccount(s), Spaces

2. Identity providers (IDP) are used for user authentication.

   Determine whether this statement is true or false.

   ☐ True

   ☐ False

3. The AppRouter is a Node.js component, distributed via the publically available SAP NPM registry.

   Determine whether this statement is true or false.

   ☐ True

   ☐ False

Unit 5

Learning Assessment - Answers

1. The current domain model consists of:

   Choose the correct answer.

   ☐ **A** Space, Account, Subaccounts

   ☐ **B** Domain Host, Spaces, Accounts

   ☒ **C** Account, Subaccount(s), Spaces

   This is correct. Correct is the answer option: Account, Subaccount(s), Spaces.

2. Identity providers (IDP) are used for user authentication.

   Determine whether this statement is true or false.

   ☒ True

   ☐ False

   This is correct. The statement is correct.

3. The AppRouter is a Node.js component, distributed via the publically available SAP NPM registry.

   Determine whether this statement is true or false.

   ☒ True

   ☐ False

   This is correct. The statement is correct.